The Maturity Insurance Channel of Monetary Transmission*

Giada Bozzelli
JOB MARKET CANDIDATE
Boston College
bozzelli@bc.edu

Francesca Barbiero
European Central Bank

06 November 2026

Click here for the latest version of the paper

Abstract

How does endogenous corporate debt maturity shape monetary transmission? We show that it depends on both the determinants of long-term interest rates and firms' refinancing needs. Using euro-area credit-registry data, we find that firms with greater refinancing needs borrow at relatively longer maturities when expected future short rates rise, but relatively shorter maturities when term premia increase. Our empirical design exploits predetermined repayment schedules, within-bank-month comparisons, and monetary-policy surprises. A heterogeneous-firm model with financial frictions reproduces these untargeted differential maturity responses and shows that maturity flexibility mitigates investment contractions and reduces default following news of higher future rates. The investment benefits of maturity flexibility are larger when expected-rate news coincides with an increase in term premia. Applying the model to the 2022–24 euro-area tightening cycle, we show that firms enter tightening with longer-maturity debt accumulated during the preceding low-rate period, while expected-rate news induces earlier maturity adjustment that reduces subsequent default without additional deleveraging.

Keywords: monetary policy transmission, debt maturity, interest rate risk, yield curve, term premium

JEL Codes: E52, G32, E60, E50, E43

*Corresponding author: Giada Bozzelli. I am indebted to my advisors Rosen Valchev, Fabio Schiantarelli, Jaromir Nosal. I also thank Enrico Sette, Caterina Mendicino, Phil Strahan, Robert Ulbricht, Susanto Basu, Pablo Guerron, Bora Durdu, Sarah Zoi, Levent Altinoglu, Mariassunta Giannetti, Luigi Falasconi, Sergio Villavalzo, Michele Modugno, Daniel Barth, Alexandros Vardoulakis, Matt Darst, Jose Berrospide, David Rapport, Jin-Wook Chang, Matteo Lembroni, Pierluigi Balduzzi and seminar and conference participants at Boston College, FED Board, WEAI CSWEP, 6th WE_ARE_IN Macroeconomics and Finance Conference, Fall Midwest Macroeconomics Conference for helpful comments. Thanks to Chengcheng Jia for discussing the paper. The views expressed are those of the authors and do not necessarily reflect those of the ECB or the Eurosystem. A substantial part of this work has been conducted while visiting the ECB Monetary Policy Division from June to December 2025. All errors are our own.

1 Introduction

A central insight of monetary transmission is that borrowers' existing financial obligations shape their exposure to changes in interest rates. Firms with substantial debt coming due must refinance at prevailing market conditions, while those with longer-maturity debt can postpone refinancing (Bernanke and Gertler, 1995; Almeida et al., 2012). Similarly, existing mortgage contracts and refinancing opportunities generate heterogeneity and path dependence in monetary transmission to households (Berger et al., 2021; Eichenbaum et al., 2022). Yet debt maturity is not merely an inherited source of exposure: it is an endogenous choice that trades off future refinancing risk against the cost of long-term borrowing (Diamond, 1991; Arellano and Ramanarayanan, 2012).

This paper asks how non-financial corporations' ability to adjust debt maturity shapes monetary transmission to the real economy. With financial frictions and idiosyncratic uncertainty, long-maturity debt provides insurance against the need to refinance under adverse future conditions. By adjusting the maturity of new borrowing, firms can reshape their future repayment obligations, with consequences for financing costs, investment, and default. Endogenous maturity adjustment therefore provides an additional channel through which monetary policy affects corporate financing and real economic activity.

The value and price of maturity insurance depend both on the determinants of long-term interest rates and firms' refinancing needs. News of higher future short rates raises the current cost of long-term borrowing, but also increases the value of avoiding refinancing under adverse future conditions. This insurance benefit is particularly important for firms with substantial debt coming due. A higher term premium, by contrast, raises the cost of obtaining maturity protection without a comparable increase in its insurance value. Consequently, rollover-exposed firms lengthen new-debt maturity relatively more following expected-rate news, but relatively less when term premia rise. Thus, the level of long-term interest rates alone is not a sufficient statistic to characterize firms' maturity outcomes or their consequences for monetary transmission. The same movement in long-term rates can have different effects depending on whether it reflects expected future short rates or term premia. We refer to this mechanism as the *maturity-insurance channel of monetary transmission*.

We establish three main findings. First, using euro-area credit-registry data, we show that firms with high rollover exposure borrow at approximately 12% longer maturities than firms with low exposure. This relative maturity response depends on the determinants of long-term interest rates: higher expected future short rates strengthen maturity extension among exposed firms, whereas higher term premia weaken it. Second, to quantify the real consequences of these adjustments, we develop a heterogeneous-firms model with financial frictions that successfully reproduces the opposing maturity responses in untargeted firm-level regressions. Counterfactuals show that maturity flexibility mitigates monetary tightening and firms with high rollover exposure value it the most. Following news of higher future rates, it preserves 2.4 percentage points of cumulative investment relative to capital and reduces cumulative default by 1.4 percentage points over three years. Third,

the investment benefits of maturity flexibility depend on the term-premium environment. When expected-rate news coincides with a higher term premium, maturity flexibility preserves an additional 1.5 percentage points of cumulative investment over three years, relative to expected-rate news alone. Thus, the determinants of long-term interest rates determines not only how firms adjust debt maturity but also how much protection this adjustment provides against monetary tightening.

We study this mechanism during the 2022–23 euro-area monetary tightening cycle, when both sides of the maturity-insurance trade-off shifted sharply. Firms entered the tightening with repayment schedules inherited from earlier borrowing decisions, while expected future policy rates rose and long-duration financing became more expensive.[1] This episode therefore provides variation in both the value of postponing future refinancing and the cost of obtaining maturity protection. The euro area is particularly informative because non-financial corporations rely heavily on bank credit, a substantial share of borrowing is fixed-rate, and repayment obligations are staggered across firms[2]. We use *AnaCredit*, the euro-area credit registry, matched to firm balance sheets from September 2018 through December 2024. The data allow us to observe repayment schedules on outstanding bank debt and the maturity of newly originated loans. We measure rollover exposure as the share of outstanding bank debt scheduled to mature within the next twelve months. Our empirical design compares firms borrowing from the same bank in the same month but facing different inherited refinancing needs. Bank-by-month fixed effects absorb contemporaneous lender-level credit conditions, while firm-by-bank fixed effects account for persistent heterogeneity within lending relationships. To strengthen identification, we exploit repayment schedules fixed before the tightening cycle and high-frequency monetary-policy surprises.

The empirical analysis establishes three results. First, firms with higher inherited rollover exposure obtain longer maturities on new borrowing in this sample period. Moving from the 25th to the 75th percentile of rollover exposure is associated with approximately 12% longer new-loan maturity, equivalent to about 7.5 months at the sample mean. This relationship remains essentially unchanged when exposure is constructed from repayment schedules fixed before the tightening cycle. High-rollover firms therefore shift future refinancing obligations further into the future.
Second, and central to the mechanism, maturity responses depend on the determinants of long-term interest rates. We decompose five-year euro-area OIS rates into expected future short rates and term premia following Joslin et al. (2011). Higher expected future short rates are associated with relatively longer new-debt maturity among high-rollover firms, whereas higher term premia are associated with relatively shorter maturity. The opposing responses reflect the trade-off between

[1] In June 2022, before the first policy-rate increase, the ECB announced its intention to raise rates in July and signaled further tightening. Around the same time, the ECB Bank Lending Survey reported that firms could bring forward borrowing in anticipation of higher rates, even as banks tightened credit standards, particularly for long-term loans. See European Central Bank, "Monetary Policy Statement," June 9, 2022, and *The Euro Area Bank Lending Survey: Second Quarter of 2022*, July 19, 2022.

[2] Fixed-rate contracts account for 61% of outstanding term-loan volume in our sample, reaching 75% in Germany and 72% in France. The average firm's share of bank debt maturing within twelve months is approximately 30%, with substantial heterogeneity across borrowers, industries, and countries.

the value and cost of maturity insurance. Firms with substantial debt coming due benefit more from postponing refinancing, but also face immediate funding needs that make costly long-term borrowing particularly burdensome.

Direct measures of expected policy rates from the ECB Survey of Professional Forecasters yield the same qualitative pattern. High-frequency monetary-policy surprises (Altavilla et al., 2019) provide complementary evidence. Timing and Forward Guidance shocks, which shift expected future policy rates, are associated with relatively longer maturity among exposed firms, while Target shocks generate relatively shorter maturity. The results are robust to alternative yield decompositions (Adrian et al., 2013) and to distinguishing monetary-policy from central-bank information shocks. Additional tests indicate that the findings are not explained by coupon repricing risk, observed borrower risk, selection into new borrowing, or substitution toward committed liquidity, consistent with active liability management.

Third, investment responses to monetary conditions depend not only on inherited rollover exposure but also on firms' previous maturity adjustments. Following increases in the cost of long-term financing, investment declines are concentrated among rollover-exposed firms that had not previously extended maturity, particularly smaller firms. Consistent with this pattern, firms with higher rollover exposure shift their overall debt portfolios toward longer maturities, but this adjustment weakens when long-term financing becomes more expensive. They also face higher interest expenses under these conditions. These findings suggest that inherited refinancing needs and earlier liability adjustments jointly help explain heterogeneity in investment responses.

The empirical analysis shows that firms with higher rollover exposure adjust debt maturity differently as monetary conditions change, but does not establish the real consequences of these adjustments for monetary transmission. We develop a heterogeneous-firms model with financial frictions to quantify how endogenous maturity adjustment shapes borrowing, investment, and default following monetary shocks. Firms jointly choose investment, borrowing, and debt maturity while entering each period with inherited repayment obligations. Aggregate financing conditions are summarized by the current short rate, news about future short rates, and a term-premium wedge. We discipline the model using loan- and firm-level moments of default, debt issuance, and investment, without targeting the differential maturity responses central to the empirical analysis. Nevertheless, the model reproduces the opposing interactions between rollover exposure and expected future short rates and term premia in simulated firm-level regressions.

We establish two quantitative results. First, we show that endogenous maturity adjustment materially affects monetary transmission. We compare the benchmark economy with a counterfactual in which new-debt maturity cannot respond directly to aggregate financing conditions, while borrowing, investment, default, and maturity adjustments driven by firms' evolving balance sheets remain endogenous. Following news of higher future short rates, maturity flexibility preserves 2.4 percentage points of cumulative investment relative to capital and reduces cumulative default by 1.4 percentage points over three years. Importantly, this protection does not necessarily require longer debt maturity in aggregate. When long-term financing becomes expensive, firms can ben-

efit from borrowing at shorter maturities, accepting greater future refinancing exposure to avoid paying for costly long-term debt today. Moreover, this is distinct from interest-rate fixation: even when coupons reset with the short rate, maturity flexibility continues to reduce default by limiting principal refinancing risk.

Second, the real effects of expected-rate news depend on the term-premium environment. Holding the monetary innovation and firms' initial balance sheets fixed, a higher prevailing term premium amplifies the three-year investment contraction by 3.2 percentage points. Firms shorten maturity more sharply, increasing their subsequent refinancing exposure. Yet higher term premia also increase the investment benefit of maturity flexibility. When expected-rate news coincides with a term-premium increase, flexibility preserves 3.9 percentage points of cumulative investment over three years, compared with 2.4 points under expected-rate news alone. The additional 1.5-percentage-point benefit is statistically significant. Thus, more expensive long-term financing amplifies monetary tightening, but the ability to adjust toward shorter maturities mitigates the resulting investment losses relative to an economy in which maturity cannot respond to aggregate financing conditions.

Finally, we apply the model to the 2022–24 euro-area monetary tightening sequence, initializing the economy in 2018 to allow firms' liability structures to evolve before tightening begins. The preceding period of negative short rates and low term premia induces firms to accumulate longer-maturity debt, allowing them to enter tightening with inherited maturity protection. Expected-rate news then induces earlier adjustment of new-debt maturity, reducing subsequent default without additional deleveraging relative to the short-rate-only path. Along the full simulated monetary path, maturity flexibility mitigates cumulative investment losses by 5.0 percentage points and reduces cumulative default by 4.9 points relative to restricted maturity adjustment.

Taken together, our findings show that the expected path of policy rates alone is not sufficient to characterize monetary transmission through corporate balance sheets. The same revision to expected future short rates can generate different maturity and investment responses depending on the prevailing term premium and firms' refinancing needs. For central banks, this highlights the importance of monitoring corporate repayment schedules and maturity choices alongside the decomposition of long-term interest rates into expected rates and term premia.

Related literature. Our paper contributes to three strands of literature and is related to a fourth. First, we contribute to research on financial amplification and heterogeneous monetary transmission through corporate balance sheets. A large literature shows that monetary policy affects firms through borrower net worth, external-finance premia, and the supply of bank credit (Gertler and Gilchrist, 1994; Bernanke and Gertler, 1995; Kashyap and Stein, 2000; Jiménez et al., 2012). More recent work emphasizes that transmission depends on firms' pre-existing financial positions. Floating-rate debt exposes interest expenses directly to policy-rate changes (Ippolito et al., 2018), while heterogeneity in financial positions and credit spreads generates differential transmission of monetary tightening (Ottonello and Winberry, 2020; Anderson and Cesa-Bianchi,

2024). Refinancing needs are particularly important when external financing conditions deteriorate: firms with debt coming due reduce investment more sharply during credit contractions (Almeida et al., 2012; Duchin et al., 2010), and rollover risk amplified the decline in investment during the European sovereign debt crisis (Kalemli-Özcan et al., 2022).

Closest to our paper, Deng and Fang (2022) show that inherited debt maturity shapes firms' investment responses to monetary policy through default risk and external financing costs, while Jungherr et al. (2024) emphasize rollover risk and nominal debt overhang in a model with endogenous maturity, showing that conventional and unconventional policies have different effects. We complement these studies by focusing on how firms adjust their refinancing exposure in response to monetary conditions. Using credit-registry data, we identify how new-loan maturity responds to expected future short rates and term premia conditional on inherited rollover exposure, exploiting within-bank-month comparisons, predetermined repayment schedules, and monetary-policy surprises. A quantitative model reproduces these differential responses without targeting them and shows that the investment and default effects of a given monetary innovation depend on the prevailing term premium and firms' inherited liability structures. Importantly, maturity flexibility can mitigate investment losses even when firms shorten rather than extend debt maturity. Thus, the contribution of endogenous maturity adjustment to monetary transmission depends not only on firms' ability to manage refinancing risk, but also on the cost of obtaining maturity protection.

Second, our paper contributes to research on expectations and the timing of monetary transmission. Monetary-policy announcements convey news about both the current stance and the expected path of policy rates (Gürkaynak et al., 2005; Altavilla et al., 2019), and expected future rates can affect current decisions before they are realized (McKay et al., 2016). Related work shows that the effects of forward guidance depend on the macroeconomic and financial environment (Keen et al., 2017; Lunsford, 2020; Bauer et al., 2022). We identify an additional source of state dependence operating through corporate debt maturity. Because long-term rates reflect both expected future short rates and term premia (Hanson and Stein, 2015), the cost of maturity protection depends on the determinants of long-term financing costs (Kozlowski, 2021). We show that the same news about future policy rates can generate different maturity and investment responses depending on the prevailing term premium. In particular, higher term premia amplify investment contractions while increasing the value of maturity flexibility. Together with our empirical evidence on inherited rollover exposure, these findings establish that monetary transmission depends on both the determinants of long-term interest rates and firms' existing repayment obligations.

Third, we contribute to the literature on endogenous corporate debt maturity. Classic theories emphasize debt overhang, asymmetric information, rollover risk, and the relative cost of short- and long-term finance (Myers, 1977; Flannery, 1986; Diamond, 1991; He and Xiong, 2012; Brunnermeier and Oehmke, 2013). Empirical studies show that firms adjust debt maturity and interest-rate exposure in response to aggregate financing conditions (Baker et al., 2003; Faulkender, 2005; Greenwood et al., 2010), while quantitative models study maturity choices jointly with borrowing, investment, and default (Jungherr and Schott, 2021; Poeschl, 2023; Ferrante et al., 2025). More closely related

to our analysis, Bräuning et al. (2020) document the role of term premia in aggregate corporate bond maturity, while Fabiani et al. (2024) examine how monetary policy affects corporate debt maturity through the relative cost of long-term financing. We complement these studies by showing that firms' maturity responses depend on both the determinants of long-term interest rates and their inherited rollover exposure. Firms with greater rollover exposure extend new-debt maturity relatively more when expected future short rates rise, but relatively less when term premia increase. These findings show that maturity choices reflect not only the cost of long-term borrowing, but also the value of insurance against future refinancing risk.

Finally, our paper is related to work on firms' intertemporal management of financing risk. Dynamic corporate theories emphasize that firms time external finance, accumulate liquidity, and manage risk in anticipation of future funding needs (Bolton et al., 2013; Eisfeldt and Muir, 2016; Acharya et al., 2020). Empirically, firms refinance early when credit conditions are favorable (Mian and Santos, 2018), while Xu (2018) documents precautionary maturity extension in corporate bond markets. We add to this literature by considering debt maturity as a liability-side tool for managing future financing risk. By changing the maturity of new borrowing, firms determine when principal must next be refinanced and therefore how much external funding they will need to raise in future states in which financing may be costly. Maturity choice thus allows firms to reshape the timing of their exposure to future monetary and credit conditions.

The remainder of the paper proceeds as follows. Section 2 presents the data and motivating facts. Section 3 describes the loan-level empirical design and results. Sections 4–5 validate the mechanism and describe the real effects. Section 6 presents the quantitative model, and Sections 7–8 present the model results and counterfactuals. Section 9 concludes.

2 Data

Loan-level data. The main data source is *AnaCredit*, the euro-area credit registry, which covers the universe of credit transactions above a reporting threshold of EUR 25,000 at origination. In our analysis, we consider bank loans to non-financial corporations from September 2018 to December 2024. AnaCredit includes rich loan-level information such as the loan amount, rate, maturity, collateral, borrower, and lender identifiers for both outstanding balances and newly originated loans. AnaCredit is especially well-suited for our analysis because we can observe both the stock of short-term debt—our proxy for rollover exposure—and the terms of newly originated loans [3].

The main outcome for the loan-level analysis is loan maturity at origination. New loan maturity is highly dispersed: in the regression sample, the mean is 1,895 days (5.2 years), the median 737 days (2.0 years), the interquartile range runs from 113 to 1,836 days, and the upper tail reaches 7,330 days (20.1 years). The maturity dispersion is present both across borrowers and within lending relationships over time. Appendix Figure 14 plots the evolution of average new-loan maturity

[3] Appendix Table 25 extends the new loan sample back to 2014 using Orbis balance-sheet proxy of rollover exposure. AnaCredit only allows tracking outstanding loans beginning in September 2018.

between 2018 and 2024, while Appendix Figure 13 documents a clear positive relationship between new-loan maturity and the 5Y–1Y yield curve slope, after absorbing firm and bank fixed effects.
New borrowing is infrequent and relationship-concentrated. Between 2018 and 2024, 5.8% of indebted non-financial corporations originated a new loan in a given month, and about 32.5% of borrowing firms obtained the new credit from their main bank [4].
Regarding the rate type composition of new lending, fixed-rate contracts are 71 percent of new loans, 51 percent of new credit (including credit lines), but floating-rate loans are on average 2.4 times larger. Appendix Figure 9 reports the distribution of new credit across original-maturity buckets separately for floating- and fixed-rate loans, and shows that fixed-rate credit is disproportionately concentrated at very short and very long maturities. On the outstanding debt stock, fixed-rate contracts account for about 61 percent of term-loan volume, a share stable across 2018–2024, with wide cross-country dispersion: fixed-rate contracts account for 75 percent of term-loan volume in Germany and 72 percent in France against 39 percent in Italy and 14 percent in Finland [5].

Firm-level and macro-level data. Annual firm-level balance sheets come from Bureau van Dijk ORBIS, which covers listed and unlisted firms with good coverage of small and medium-sized enterprises. We follow Kalemli-Özcan et al. (2024) in cleaning ORBIS and obtain time-varying controls for total assets, tangibility, liquidity, leverage, and sales growth. Yields and policy rates are from the ECB Statistical Data Warehouse; inflation swaps from Bloomberg.
The final sample runs monthly from September 2018 to December 2024 and comprises 1.67 million firm–bank–month observations covering 811 banks and 109,051 firms, with about 380 banks originating a new loan in a typical month. Bank×month cells have a median of 58 borrowing firms. The loan- and firm-level variables are winsorized at 1%. The loan-level analysis is at the firm–bank–month level, while the firm-level analysis is conducted at the firm–year level; summary statistics are reported in the Appendix Table 15.

2.1 Stylized Facts

This section documents three stylized facts that motivate the analysis.

Fact 1: Long-term rates move through expected future short rates and term premia. The slope of the yield curve varies substantially over monetary-policy cycles. We measure the euro-area yields at a monthly frequency using market-implied zero-coupon rates from AAA euro area government bonds and overnight index swaps (OIS) at maturities ranging from three months to ten years. The yield curve (YC) slope is defined as the difference between a long-term yield and a short-term risk-free rate, such as the 10-year government bond yield and the 2-year OIS rate, or

[4]The main bank is defined as the lender holding the largest share of the firm's outstanding credit in the previous month.

[5]Because our sample excludes credit lines, which are predominantly floating rate, this share is higher than the evidence reported by Core et al. (2025), who find that non-floating debt accounts for about 40% of total credit volume in December 2021.

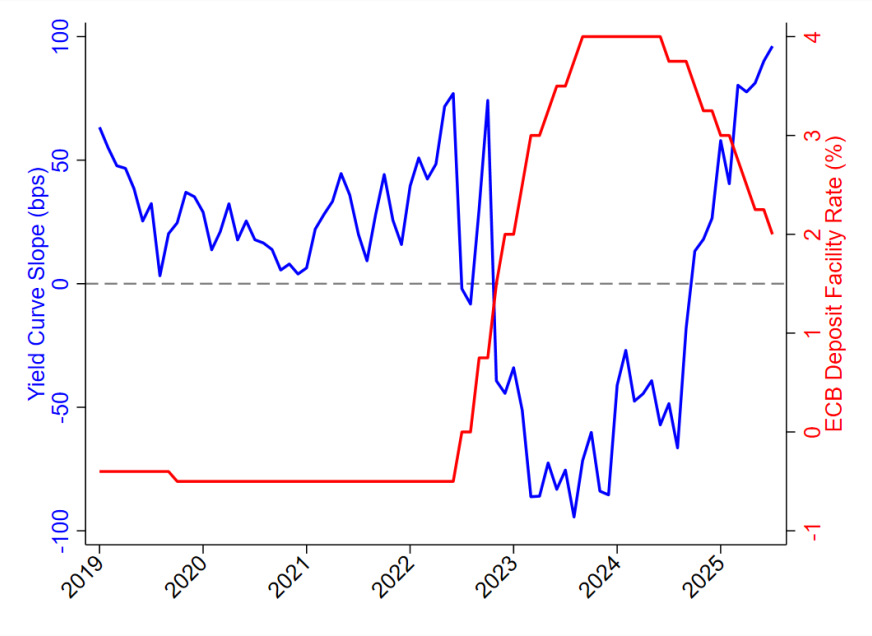


Figure 1: Monetary Policy Rate and the Yield Curve Slope

Notes: The figure plots the time series of the ECB Deposit Facility Rate (in red) and the Yield Curve Slope between 10-year AAA government bond yields and 2-year €STR OIS rates (in blue) between 2019 and 2025. Source: ECB Statistical Data Warehouse.

the 5-year yield and the 1-year OIS rate. Figure 1 plots the spread between the 10-year government bond yield and the 2-year OIS rate together with the ECB deposit facility rate between 2019 and 2025. The slope exhibits substantial time-series variation. Its average level is 0.18 percentage points, with a standard deviation of 0.53 percentage points. The curve flattens sharply during the 2022-23 tightening cycle, reaching a minimum of -0.94 percentage points, and subsequently steepens in late 2024 as financial conditions normalize. To interpret these movements, we decompose OIS rates into expected short rates and term premia using the standard affine term structure model of Joslin et al. (2011); as a robustness check, we also employ the yield decomposition following the methodology of Adrian et al. (2013). See Appendix D for estimation details.

Figure 2 reports the decomposition across yields at maturities of 1,2,5 and 10 years. Two results stand out. First, variation in short-maturity yields is almost entirely driven by changes in expected policy rates. The increase in one- to two-year yields in 2022 reflects a sharp upward revision in the expected path of ECB policy, followed by a gradual decline as expectations of future easing emerge. Second, variation in longer-maturity yields is partially accounted for by movements in term premia. The term premium increases from negative values prior to 2022 to persistently positive levels, accounting for a substantial share of the increase in long-term yields during this period. Thus, these patterns indicate that recent changes in the slope of the yield curve reflect not only revisions in expected policy, but also a repricing of duration risk.

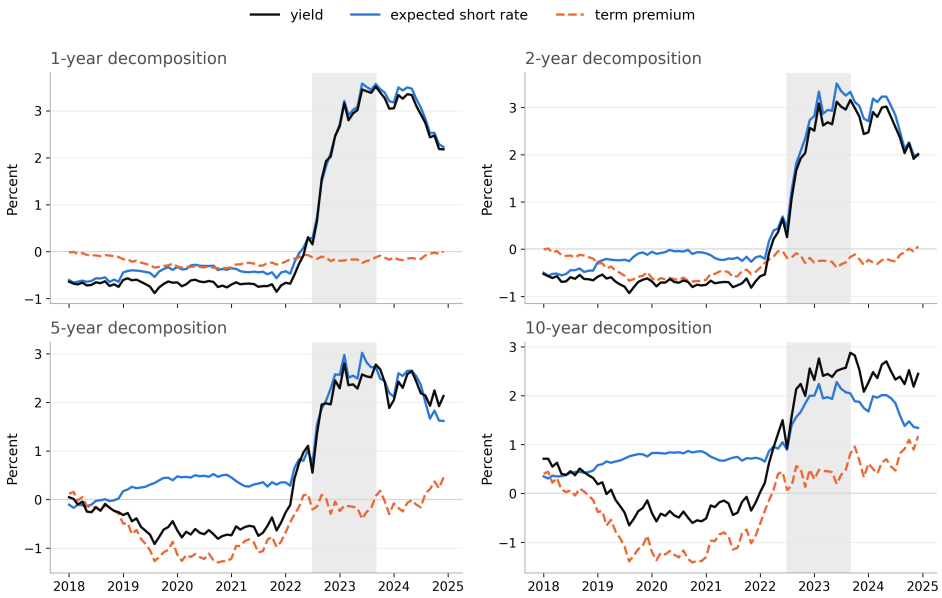


Figure 2: Yield Decomposition into Expected Short Rates and Term Premia

Notes: This figure reports the decomposition of euro-area risk-free OIS zero-coupon rates into expected future short-rate path and term premia at horizons 1,2,5,10 years. The decomposition uses the affine term-structure model of Joslin et al. (2011). Expected short rates (in blue) capture the anticipated path of future policy rates, while the term-premium (in orange) captures the risk investors want to be compensated to hold long-term assets. The shaded grey region denotes the ECB monetary policy tightening cycle beginning in 2022. Source: authors' calculation, ECB Statistical Data Warehouse.

Fact 2: Corporate rollover exposure is large and highly dispersed. We proxy for rollover exposure with the share of short-term debt, defined as outstanding loans with residual maturity of up to 12 months, excluding credit lines, over total outstanding bank debt. Euro-area non-financial firms rely heavily on short-term debt. On average, approximately 30% of firms' outstanding bank debt matures within twelve months over 2018–2024, with substantial heterogeneity across countries and sectors. Short-term debt is particularly high in Spain, Portugal, Slovenia, and Belgium, as well as in sectors such as mining, trade, wholesale, and manufacturing. In our data, Figure 3 shows that 35% of firm–month observations carry little short-term debt, while 16% of firm–month observations have their entire debt due within the year. We exploit this cross-sectional variation in the empirical design. Because firms differ in the share of their debt coming due, a given movement in the term structure generates heterogeneous responses across borrowers facing the same credit conditions.

Rollover exposure could in principle proxy for credit quality. Appendix Table 16 compares the difference in group means between the bottom and top quartiles of short-term debt. High-exposure firms have lower tangibility and are smaller in terms of fixed assets, but on standard measures of financial health they look, if anything, stronger—they hold lower leverage, substantially higher interest coverage, and indistinguishable sales growth. On observable characteristics, rollover exposure is therefore not a proxy for weaker borrowers; the comparison cannot rule out unobserved differences in credit quality or credit demand, which the fixed-effects design and the legacy-contract measure address only in part. These firm-level characteristics are controlled for throughout, and the

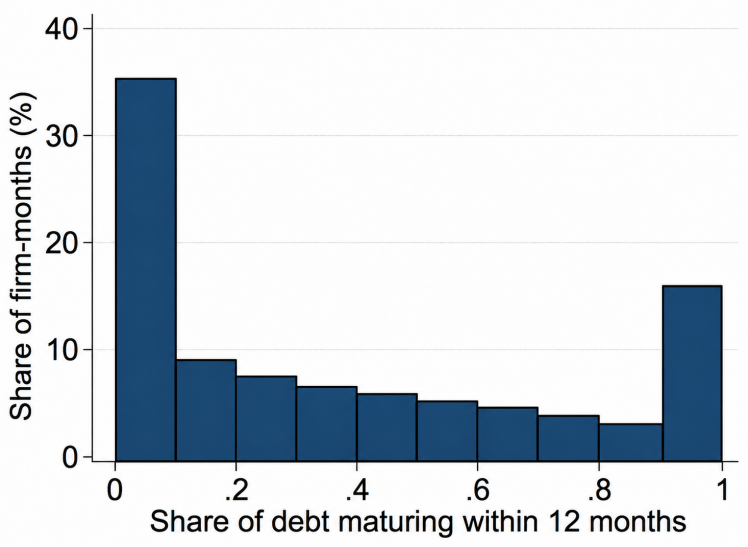


Figure 3: Rollover Exposure across Non-Financial Firms

Notes: This figure reports the distribution of short-term debt across firm-months. Short-term debt is the share of outstanding debt with residual maturity up to 12 months, excluding credit lines, collapsed at the firm-time level. The figure shows that 35 percent of firm-month observations carry little short-term debt and 16 percent of firm-month observations have their entire outstanding debt maturing within twelve months. All firms have positive outstanding bank debt; zero denotes no debt scheduled to mature within 12 months. Source: AnaCredit, 2018–2024.

legacy-contract measure introduced below addresses the concern that unobserved borrower quality and maturity choice are jointly determined.

Fact 3: A steeper yield curve raises the relative price of long-term over short-term loans. The slope of the yield curve is associated with firms' relative borrowing costs across maturities. Figure 4 shows that a steeper 5Y–1Y yield curve slope is associated with a larger loan spread between loan rates at origination on long-term loans (with maturity ≥ 3 years) and short-term loans (with maturity ≤ 1 year). Because the loan rate spread is residualized with respect to firm and bank fixed effects, this relationship reflects time-series variation in relative borrowing costs rather than persistent cross-sectional differences. This is pricing evidence consistent with the term structure affecting the relative prices of long- and short-term borrowing, rather than an estimate of the pass-through of the yield curve into loan spreads, since residualizing on firm and bank fixed effects does not remove time-varying differences in loan composition[6].

[6]Complementary pricing evidence points the same way. Among newly originated fixed-rate loans, with spreads measured relative to maturity-matched OIS rates, firms with greater rollover exposure pay modestly higher spreads when the long end of the yield curve steepens, consistent with maturity insurance becoming more costly when long-term funding is expensive.

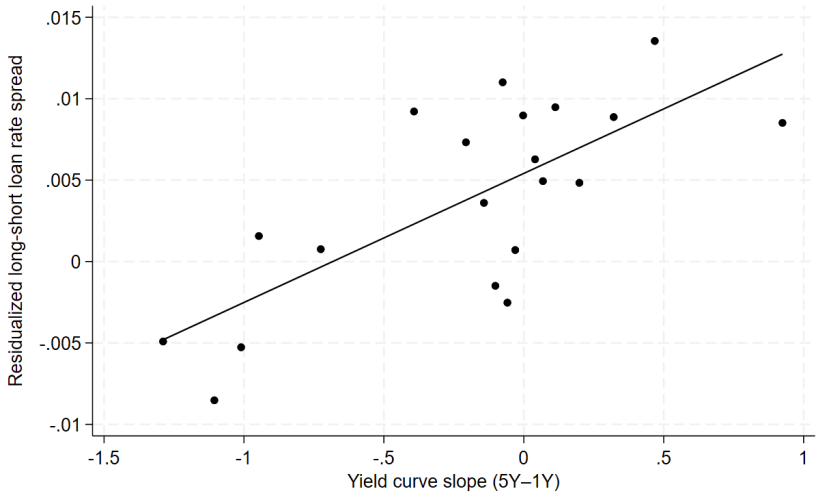


Figure 4: The Yield Curve Slope and Corporate Loan Spreads

Notes: This figure plots the monthly loan rate spread between residualized interest rates on long-term loans (new maturity $\geq$ 3 years) and short-term loans (new maturity $\leq$ 1 year) against the 5Y–1Y yield curve slope. Loan rates at origination are residualized using firm and bank fixed effects. The solid line shows the linear fitted relationship. Source: AnaCredit, 2018–2024.

3 Empirical Specification and Main Results

The loan-level analysis proceeds in three steps. We first document how the maturity of newly originated loans responds to the slope of the yield curve, allowing the response to vary with firms' predetermined rollover exposure. We then exploit high-frequency monetary policy shocks to identify maturity responses to exogenous shifts in the term structure (Altavilla et al., 2019). Finally, we decompose long-term yields into expected future short rates and term premia using standard affine term-structure models (Joslin et al., 2011; Adrian et al., 2013), allowing us to assess which component of the yield curve drives firms' maturity adjustment.

3.1 Rollover-exposed firms lengthen new loan maturity

We begin by estimating at the bank b, firm f, and month t level the following specification:

$$\log(\text{Maturity})_{bft} = \alpha_{bt} + \alpha_{fb} + \beta_1 \text{ST Debt}_{f,t-1} + \beta_2 \left(\text{Yield Curve Slope}_{t-1}^{5Y-1Y} \times \text{ST Debt}_{f,t-1}\right) + \Gamma X_{f,t-1} + \varepsilon_{bft} \tag{1}$$

The dependent variable is the logarithm of the maturity of newly originated loans at the firm–bank–month level, aggregated within each relationship-month using new credit as weights. ST Debt$_{f,t-1}$ is the lagged share of outstanding bank debt with residual maturity of 12 months or less, excluding credit lines. Our baseline measure of the term structure, Yield Curve Slope^{5Y-1Y}, is the difference between the yield on 5-year AAA-rated euro-area government bonds and the 1-year OIS rate. We use the 5-year endpoint because it closely matches the average maturity of newly originated loans in

Table 1: Rollover Exposure, the Term Structure, and New Loan Maturity

$\log(\text{Maturity})_{b,f,t}$	(1)	(2)	(3)	(4)	(5)	(6)
ST Debt$_{f,t-1}$	0.226***	0.233***	0.228***	0.236***	0.252***	0.269***
	(0.0043)	(0.0041)	(0.0049)	(0.0044)	(0.0059)	(0.0064)
YC (5Y–1Y)$_{t-1}\times$ ST Debt$_{f,t-1}$	-0.0210***	-0.0148**				
	(0.0043)	(0.0053)				
YC (10Y–2Y)$_{t-1}\times$ ST Debt$_{f,t-1}$				-0.0449***	-0.0812***	
				(0.0050)	(0.0056)	
1Y Rate$_{t-1}\times$ ST Debt$_{f,t-1}$			0.129***			
			(0.0055)			
5Y Rate$_{t-1}\times$ ST Debt$_{f,t-1}$			-0.050***			
			(0.0068)			
2Y Rate$_{t-1}\times$ ST Debt$_{f,t-1}$						0.150***
						(0.0140)
10Y Rate$_{t-1}\times$ ST Debt$_{f,t-1}$						-0.095***
						(0.0097)
Observations	1,674,254	1,041,988	1,674,254	1,674,254	1,041,988	1,674,254
Adjusted R^2	0.620	0.620	0.620	0.620	0.620	0.620
Bank × Month FE	Yes	Yes	Yes	Yes	Yes	Yes
Firm × Bank FE	Yes	Yes	Yes	Yes	Yes	Yes
Firm Controls	Yes	Yes	Yes	Yes	Yes	Yes
Maturity Wall	No	Yes	No	No	Yes	No

Notes: In this table the dependent variable is the logarithm of maturity of newly originated loans. Columns (1)–(3) use the 5Y–1Y yield curve slope, while columns (4)–(6) use the 10Y–2Y slope as robustness. Columns (1) and (4) report the baseline specification in eq. 1. Columns (2) and (5) replace short-term debt (ST Debt) with the predetermined maturity wall constructed from debt outstanding as of December 2019. Columns (3) and (6) split the corresponding yield curve slope into its short- and long-rate components. All specifications include lagged firm controls, bank×month fixed effects, and firm×bank fixed effects. Standard errors are clustered at the firm and month level. *, **, and *** denote significance at the 10%, 5%, and 1% levels, respectively.

our sample. As a robustness check, we use the corresponding $10Y - 2Y$ spread. Bank-by-time fixed effects, α_{bt}, absorb time-varying bank-level credit-supply conditions as well as aggregate shocks (Khwaja and Mian, 2008), while firm-by-bank fixed effects, α_{fb}, absorb persistent relationship characteristics and time-invariant differences in maturity choice across lending relationships. The results are robust to more granular credit-supply controls, including bank–time–industry–location fixed effects (Degryse et al., 2019). The vector $X_{f,t-1}$ includes time-varying firm characteristics such as size, tangibility, the interest coverage ratio, sales growth, and cash over total assets.

The first coefficient of interest is β_1, which captures the relationship between rollover exposure and new-loan maturity at the average yield curve slope.[7] A higher rollover exposure does not mechanically imply longer new maturity and the sign of β_1 is a priori ambiguous. Firms may lengthen new maturity to reduce future refinancing and interest-rate risk, or they may instead borrow short if long-duration financing is costly or lenders are unwilling to supply long-term credit.

[7]We demean the 5Y–1Y yield curve slope so that β_1 measures this relationship at the average slope rather than at a slope of zero.

Our dependent variable is the maturity of newly originated loans, conditional on borrowing, rather than the incidence or volume of refinancing. Moreover, firm-by-bank fixed effects absorb persistent differences in maturity across lending relationships, so β_1 is identified from variation within a given firm–bank relationship rather than from exposed firms borrowing from banks that systematically offer longer-maturity credit. A positive β_1 indicates that greater inherited rollover exposure is associated with longer-maturity new borrowing within the same lending relationship.

The second coefficient of interest is β_2, which measures how the maturity adjustment varies with the term structure. Because the yield curve slope varies only over time, its direct effect is absorbed by bank-by-month fixed effects[8] . Identification therefore comes from comparing firms borrowing from the same bank in the same month, and hence facing the same contemporaneous lender conditions, but entering the period with different inherited rollover exposure. The specification asks whether the same lender allocates relatively longer or shorter maturities to more exposed firms as the term structure changes. Bank-by-month fixed effects also absorb shifts in lenders' aggregate willingness to supply duration. For example, if a bank generally prefers shorter maturities when expected future short rates rise, such a bank-wide response is absorbed by α_{bt} and does not contribute to the identifying variation in β_2. The specification does not rule out banks adjusting their maturity menu differentially across borrowers as a function of rollover exposure. We therefore interpret β_2 as the equilibrium response of the maturity of credit allocated to exposed firms.

A potential concern with the baseline measure of rollover exposure is that short-term debt may itself reflect contemporaneous maturity management. To address this concern, we construct an alternative measure of rollover exposure using only debt contracts outstanding as of December 2019, following the approach of Almeida et al. (2012). For each firm f and month t, we define

$$\text{Maturity Wall}_{f,t}^{2019} = \frac{\text{debt outstanding as of December 2019 scheduled to mature in } [t, t+12]}{\text{total debt outstanding as of December 2019}}$$

This measure fixes the firm's debt portfolio as of December 2019 and uses only the contractual maturity dates of those legacy obligations. It therefore predates both the 2022 monetary-policy tightening cycle and the COVID credit support programs. Although the maturity structure of the December 2019 debt stock reflects earlier financing decisions, its contractual repayment schedule cannot respond to subsequent movements in the yield curve or to contemporaneous credit conditions. Time variation in $\text{Maturity Wall}_{f,t}^{2019}$ is instead generated by the predetermined timing at which these legacy obligations come due. We replace $\text{ST Debt}_{f,t-1}$ in equation (1) with $\text{Maturity Wall}_{f,t}^{2019}$. This specification does not rely on contemporaneous short-term debt being exogenous. Rather, it asks whether firms facing refinancing needs predetermined by their pre-tightening cycle debt contracts adjust the maturity of new borrowing differentially as the term

[8]Figure 13 complements this evidence by documenting the corresponding aggregate pattern. Average new-loan maturity is strongly positively correlated with the slope of the yield curve. This relationship is descriptive because the term structure co-moves with aggregate macroeconomic and credit conditions. Its role is therefore not identification, but to establish that maturity choice moves systematically with the term structure in the aggregate, while equation (1) isolates the differential response associated with inherited rollover exposure.

structure changes. The identifying assumption is that, conditional on the fixed effects and controls, the contractual timing of maturities in the December 2019 debt stock is orthogonal to subsequent time-varying shocks that differentially affect maturity choice within a firm–bank relationship.

Table 1 reports the results. Column (1) shows the β_1 coefficient on lagged short-term debt is 0.226 and statistically significant. Moving from the 25th to the 75th percentile of rollover exposure—a 50 percentage-point increase in short-term debt in our sample—raises new-loan maturity by approximately 12% ($e^{0.226\times0.5} - 1$), which corresponds to roughly 7.5 months when evaluated at the sample mean maturity of 5.2 years; the conversion to months is an approximation at the mean, not a separately estimated level effect. Thus, firms with more debt coming due do not simply refinance mechanically: conditional on new borrowing, they push the next refinancing date further into the future. Column (4) confirms the finding with the 10Y-2Y yield curve slope.

A natural concern is that contemporaneous rollover exposure may itself reflect recent maturity choices. Firms anticipating future financing conditions could simultaneously alter both the share of debt coming due and the maturity of new borrowing. Columns (2) and (5) therefore replace current rollover exposure with the predetermined maturity wall implied by loans already outstanding as of December 2019. Because the repayment dates of these legacy contracts were fixed before the subsequent tightening cycle, this measure isolates inherited refinancing needs from contemporaneous maturity management. The estimated coefficients remain positive, statistically significant, and similar in magnitude to the baseline estimates. Hence, the result survives when exposure is fixed before the tightening: firms facing greater inherited refinancing needs subsequently choose longer maturities on new borrowing.

Next, we examine whether this maturity adjustment depends on the term structure of interest rates. In column (1), β_2, the interaction between the 5Y–1Y yield curve slope and rollover exposure, is negative and statistically significant. A one percentage point yield curve steepening, when long-term rates are high relative to short-term rates, makes firms with greater rollover exposure extend maturity by 9% less. The same pattern is stronger using the 10Y–2Y slope in column (4), with a reduction of about 19%. This is consistent with a trade-off between the benefit of reducing future rollover needs and the current cost of locking in long-duration funding. Finally, Columns (3) and (6) provide a first look at the two sides of this trade-off by entering the short- and long-rate components of each slope separately. In both specifications, higher short rates are associated with greater maturity extension among rollover-exposed firms, whereas higher long rates are associated with less extension. For the 5Y–1Y specification, the interaction coefficients are 0.129 for the 1-year rate and -0.050 for the 5-year rate; the corresponding coefficients for the 2-year and 10-year rates are 0.150 and -0.095. These opposing responses motivate the yield decomposition below, that splits long rates into expected future short rates and term premia movements.

3.2 Expected future rates vs term premia

We decompose long-term OIS risk-free rates to distinguish the value and price of maturity insurance. The previous results show that the maturity choices of rollover-exposed firms depend on the term structure of interest rates. However, long-term yields combine two economically distinct components with opposing implications for maturity choice for firms with high refinancing needs. To separate these forces, we use euro-area OIS zero-coupon rates and decompose a n-year yield into an expected-rate component and a term premium as

$$y_t^n = \frac{1}{n}\sum_{j=0}^{n-1}\mathbb{E}_t[r_{t+j}] + \text{Term Premium}_t^n$$

Our baseline decomposition follows Joslin et al. (2011); we use Adrian et al. (2013) as an alternative decomposition.[9] We use OIS rates because they provide the risk-free policy path rather than sovereign yields, and they are the benchmark for loan pricing. We focus on 5- and 10-year yields, where the distinction between expected rates and term premia is economically meaningful, and term-premium variation is more pronounced as shown in Figure Figure 2. For horizons $n \in \{5, 10\}$ years, we estimate

$$\begin{aligned}\log(\text{Maturity})_{bft} = \alpha_{bt} + \alpha_{fb} + \beta_1 \text{ST Debt}_{f,t-1} \\ + \beta_{\text{exp}}\left(\text{Expected Rate Path}_{t-1}^n \times \text{ST Debt}_{f,t-1}\right) \\ + \beta_{\text{TP}}\left(\text{Term Premium}_{t-1}^n \times \text{ST Debt}_{f,t-1}\right) + \Gamma X_{f,t-1} + \varepsilon_{bft}\end{aligned} \tag{2}$$

As in the baseline specification, the level effects of the aggregate term-structure components are absorbed by bank-by-month fixed effects, so identification comes from their differential effect across firms with different inherited rollover exposure. The two components have opposite implications for maturity outcomes for firms with high rollover exposure. A rise in the expected path of short rates is already reflected in higher long-term prices, which results in tighter refinancing conditions. But the value of locking in maturity today rises most for firms with higher rollover exposure under idiosyncratic rollover risk and costly refinancing, implying $\beta_{\text{exp}} > 0$. By contrast, a rise in the term premium raises only the contemporaneous cost of obtaining long-duration financing without a corresponding increase in the value of maturity insurance. Long-term debt therefore becomes more expensive and dampens the maturity response, especially of high-rollover firms, implying $\beta_{\text{TP}} < 0$. These opposite sign predictions provide the paper's central mechanism test.

Table 2 reports the results. Columns (1)–(2) use the Joslin et al. (2011) decomposition, while Columns (3)–(4) use the Adrian et al. (2013) decomposition. Across both decompositions and

[9] The two approaches recover the same economic objects using different identification strategies. Joslin et al. (2011) estimates an affine no-arbitrage term-structure model and separates physical-measure forecasts of future short rates from the risk-premium component of observed long yields. Adrian et al. (2013) instead uses the predictability of bond excess returns and estimated prices of yield curve risk to recover the term premium. Appendix Figure 11 and Appendix Table 2 show that the qualitative results are robust across the two decompositions.

Table 2: Yield Decomposition and New Loan Maturity

$\log(\text{Maturity}_{bft})$	(1) 5Y	(2) 10Y	(3) 5Y	(4) 10Y
ST $\text{Debt}_{f,t-1}$	0.179***	0.158***	0.183***	0.157***
	(0.00695)	(0.00828)	(0.00666)	(0.00824)
Expected Rate Path_{t-1} × ST $\text{Debt}_{f,t-1}$	0.0293***	0.0571***	0.0292***	0.0644***
	(0.00313)	(0.00581)	(0.00312)	(0.00639)
Term Premium_{t-1} × ST $\text{Debt}_{f,t-1}$	-0.0382***	-0.0265***	-0.0399***	-0.0297***
	(0.00610)	(0.00440)	(0.00643)	(0.00462)
Observations	1,674,254	1,674,254	1,674,254	1,674,254
Adjusted R^2	0.620	0.620	0.620	0.620
Bank × Month FE	Yes	Yes	Yes	Yes
Firm × Bank FE	Yes	Yes	Yes	Yes
Firm Controls	Yes	Yes	Yes	Yes

Notes: The dependent variable is the logarithm of new loan maturity. ST $\text{Debt}_{f,t-1}$ is the share of outstanding bank debt contractually scheduled to mature within twelve months. The Expected Rate Path and Term Premium are obtained following Joslin et al. (2011) in columns (1) and (2) or Adrian et al. (2013) in columns (3) and (4). Columns (1) and (3) use the five-year decomposition, while columns (2) and (4) use the ten-year decomposition. The regressions include bank-by-month and firm-by-bank fixed effects and lagged firm-level controls. Standard errors are clustered at the firm and month level and reported in parentheses. *, **, and *** denote statistical significance at the 10%, 5%, and 1% levels, respectively.

at both five- and ten-year horizons, the interaction between rollover exposure and expected rates is positive and statistically significant, whereas the interaction with term premia is negative and statistically significant. The effects are also economically meaningful. A one-standard-deviation increase in the expected-rate component leads firms at the 75th percentile of rollover exposure to choose about 1.5–1.6% longer maturity—roughly 27–30 additional days—relative to firms at the 25th percentile. A comparable increase in the term premium instead reduces relative maturity by about 1.0–1.1%, or 18–19 days. These opposite responses distinguish the value and price of maturity insurance. When expected future short rates rise, firms with greater rollover exposure have a stronger incentive to postpone their return to the credit market if they may be hit by a bad productivity shock, making them more likely to default and not pay back their debt. When term premia rise, long-duration borrowing becomes more expensive today, reducing their incentive to extend maturity. Hence, the maturity response depends not simply on the level of long-term rates but on the source of their movement. The similarity of the estimates across the two decompositions indicates that the result is not driven by the choice of an affine term-structure model.

We complement the model-based decompositions with direct measures of expected policy rates from the ECB Survey of Professional Forecasters (SPF). For each quarterly survey round q, we construct the average expected policy-rate path over the subsequent year:

$$\text{SPF Expected Path}_q = \frac{E_q i_{3m} + E_q i_{6m} + E_q i_{9m} + E_q i_{12m}}{4}$$

Table 3: ECB Survey of Professional Forecasters (SPF) and New Loan Maturity

$\log(\text{Maturity})_{b,f,t}$	(1)	(2)	(3)
ST $\text{Debt}_{f,t-1}$	0.225***	0.226***	0.224***
	(0.00443)	(0.00438)	(0.00447)
SPF Expected $\text{Path}_{q-1}\times$ ST $\text{Debt}_{f,t-1}$	0.00776***		
	(0.00228)		
SPF 12 months $\text{Ahead}_{q-1}\times$ ST $\text{Debt}_{f,t-1}$		0.00682**	
		(0.00298)	
Δ SPF Expected $\text{Path}_{q-1}\times$ ST $\text{Debt}_{f,t-1}$			0.0284***
			(0.00718)
Observations	1,674,254	1,674,254	1,674,254
Adjusted R^2	0.620	0.620	0.620
Bank $\times$ Month FE	Yes	Yes	Yes
Firm $\times$ Bank FE	Yes	Yes	Yes
Firm Controls	Yes	Yes	Yes

Notes: The dependent variable is the logarithm of the maturity of newly originated loans. The ECB Survey of Professional Forecasters (SPF) is used as an alternative measure of expectations. Column (1) interacts firms' rollover exposure with the average expected policy-rate path over the subsequent 12 months. Column (2) uses the SPF 12-month-ahead expected policy rate. Column (3) uses the quarter-to-quarter revision in the expected policy-rate path. All specifications include lagged firm controls, bank×month fixed effects, and firm×bank fixed effects. Standard errors are clustered at the firm and month level. Note that the survey is conducted quarterly. *, **, and *** denote significance at the 10%, 5%, and 1% levels, respectively.

as well as the 12-month-ahead expected rate and the revision in the expected path as $\Delta\text{SPF Expected Path}_q = \text{SPF Expected Path}_q - \text{SPF Expected Path}_{q-1}$. Then, we interact each measure with firms' lagged rollover exposure in the same loan-level specification as equation 1.

Table 3 shows that higher expected future policy rates are consistently associated with greater maturity extension among rollover-exposed firms. Columns (1)–(2) use the average expected policy-rate path over the subsequent 12 months and the 12-month-ahead expected policy rate, respectively. Column (3) focuses on revisions in expectations and suggests that a one-percentage-point upward revision in the expected path leads firms at the 75th percentile of rollover exposure to choose about 1.4% longer maturity—approximately 26 additional days—relative to firms at the 25th percentile. This specification rules out persistent differences in firms' or banks' expectations, which are captured by firmxbank fixed effects. Since the ECB SPF measures expectations directly, the survey evidence confirms that the positive expected-rate response is not an artifact of the yield curve decomposition.

Table 4: High-Frequency Monetary Policy Shocks and New Loan Maturity

$\log(\text{Maturity})_{b,f,t}$	(1)	(2) Pure Monetary Policy	(3) Information
ST Debt$_{f,t-1}$	0.228***	0.229***	0.229***
	(0.00429)	(0.00429)	(0.00428)
Target$_{t-1}\times$ ST Debt$_{f,t-1}$	-0.00778***	-0.0265***	0.000266
	(0.00199)	(0.00531)	(0.00172)
Timing$_{t-1}\times$ ST Debt$_{f,t-1}$	0.0175***	0.0331***	0.00574
	(0.00377)	(0.00640)	(0.00670)
Forward Guidance$_{t-1}\times$ ST Debt$_{f,t-1}$	0.00819***	0.0131***	0.00617***
	(0.00172)	(0.00444)	(0.00205)
Quantitative Policy$_{t-1}\times$ ST Debt$_{f,t-1}$	-0.000678	-0.00801	-0.00278
	(0.00250)	(0.00579)	(0.00301)
Observations	1,674,254	1,674,254	1,674,254
Adjusted R^2	0.620	0.620	0.620
Bank × Month FE	Yes	Yes	Yes
Firm × Bank FE	Yes	Yes	Yes
Firm Controls	Yes	Yes	Yes

Notes: The dependent variable is the logarithm of maturity of newly originated loans. ST Debt$_{f,t-1}$ is the firm's lagged short-term debt share, and Shock$_{k,t-1}$ is the cumulated, standardized high-frequency policy-surprise factor following Altavilla et al. (2019). The four factors are Target, Timing, Forward Guidance, and Quantitative Policy. Column (1) uses the high-frequency monetary policy factors directly. Columns (2)–(3) separate the surprises into pure monetary policy and central-bank information components using the high-frequency comovement between short-term OIS rates and equity prices (Jarociński and Karadi, 2020). Shocks are standardized by their monthly standard deviation. All specifications include lagged firm controls, bank×month fixed effects, and firm×bank fixed effects. Standard errors are clustered at the firm and month level. *, **, and *** denote significance at the 10%, 5%, and 1% levels, respectively.

3.3 Future-rate policy news drive maturity extension

A concern with the baseline specification is that movements in the yield curve may reflect macroeconomic and financial conditions that independently affect firms' maturity choices. We therefore exploit monetary-policy surprises measured in narrow windows around ECB announcements using the Euro Area Monetary Policy Event-Study Database of Altavilla et al. (2019). Let X_τ denote the vector of high-frequency changes in risk-free yields around announcement τ. The surprises satisfy

$$X_\tau = \Lambda F_\tau + u_\tau$$

where the factors are extracted by principal component analysis from intraday yield changes and rotated to identify policy news affecting different segments of the term structure[10].

We obtain four factors with economic interpretation: *Target* captures innovations concentrated at the short end; *Timing* captures revisions to the expected timing of near-term policy changes; *Forward Guidance* captures revisions to the expected policy path at intermediate horizons; and *Quantitative Policy* captures variation associated with asset-purchase policies and longer-horizon rates. Appendix Figure 12 reports the resulting standardized shock series and Appendix Table 19 validates the series. After constructing the shocks, we cumulate announcement-level surprises within calendar months, assigning zero to months without ECB announcements. Then, we estimate:

$$\log(\text{Maturity})_{bft} = \alpha_{bt} + \alpha_{fb} + \beta_1 \,\text{ST Debt}_{f,t-1} + \sum_{k \in \mathcal{K}} \beta_k \left(\text{Shock}_{k,t-1} \times \text{ST Debt}_{f,t-1}\right) + \Gamma X_{f,t-1} + \varepsilon_{bft} \tag{3}$$

where $\mathcal{K}$ contains the four policy factors. Because the shocks vary only over time, their direct effects are absorbed by bank-by-month fixed effects. Identification of β_k therefore comes from comparing firms borrowing from the same bank in the same month, but entering the period with different inherited rollover exposure, following the same policy surprise. The fixed effects strategy and firm controls are the same as equation 1. The coefficients β_k measure whether plausibly exogenous policy-induced movements in interest rates generate a differential maturity response among firms with high rollover exposure. The high-frequency identification makes the shocks unlikely to reflect contemporaneous changes in firm-level credit demand or borrower fundamentals. We acknowledge that high-frequency rate surprises may also contain information revealed by the central bank about the economic outlook. Following Jarociński and Karadi (2020), we distinguish pure monetary-policy shocks from central-bank information shocks using the high-frequency co-movement of the 1-month OIS surprise and Euro Stoxx 50 equity prices. Announcements in which interest rates and equity prices move in opposite directions are classified as *pure monetary-policy shocks*; announcements in which they move in the same direction are classified as *central-bank information shocks*. Appendix Figure 12 reports the series, and we complement the above-described analysis.

Table 4 reports the results. Column (1) shows that Timing shocks generate the largest maturity extension among high rollover-exposed firms. A one-standard-deviation Timing shock lengthens new-loan maturity by about 0.9% more for firms at the 75th than at the 25th percentile of rollover exposure, or roughly 17 days at the mean maturity. Forward Guidance shocks raise relative maturity by about 0.4%, or 8 days. Target shocks instead reduce it by about 0.4%, or 7 days. This sign difference is consistent with the maturity-insurance mechanism. Target shocks raise near-term refinancing costs precisely when exposed firms must borrow, whereas Timing and Forward Guidance shocks raise expected rates farther ahead, increasing the value of postponing the next refinancing

[10]Following Altavilla et al. (2019), the rotation matrix is restricted so that the non-target factors have zero loading on the 1-month OIS surprise. The quantitative-policy factor is further selected, among the admissible orthogonal rotations, to minimize its variance over the pre-crisis period (January 2002–August 2008). The first restriction separates news about future policy from contemporaneous target-rate surprises, while the second identifies a factor that is largely inactive before unconventional monetary policy.

date. The Quantitative Policy interaction is negative but statistically insignificant. Columns (2)–(3) separate pure monetary-policy and central-bank information components following Jarociński and Karadi (2020). The results strengthen for pure policy shocks: the Timing effect rises to about 1.7% (31 days), the Forward Guidance effect to 0.7% (12 days), and the Target effect to −1.3% (25 fewer days), while Quantitative Policy remains insignificant. By contrast, the information components have little differential effect on maturity, apart from a small positive Forward Guidance coefficient. Taken together, the high-frequency evidence reinforces the maturity-insurance interpretation: firms with greater inherited rollover exposure extend maturity primarily when policy news raises expected future rates, whereas unexpected tightening pushes maturity in the opposite direction.

3.4 Robustness

We assess whether the baseline results are robust to stricter time-varying credit supply fixed effects, the functional form of the maturity outcome, inflation compensation, or the particular sample period. Appendix Table 18 reports a series of robustness checks, Column (1) reports the baseline.

Credit supply and firm fundamentals. A first concern is that the interaction between rollover exposure and the yield curve reflects lender behavior rather than borrowers' refinancing needs. The baseline specification already absorbs shocks common to all borrowers of a given bank in a given month through bank×month fixed effects. Column (2) further replaces these with bank×industry×location×month fixed effects (Degryse et al., 2019). The comparison is therefore among firms borrowing from the same lender, in the same industry and region, in the same month, absorbing credit-supply policies targeted to particular sectors or geographic markets. The interaction remains essentially unchanged, at −0.0219 compared with −0.0210 in the baseline, although the sample declines substantially as more fixed-effect cells become singletons. Then, Column (3) controls directly for total new credit, addressing the possibility that maturity adjustment simply reflects changes in loan size or overall credit availability. The interaction remains negative and similar in magnitude (−0.0175). Column (4) uses the 10Y–2Y slope and yields a larger negative interaction (−0.0812), indicating that the result is not specific to the 5Y–1Y measure of the term structure. Column (5) additionally interacts rollover exposure with the lagged ECB deposit facility rate. The yield curve interaction remains negative (−0.0180), while the policy-rate interaction is small and statistically insignificant. Column (6) replaces bank×month fixed effects with observable time-varying lagged bank controls, while column (7) instead adds firm×year fixed effects, absorbing annual firm-level changes in investment plans, asset composition, and cash-flow expectations. The interaction remains negative and significant in both cases. Taken together, these exercises indicate that the baseline result is not driven by observable lender characteristics, targeted credit supply, or changes in firm fundamentals.

Functional form of the maturity outcome. A second concern is that the baseline log-maturity specification may be influenced by a small number of very long-duration contracts. Appendix Table 22 replaces the logarithm of maturity with indicators for whether a newly originated loan maturity exceeds one, two, three, five, and seven years. Rollover-exposed firms are more likely to

borrow above every maturity threshold: the coefficient on short-term debt declines monotonically from 0.076 at the one-year threshold to 0.008 at the seven-year threshold. The interaction with the yield curve slope is negative and significant at the one-, two-, and three-year thresholds, but small and statistically insignificant at five and seven years. Hence, the baseline interaction is not driven by the extreme upper tail of the maturity distribution; it is concentrated at maturities where most loan contracts are observed.

Inflation compensation. A third concern is that the 2022–2024 period combined rapid monetary tightening with unusually high inflation, raising the possibility that nominal yield curve movements capture changes in inflation compensation rather than financing conditions relevant for maturity choice. This distinction is particularly relevant given the evidence of Ropele et al. (2022), who show that firms adjust their debt maturity in response to inflation expectations, including by refinancing existing loans to lock in borrowing rates. In Appendix Table 24 we decompose the nominal slope into real-rate and inflation-compensation components following Gürkaynak et al. (2010). The interaction between rollover exposure and the real slope remains negative and highly significant: -0.079 for the 10Y–2Y slope and -0.071 for the 5Y–1Y slope. The interaction with inflation compensation is also negative and significant. Thus, the negative slope interaction is not solely attributable to inflation compensation; it is present in the real-rate component as well.

Extended sample period. We next examine whether the result is specific to the 2022–2023 tightening episode. Appendix Table 25 extends the new loan analysis back to 2014 using Orbis balance-sheet data. Because outstanding debt is not available before 2018 in Anacredit, we proxy rollover exposure by current liabilities relative to total debt from Orbis. This measure is substantially coarser than the AnaCredit measure and includes trade credit and other short-term obligations. Nevertheless, the interaction between the yield curve slope and rollover exposure remains negative and significant under both the 5Y–1Y and 10Y–2Y yield curve slopes as reported in Columns (1)-(2). Column (3) further allows both the level effect of rollover exposure and its interaction with the yield curve to differ after 2020. Neither the post-2020 interaction with rollover exposure nor the triple interaction with the yield curve is statistically significant. Thus, although the extended-sample measure is considerably noisier, we find no evidence of a structural break in the maturity response after 2020. Taken together, these results indicate that the mechanism is not specific to the most recent ECB tightening cycle or to the COVID-era sample period.

Extended event set for policy surprises. Finally, we assess whether the high-frequency results depend on restricting policy surprises to scheduled ECB Governing Council meetings. The baseline high-frequency analysis uses policy surprises around ECB announcement windows following Altavilla et al. (2019), but this may omit economically relevant policy news communicated between meetings. We therefore repeat the analysis using the extended event-study database of Altavilla et al. (2026), which broadens the event set to include policymakers' intermeeting communication[11]. This extension is particularly useful during periods of frequent policy communication, when re-

[11] In this case, *Target* is the 1-month OIS surprise; *Timing* is the component of the 6-month OIS surprise orthogonal to Target; *Forward Guidance* is the component of the 2-year OIS surprise orthogonal to Target and Timing; and *Quantitative Policy* is the residual 10-year surprise orthogonal to the three shorter-horizon components.

stricting measurement to scheduled Governing Council meetings may omit economically relevant policy news. We aggregate the event-level surprises to the monthly frequency and re-estimate equation (3). Appendix A.3 reports the resulting shock series, their loadings across the OIS curve, and the corresponding loan-level maturity estimates. Appendix Table 30 shows that the negative Target response and positive Timing response remain robust in the broader event sample. The Forward Guidance coefficient becomes small and statistically insignificant, while Quantitative Policy shocks are associated with shorter maturity. The key result that near-term versus future-rate policy news generates different maturity responses therefore survives the broader event definition.

4 Validating the Mechanism and Alternative Explanations

The baseline results are consistent with firms using debt maturity to manage interest rate and refinancing risk. This section provides additional evidence on that interpretation. We first distinguish rollover risk—the need to refinance principal—from repricing risk—the exposure of outstanding debt to changes in short rates. We then ask whether firms adjust the type of interest rate of new borrowing alongside maturity. Finally, we examine alternative explanations based on asset-maturity matching, borrower risk, liquidity substitution, and selection into new borrowing.

4.1 Rollover Risk, Repricing Risk, and Contract Design

A firm can be exposed to future interest-rate changes without having principal coming due, and it can have principal coming due without facing an interest-rate reset. The former is repricing exposure: the interest rate on outstanding debt resets, as on floating-rate loans, while the principal remains outstanding. The latter is rollover exposure: principal must be refinanced at prevailing market conditions, irrespective of the interest-rate structure of the existing contract. We measure repricing exposure by the share of outstanding bank debt whose interest rate resets within the next 12 months:

$$\text{Reset Share}_{f,t-1} = \frac{\text{Bank debt with an interest-rate reset within 12 months}}{\text{Total bank debt}} \tag{4}$$

Interestingly, the correlation between reset and rollover exposure is only -0.04. Then, we estimate

$$\begin{aligned} \log(\text{Maturity})_{bft} = \alpha_{bt} + \alpha_{fb} + \beta_1 \text{ST Debt}_{f,t-1} + \beta_2 \text{Reset Share}_{f,t-1} \\ + \beta_3 \left(\text{ST Debt}_{f,t-1} \times \text{Reset Share}_{f,t-1}\right) \\ + \beta_4 \left(\text{YC}_{t-1} \times \text{ST Debt}_{f,t-1}\right) + \beta_5 \left(\text{YC}_{t-1} \times \text{Reset Share}_{f,t-1}\right) \\ + \beta_6 \left(\text{YC}_{t-1} \times \text{ST Debt}_{f,t-1} \times \text{Reset Share}_{f,t-1}\right) + \Gamma X_{f,t-1} + \varepsilon_{bft} \end{aligned} \tag{5}$$

Table 5 reports the results. Controlling for repricing exposure, firms with greater rollover exposure continue to borrow at substantially longer maturities. At the average yield curve, the coefficient on Short-Term Debt is positive and statistically significant: 0.229 with the 5Y–1Y slope

Table 5: Rollover Exposure, Repricing Exposure, and New Loan Maturity

$\log(\text{Maturity})_{b,f,t}$	(1) 5Y–1Y	(2) 10Y–2Y
ST Debt$_{f,t-1}$	0.229***	0.236***
	(0.00549)	(0.00563)
Reset Share$_{f,t-1}$	-0.0205***	-0.0284***
	(0.00777)	(0.00794)
ST Debt$_{f,t-1}\times$ Reset Share$_{f,t-1}$	-0.0128	-0.00215
	(0.0108)	(0.0110)
YC$_{t-1}\times$ ST Debt$_{f,t-1}$	-0.00282	-0.0285***
	(0.00599)	(0.00708)
YC$_{t-1}\times$ Reset Share$_{f,t-1}$	0.0312***	0.0343***
	(0.00835)	(0.00993)
YC$_{t-1}\times$ ST Debt$_{f,t-1}\times$ Reset Share$_{f,t-1}$	-0.0564***	-0.0455***
	(0.0133)	(0.0152)
Observations	1,659,220	1,659,220
Adjusted R^2	0.617	0.617
Bank $\times$ Month FE	Yes	Yes
Firm $\times$ Bank FE	Yes	Yes
Firm Controls	Yes	Yes

Notes: The dependent variable is the logarithm of the maturity of newly originated loans. Rollover exposure is proxied by Short-term Debt, which is calculated as the share of outstanding bank debt maturing within 12 months. Repricing exposure is proxied by the Reset Share, which is the share of outstanding bank debt whose interest rate resets within 12 months. Column (1) uses the 5Y–1Y yield curve slope and column (2) uses the 10Y–2Y slope as robustness. All specifications include lagged firm controls, bank×month fixed effects, and firm×bank fixed effects. Standard errors are clustered at the firm and month level. *, **, and *** denote significance at the 10%, 5%, and 1% levels, respectively.

and 0.236 with the 10Y–2Y slope. By contrast, the coefficient on Reset Share is negative and statistically significant, at -0.0205 and -0.0284, respectively. The interaction between rollover and repricing exposure is small and statistically insignificant in both specifications. These results indicate that the positive relationship between rollover exposure and new-loan maturity persists after accounting for differences in interest-rate repricing.

The key finding is that repricing exposure amplifies the sensitivity of maturity choices to the yield curve. The triple interaction between the yield-curve slope, rollover exposure, and repricing exposure is negative and statistically significant under both slope measures, with coefficients of -0.0564 and -0.0455, respectively. With the 5Y–1Y slope, the interaction between the yield curve and rollover exposure is only -0.0028 for firms with no debt repricing within 12 months, compared with -0.0592 for firms whose outstanding debt fully reprices within the year. Using the 10Y–2Y slope, the corresponding effects are -0.0285 and -0.0740. These differences are economically meaningful. For the 50-percentage-point difference in rollover exposure between the 25th and

Table 6: New Loan Maturity and Asset Maturity Matching

$\log(\text{Maturity})_{b,f,t}$	(1) Full Sample	(2) No Real Estate	(3) No Working Capital	(4) Investment Only
ST $\text{Debt}_{f,t-1}$	0.179***	0.105***	0.153***	0.291***
	(0.0038)	(0.0047)	(0.0046)	(0.0527)
YC (5Y–1Y)$_{t-1}\times$ ST $\text{Debt}_{f,t-1}$	-0.0359***	-0.0183***	-0.0190***	-0.0003
	(0.0042)	(0.0052)	(0.0049)	(0.0613)
Observations	2,030,234	1,131,837	833,305	20,480
Adjusted R^2	0.638	0.724	0.716	0.630
Bank × Month FE	Yes	Yes	Yes	Yes
Firm × Bank FE	Yes	Yes	Yes	Yes
Firm Controls	Yes	Yes	Yes	Yes

Notes: The dependent variable is the log maturity of newly originated loans. Column (1) reports the full sample. Note that the number of observations is different from the baseline regression because we include only tangibility, size, and leverage as firm controls to maximize the number of observations. Column (2) excludes loans with a real-estate purpose, column (3) excludes working-capital loans, and column (4) restricts the sample to loans declared to be used for investment purposes. All specifications include lagged firm controls, bank×month fixed effects, and firm×bank fixed effects. Standard errors are clustered at the firm and month level. *, **, and *** denote significance at the 10%, 5%, and 1% levels, respectively.

75th percentiles, a one-percentage-point steepening of the 5Y–1Y slope reduces relative new-loan maturity by only about 0.1% when Reset Share is zero, compared with approximately 3.0% when Reset Share equals one. For the 10Y–2Y slope, the corresponding effects are 1.4% and 3.7%. Thus, although rollover and repricing exposure capture distinct contractual risks, they interact in shaping maturity decisions: firms facing both near-term refinancing obligations and interest-rate resets exhibit substantially stronger maturity responses to the term structure of interest rates.

Additional evidence confirms that maturity adjustment is not simply a means of fixing interest rates. Appendix Table 21 estimates the baseline specification separately for fixed-rate and floating-rate new loans. Rollover exposure predicts longer new-loan maturity in both samples, and the interaction with the yield curve remains negative. Importantly, among fixed-rate originations, the interest rate is already fixed, so extending maturity also postpones the next principal refinancing date. The persistence of the maturity response in this subsample is therefore consistent with a refinancing-insurance mechanism distinct from coupon fixation.

Finally, firms also adjust the interest-rate structure of new borrowing. Appendix Table 17 uses the floating-rate share of newly originated loans as the dependent variable. The interaction between the yield curve and rollover exposure is negative under both slope measures. For the 5Y–1Y slope, a one-percentage-point steepening reduces the floating-rate share of new borrowing by about 0.6 percentage points more for a firm at the 75th than at the 25th percentile of rollover exposure; the corresponding effect for the 10Y–2Y slope is approximately 0.7 percentage points. Taken together, these findings show that firms adjust both debt maturity and interest-rate structure in response to financing conditions. While the two margins interact, maturity adjustment provides a distinct means of managing principal refinancing risk beyond the fixation of interest rates.

4.2 Alternative Explanations

Next, we examine whether the maturity response can instead be explained by asset-maturity matching, borrower risk, substitution toward committed credit lines, or selection into new borrowing.

Asset-maturity matching. One possibility is that firms choose longer maturity debt simply because the assets being financed have longer duration. Table 6 re-estimates the baseline specification across loan-purpose subsamples. The yield curve interaction remains negative after excluding real-estate loans and after excluding working-capital loans. In the investment-only sample the point estimate is close to zero. More directly, column (7) of Appendix Table 18 includes firm×year fixed effects, which absorb annual changes in investment plans and asset composition. The yield curve interaction remains -0.0163, close to the baseline estimate. These results provide little support for asset-maturity matching as the source of the maturity adjustment.

Borrower risk. A Diamond (1991) mechanism predicts that riskier firms may extend maturity to lock in funding before their expected credit quality deteriorates. If this alternative mechanism explains the baseline results, the sensitivity of maturity to the yield curve should be stronger among firms with higher default risk. To test this prediction, we augment the baseline specification with the probability of default assigned by the lending bank and interact it with both the yield curve and short-term debt exposure. Table 7 controls for the lender-assigned probability of default and allows the entire maturity response to vary with default risk. The yield curve interaction with rollover exposure remains negative and statistically significant (-0.0248), while the triple interaction with probability of default is statistically insignificant. Borrower risk therefore affects maturity levels but does not account for the differential term-structure response of rollover-exposed firms.

New bank credit. An alternative explanation is that rollover-exposed firms adjust the quantity of new borrowing, rather than its maturity, when financing conditions change. Appendix Table 28 examines this possibility using the volume of newly originated credit as the dependent variable. While firms with greater rollover exposure obtain more new credit on average, the interaction between rollover exposure and the yield curve is small and statistically insignificant for both total new credit and term loans. These results suggest that the differential maturity response is not accompanied by a corresponding adjustment in new credit volumes, conditional on origination. The evidence is therefore consistent with firms adjusting the contractual maturity of new borrowing rather than simply changing its amount.

Committed credit lines. Committed credit lines could provide an alternative way to insure against future refinancing needs. If they substitute for maturity insurance, exposed firms should increase their reliance on committed lines when long-duration borrowing becomes expensive. Column (6) of Table 8 provides no evidence of such substitution. The interaction between the yield curve and rollover exposure is negative, indicating that exposed firms do not offset shorter maturity by expanding committed credit lines. Hence, the baseline maturity response is not explained by substitution toward pre-committed liquidity.

Pledged collateral. Banks may respond to greater refinancing risk or changes in financing

Table 7: Rollover Exposure and Borrower Riskiness

$\log(\text{Maturity})_{b,f,t}$	(1)	(2)
ST $\text{Debt}_{f,t-1}$	0.192***	0.202***
	(0.00782)	(0.00804)
YC (5Y–1Y)$_{t-1}\times$ ST $\text{Debt}_{f,t-1}$	-0.0170**	-0.0248***
	(0.00785)	(0.00833)
$\text{PD}_{f,t-1}$	0.160***	0.252***
	(0.0311)	(0.0452)
YC (5Y–1Y)$_{t-1}\times$ $\text{PD}_{f,t-1}$		-0.126*
		(0.0748)
ST $\text{Debt}_{f,t-1}\times$ $\text{PD}_{f,t-1}$		-0.251***
		(0.0587)
YC (5Y–1Y)$_{t-1}\times$ ST $\text{Debt}_{f,t-1}\times$ $\text{PD}_{f,t-1}$		0.149
		(0.109)
Observations	559,105	559,105
Adjusted R^2	0.642	0.642
Bank × Month FE	Yes	Yes
Firm × Bank FE	Yes	Yes
Firm Controls	Yes	Yes

Notes: The dependent variable is the logarithm of maturity of newly originated loans. Borrower riskiness is proxied by the firm's lagged probability of default (PD), which is assigned by the lender. Column (1) controls for borrower risk directly, while column (2) allows the relationship between rollover exposure, the yield curve, and loan maturity to vary with borrower risk. All specifications include lagged firm controls, bank×month fixed effects, and firm×bank fixed effects. Standard errors are clustered at the firm and month level. *, **, and *** denote significance at the 10%, 5%, and 1% levels, respectively.

conditions by adjusting collateral requirements on new loans, potentially providing an alternative margin of contract adjustment to debt maturity. Appendix Table 20 examines this margin using the reported value of collateral associated with new bank credit. Loans to firms with greater rollover exposure are associated with significantly higher pledged collateral values on average. However, the interaction between rollover exposure and the yield curve is economically negligible and statistically insignificant. Thus, while collateral pledged varies systematically with firms' refinancing exposure, we find no evidence of differential collateral adjustments across firms with greater rollover exposure as the yield curve changes.

Selection into new borrowing. Finally, the maturity regressions condition on firms obtaining new credit. Changes in the composition of borrowers could therefore generate the estimated response if rollover-exposed firms selectively exit the new-loan sample when the yield curve changes. Appendix Table 23 estimates firm-month linear probability models for obtaining a new loan. The evidence goes in the opposite direction. A 10 percentage-point increase in rollover exposure raises the monthly probability of new borrowing by about 0.7 percentage points, and the interaction between rollover exposure and the yield curve is positive and statistically significant.

Thus, rollover-exposed firms become more, rather than less, likely to appear in the new-loan sample when the curve steepens. These extensive-margin results do not support the specific hypothesis that exposed firms systematically exit the origination sample; a higher probability of borrowing does not by itself rule out changes in the characteristics of the exposed firms that do borrow, which the borrower-risk and loan-purpose tests above address. Overall, the evidence supports an active liability-management interpretation. Rollover and repricing exposure have distinct effects, firms adjust both maturity and coupon structure, and the results are not explained by asset-maturity matching, borrower risk, liquidity substitution, or selection into borrowing.

5 Real Effects

Does maturity adjustment affect firms' balance sheets and real activity? Financing choices are the transmission margin; real outcomes reveal whether that margin matters. In this section, we show that exposed firms rebalance toward longer-term bank debt but do so less when the yield curve steepens; that they then face broader balance-sheet tightening and higher debt-service pressure; and that investment declines are concentrated among firms that had *not* previously obtained maturity insurance, especially smaller ones.

5.1 Empirical Specification

We estimate at the firm-annual frequency:

$$\Delta Y_{f,t} = \alpha_f + \alpha_{c,s,t} + \beta_1 \text{ST Debt}_{f,t-1} + \beta_2 \left(\text{YC}_{t-1} \times \text{ST Debt}_{f,t-1}\right) + \Gamma X_{f,t-1} + \varepsilon_{f,t} \tag{6}$$

where $\Delta Y_{f,t}$ is the change between $t-1$ and t in a firm-level outcome, $\text{ST Debt}_{f,t-1}$ is the share of bank loans maturing within 12 months, and YC_{t-1} is the lagged 5Y–1Y slope. Firm fixed effects and country–sector–year fixed effects absorb time-invariant heterogeneity and shocks common to firms in the same country–sector–year cell; time-varying firm controls $X_{f,t-1}$ includes total assets, tangibility, interest coverage, sales growth, and cash over assets. The coefficient of interest is β_2: negative for borrowing and investment outcomes means exposed firms contract more when the curve steepens; positive for interest expenses outcomes means they face tighter debt-service pressures on their balance sheet.

5.2 Liability Restructuring and Debt-Service Pressure

Table 8 shows that higher rollover exposure is associated with substantial balance-sheet adjustment. Panel A documents total liability restructuring. Moving from the 25th to the 75th percentile of rollover exposure—a 20 percentage-point increase in this sample—raises average bank-debt maturity by about five months and the share of long-term bank loans by 15.8 percentage points. Bank debt increases by about 5.9%, while total debt declines by about 1.0 percentage point. Thus,

Table 8: Rollover Exposure, Firms' Balance Sheet, and the Term Structure

Panel A. Liability Restructuring

	(1) Δ Bank-Debt Maturity	(2) Δ Long Term Bank Loans	(3) Δ Bank Debt	(4) Δ Total Debt
ST Debt$_{f,t-1}$	2.219***	0.792***	0.295***	-0.0483***
	(0.0345)	(0.00583)	(0.00637)	(0.00193)
YC (5Y–1Y)$_{t-1}$× ST Debt$_{f,t-1}$	-0.335***	-0.654***	-0.0339***	-0.0466***
	(0.0554)	(0.00890)	(0.00703)	(0.00578)
Observations	594,168	594,168	594,168	594,168
Adjusted R^2	0.0506	0.0357	0.0416	0.107
Industry × Country × Time FE	Yes	Yes	Yes	Yes
Firm FE	Yes	Yes	Yes	Yes
Firm Controls	Yes	Yes	Yes	Yes

Panel B. Liquidity and Interest Expenses

	(5) Δ Cash /TA	(6) Δ Credit Lines	(7) Interest Exp. / EBITDA	(8) Debt-Service Ratio
ST Debt$_{f,t-1}$	0.153***	0.00287	-0.00111	-0.00589***
	(0.0405)	(0.00507)	(0.0008)	(0.000495)
YC (5Y–1Y)$_{t-1}$× ST Debt$_{f,t-1}$	-0.444***	-0.0179**	0.0151***	0.0512***
	(0.115)	(0.00739)	(0.0024)	(0.00156)
Observations	594,168	69,785	594,168	594,168
Adjusted R^2	0.0297	0.0368	0.673	0.822
Industry × Country × Time FE	Yes	Yes	Yes	Yes
Firm FE	Yes	Yes	Yes	Yes
Firm Controls	Yes	Yes	Yes	Yes

Notes: This table reports firm-level estimates of how rollover exposure interacts with the term structure to affect firms' total liability, liquidity, and debt-service burden. Panel A reports changes in bank-debt maturity, the share of long-term bank loans, bank debt, and total debt. Panel B reports cash growth, changes in committed credit lines, interest expenses relative to EBITDA, and the debt-service ratio. Short-term Debt is the lagged share of outstanding bank debt with residual maturity within 12 months. YC (5Y–1Y) is the lagged difference between the 5-year AAA euro-area government bond yield and the 1-year OIS rate. All specifications include lagged firm controls, firm fixed effects, and industry×country×time fixed effects. Standard errors are clustered at the firm level and reported in parentheses. *, **, and *** denote statistical significance at the 10%, 5%, and 1% levels, respectively.

exposed firms shift their liability structure toward longer-term bank financing without increasing overall leverage. This adjustment weakens when the yield curve steepens. A one percentage-point increase in the 5Y–1Y slope reduces the maturity extension by about one month and the increase in the long-term bank-loan share by 13.1 percentage points. It also lowers bank debt and total debt, indicating that firms do not fully replace the lost maturity adjustment by expanding other types borrowing. Panel B shows that the same firms experience tighter liquidity and greater debt-service pressure. A one percentage-point steepening reduces cash growth by about 8.9 percentage points for firms at the 75th relative to the 25th percentile of rollover exposure. At the same time, interest expenses relative to EBITDA rise by about 0.30 percentage points and the debt-service ratio rises by about 1.0 percentage point. Credit-line commitments also decline rather than expand. Taken together, the decline in debt and cash alongside higher debt-service burdens indicates broader balance-sheet tightening, rather than a simple reshuffling across liability categories.

5.3 Investment and Maturity Adjustment

Investment is a demanding outcome, since it responds directly to demand expectations and to information revealed by policy. We next examine whether the relationship between rollover exposure and investment depends on firms' prior maturity adjustment. We estimate

$$\begin{aligned}\text{Investment}_{f,t} = \alpha_f + \alpha_{c,s,t} + \beta_1 \text{ST Debt}_{f,t-1} + \beta_2 \left(\text{YC}_{t-1} \times \text{ST Debt}_{f,t-1}\right) \\ + \beta_3 \text{Adjust}_{f,t-1} + \beta_4 \left(\text{YC}_{t-1} \times \text{Adjust}_{f,t-1}\right) + \beta_5 \left(\text{ST Debt}_{f,t-1} \times \text{Adjust}_{f,t-1}\right) \\ + \beta_6 \left(\text{YC}_{t-1} \times \text{ST Debt}_{f,t-1} \times \text{Adjust}_{f,t-1}\right) + \Gamma X_{f,t-1} + \varepsilon_{f,t}\end{aligned} \tag{7}$$

$\text{Adjust}_{f,t-1}$ captures predetermined liability adjustment, measured in two ways: an indicator dummy for having increased debt maturity in the previous year, $t-1$, or the lagged change in long-term bank loans. We expect $\beta_2 < 0$ and $\beta_6 > 0$—rollover exposed firms cut investment more when the curve steepens, but less so if they had already lengthened. Table 9 reports the results. First, Column (1) shows that rollover-exposed firms cut investment more when the curve steepens: the β_2 slope–exposure interaction is -0.040 and statistically significant. A one-percentage-point steepening of the 5Y–1Y slope lowers the investment rate by about 0.8 percentage points more for a firm at the 75th than at the 25th percentile of rollover exposure, a 20-percentage-point gap in the firm-year sample. Second, investment declines concentrate among non-adjusters. In column (2) the β_2 slope–exposure interaction is -0.150 and statistically significant, and the β_6 triple interaction with the maturity-extension indicator is $+0.149$ and statistically significant. Among firms that had *not* extended maturity, a 1 percentage point steepening lowers investment by about 3.0 percentage points more at the 75th than at the 25th percentile of rollover exposure, while among firms that had extended, this differential is almost fully offset. Column (3) reproduces the pattern using the lagged change in long-term bank loans.

Because maturity adjustment is lagged, the result is not a mechanical contemporaneous relation. However, maturity adjustment is itself endogenous, so we read these estimates as evidence on where the real effects of rollover exposure are concentrated, not as the causal effect of maturity extension. As a robustness check, the results are robust when excluding firms with a high probability of default and hence may have extended maturity in anticipation of a deterioration in their credit quality.

Firm size. Finally, we examine how the channel varies with firm size. We define small firms as those in the bottom quartile of total assets and interact this indicator with rollover exposure and the yield curve slope, with the outcomes of Tables 1 and 8 (Appendix Table 27). The maturity response is substantially stronger for small firms: when the yield curve steepens, small rollover-exposed firms shorten new-loan maturity more than twice as much as larger firms with the same exposure, and the reductions in bank-debt maturity and in the share of long-term bank loans are 66% and 40% larger. Their interest burden also rises more. The investment response of rollover-exposed firms is about twice as large among small firms.

Table 9: Rollover Exposure, Investment and Maturity Adjustment

$\text{Investment}_{f,t}$	(1)	(2)	(3)
$\text{ST Debt}_{f,t-1}$	0.0528***	0.00747	-0.00783
	(0.00454)	(0.00531)	(0.00834)
Yield Curve 5Y–1Y$_{t-1}\times$ $\text{ST Debt}_{f,t-1}$	-0.0404***	-0.150***	-0.0502**
	(0.0124)	(0.0192)	(0.0222)
$\text{Extension Dummy}_{f,t-1}$		-0.195***	
		(0.00220)	
$\text{Short-term Debt}_{f,t-1}\times$ $\text{Extension Dummy}_{f,t-1}$		0.0392***	
		(0.00636)	
YC 5Y–1Y$_{t-1}\times$ $\text{Extension Dummy}_{f,t-1}$		-0.0703***	
		(0.00862)	
YC 5Y–1Y$_{t-1}\times$ $\text{Short-term Debt}_{f,t-1}\times$ $\text{Extension Dummy}_{f,t-1}$		0.149***	
		(0.0261)	
Δ $\text{Long Term Bank Loans}_{f,t-1}$			0.0584***
			(0.00330)
$\text{Short-term Debt}_{f,t-1} \times \Delta$ $\text{LT Bank Loans}_{f,t-1}$			0.0495***
			(0.00909)
YC 5Y–1Y$_{t-1} \times \Delta$ $\text{LT Bank Loans}_{f,t-1}$			0.0274
			(0.0174)
YC 5Y–1Y$_{t-1}\times$ $\text{Short-term Debt}_{f,t-1} \times \Delta$ $\text{LT Bank Loans}_{f,t}$			0.113**
			(0.0503)
Observations	594,168	594,168	594,168
Adjusted R^2	0.134	0.146	0.168
Industry-Country-Time FE	Yes	Yes	Yes
Firm FE	Yes	Yes	Yes
Firm controls	Yes	Yes	Yes

Notes: The dependent variable is the firm investment rate in fixed assets. Short-term debt is the share of outstanding bank debt with residual maturity of 12 months. The yield curve is the 5Y–1Y slope. The extension dummy equals one when the firm increased debt maturity in the previous year. Δ LT Bank Loans is the change in long-term bank loans. All specifications include firm fixed effects, industryxcountryxtime fixed effects and firm controls. Standard errors clustered at the firm level are in parentheses. *, **, and *** denote significance at the 10%, 5%, and 1% levels, respectively.

Taking stock:

Four findings summarize the empirical analysis. *First*, firms actively manage interest rate and refinancing risk through debt maturity. Moving from the 25th to the 75th percentile of rollover exposure increases new-loan maturity by about 7.5 months, and the effect persists when exposure is measured using repayment schedules fixed before the tightening cycle. *Second*, the response depends on the source of interest-rate movements. Higher expected future short rates induce exposed firms to lengthen maturity, whereas higher term premia induce them to shorten it. High-frequency ECB shocks reinforce this distinction: news about higher future policy rates lengthens maturity, while unexpected tightening concentrated at the current policy rate compresses it. *Third*, maturity adjustment shapes firms' balance-sheet and real responses. Exposed firms shift toward

longer-term bank debt, but this adjustment weakens when long-duration funding becomes relatively costly. Firms that do not extend maturity subsequently experience larger investment declines, with stronger effects among smaller firms. *Fourth*, the evidence is not explained by asset-maturity matching, borrower risk, unusually long contracts, substitution toward committed credit lines, or selection into new borrowing.

6 Quantitative Model

The empirical analysis shows that firms with higher rollover exposure adjust their debt maturity differently as aggregate monetary conditions change. These cross-sectional responses provide evidence of the maturity insurance channel but do not establish its aggregate consequences for monetary transmission. We develop a quantitative model to assess how endogenous maturity shapes the responses of debt, investment, and default to monetary shocks. The model is annual and builds on standard quantitative models of firm financing with financial frictions (Cooley and Quadrini, 2001; Gomes, 2001; Hennessy and Whited, 2005) and endogenous debt maturity (Crouzet, 2017; Jungherr and Schott, 2021; Poeschl, 2023). Firms differ in productivity, capital, debt, and inherited repayment obligations. Aggregate financing conditions are characterized by the current short rate, news about future short rates, and a term premium wedge.

6.1 Environment and aggregate states

Time is discrete, and the horizon is infinite. Firm i produces using capital (K) according to

$$\pi(z_{it}, K_{it}) = \bar{z} z_{it} K_{it}^{\alpha} - F, \qquad \alpha \in (0,1) \tag{8}$$

where z_{it} is firm-level productivity, $\bar{z}$ scales the level of production, and F is a fixed operating cost. Decreasing returns imply a well-defined optimal firm size, while the fixed operating cost generates negative cash flow at sufficiently low productivity, increasing firms' exposure to adverse productivity realizations.

Firm-level productivity follows an AR(1) process:

$$\log z_{i,t+1} = \rho_z \log z_{it} + \sigma_z \varepsilon^z_{i,t+1}, \qquad \varepsilon^z_{i,t+1} \sim \mathcal{N}(0,1) \tag{9}$$

Capital evolves according to

$$K_{i,t+1} = (1 - \delta_K) K_{it} + I_{it} \tag{10}$$

where I_{it} denotes gross investment and δ_K denotes the capital depreciation rate. Investment is subject to the convex adjustment cost:

$$\Phi_K(I_{it}, K_{it}) = \frac{\psi_K}{2} \left(\frac{I_{it}}{K_{it}} \right)^2 K_{it} \tag{11}$$

Capital adjustment costs make capital accumulation gradual and link current investment to the continuation value of installed capital. Investment is fully reversible: capital trades at a unit price in both directions, with disinvestment limited only by the convex cost and a minimum operating scale $K_{i,t+1} \geq \underline{K}$. This feature gives constrained firms an additional liquidity margin: capital sales can substitute for external refinancing when debt comes due, potentially attenuating the maturity-insurance mechanism.[12]

Aggregate financing conditions are summarized by

$$A_t = (r_t, f_t, \omega_t) \tag{12}$$

where r_t is the current short rate, f_t carries news about future short rates, and ω_t captures the relative cost of supplying long-duration credit. The distinction between current-rate and future-policy-path innovations follows Gürkaynak et al. (2005). Long-term yields also reflect term premia, which can vary for reasons other than revisions to expected future short rates. We represent this additional source of variation through a persistent term-premium wedge ω_t. The three states follow the following processes:

$$r_{t+1} = (1 - \rho_r)\bar{r} + \rho_r r_t + f_t + \sigma_r \varepsilon^r_{t+1} \tag{13}$$

$$f_{t+1} = \rho_f f_t + \sigma_f \varepsilon^f_{t+1} \tag{14}$$

$$\omega_{t+1} = (1 - \rho_\omega)\bar{\omega} + \rho_\omega \omega_t + \sigma_\omega \varepsilon^\omega_{t+1} \tag{15}$$

where the innovations are mutually independent standard normals.[13]

The short rate determines the risk-free discount rate for one-period debt. A one-unit increase in f_t raises $\mathbb{E}_t[r_{t+1}]$ one for one without changing the current short rate, with longer-horizon effects governed by ρ_r and ρ_f. We refer to ω_t as a term-premium wedge: an increase in ω_t reduces the proceeds from newly issued long-term debt without directly changing the expected short-rate path. It is an exogenous to isolate how the same monetary innovation affects firms' maturity choices, investment, and default under different term premium environments, and it is distinct from the compensation for default risk and recovery that enters the reference price. Its process is directly disciplined by movements in the empirical term premium component of long rates, as detailed in Appendix D.3.[14]

6.2 Debt structure and rollover exposure

Firm i enters period t with short-term debt B^S_{it} and long-term debt B^L_{it}. Short-term debt matures fully within the period. Long-term debt pays a fixed coupon c_L and amortizes geometrically: a

[12] In the simulated stationary economy, capital sales finance approximately 12% of principal coming due across the rollover-exposure distribution. Firms default only after liquidating capital to the minimum operating scale.

[13] We specify independent innovations to isolate the effects of each financing component. Section 8.3 examines their joint effects by combining expected-rate news with a deterioration in the term-premium wedge.

[14] Endogenizing the term premium would require modeling the pricing of long-duration risk, as in preferred-habitat models with limited risk-bearing capacity (Vayanos and Vila, 2021; Greenwood and Vayanos, 2014).

fraction $1-\rho_L$ of principal matures each period and a fraction ρ_L survives. Geometric amortization provides a tractable representation of long-term debt (Leland, 1994; Chatterjee and Eyigungor, 2012). Because all long-term claims share the same coupon and amortization rate, their remaining promised payments are proportional to face value regardless of issuance vintage. The inherited debt structure can therefore be summarized by total face value $B_{i,t}$ and principal due within the period $G_{i,t}$ that are defined as

$$B_{it} = B_{it}^S + B_{it}^L, \qquad G_{it} \equiv B_{it}^S + (1-\rho_L)B_{it}^L \tag{16}$$

The pair (B_{it}, G_{it}) is a sufficient statistic for the inherited debt structure, with feasible states satisfying[15]

$$(1-\rho_L)B_{it} \leq G_{it} \leq B_{it} \tag{17}$$

The lower bound corresponds to a firm financed entirely with long-term debt, whereas the upper bound corresponds to a firm financed entirely with short-term debt.

For firms with positive debt, define rollover exposure $s_{i,t}$ as

$$s_{it} \equiv \frac{G_{it}}{B_{it}} \tag{18}$$

Rollover exposure summarizes the concentration of contractual principal repayments in the current period. Holding total debt fixed, a higher s_{it} implies that a larger fraction of outstanding principal must be met from current cash flow, disinvestment, or new borrowing. Thus, when external finance cannot be raised costlessly, the timing of repayment obligations matters independently of the level of debt. This intertemporal liquidity motive is analogous to the broader insurance principle that liquidity constraints increase the value of smoothing concentrated payment obligations across periods (Ericson and Sydnor, 2026). The firm state can be summarized as

$$x_{it} = (z_{it}, K_{it}, B_{it}, G_{it}) \tag{19}$$

Conditional on continuation, the firm chooses next-period capital $K_{i,t+1}$, gross new borrowing $m_{it} \geq 0$, and the long-term share of new borrowing $\ell_{it} \in [0,1]$. Because short-term debt matures fully within the period, next period's short-term debt consists entirely of new short-term borrowing, while next period's long-term debt combines surviving long-term claims with new long-term

[15] The bounds follow directly from the inversion of (16). Subtracting the two equations gives

$$B_{it} - G_{it} = \rho_L B_{it}^L$$

so that

$$B_{it}^L = \frac{B_{it} - G_{it}}{\rho_L}, \qquad B_{it}^S = \frac{G_{it} - (1-\rho_L)B_{it}}{\rho_L}$$

Since both debt components must be nonnegative, $B_{it}^L \geq 0$ implies $G_{it} \leq B_{it}$, while $B_{it}^S \geq 0$ implies $G_{it} \geq (1-\rho_L)B_{it}$. Hence (B_{it}, G_{it}) is feasible if and only if $(1-\rho_L)B_{it} \leq G_{it} \leq B_{it}$.

borrowing:

$$B^S_{i,t+1} = (1-\ell_{it})m_{it}, \qquad B^L_{i,t+1} = \rho_L B^L_{it} + \ell_{it} m_{it}. \tag{20}$$

The benchmark model abstracts from voluntary prepayment of surviving long-term debt [16].

Debt evolves according to

$$B_{i,t+1} = B_{it} - G_{it} + m_{it}, \qquad G_{i,t+1} = (1-\rho_L)B_{i,t+1} + \rho_L(1-\ell_{it})m_{it} \tag{21}$$

Two properties of (21) summarize the roles of borrowing and maturity choice. First, $B_{i,t+1} - B_{it} = m_{it} - G_{it}$, so gross new borrowing in excess of principal coming due equals net debt growth. Second, holding gross issuance fixed, $\frac{\partial G_{i,t+1}}{\partial \ell_{it}}\Big| m_{it} = -\rho_L m_{it} < 0$. This is the sense in which long-term debt provides refinancing insurance against idiosyncratic rollover risk: for a given amount of new borrowing, allocating a larger share of new debt to long maturity reduces the principal that must be refinanced next period, which is especially valuable when refinancing is costly because of high rates, low productivity, or weak internal funds.

6.3 Firm problem

Timing. Within each period, the aggregate financing state and firm productivity are realized, and the firm produces using its installed capital. Inherited principal and coupon obligations become due. Competitive lenders quote state-contingent price schedules for newly issued short- and long-term debt, reflecting the repayment risk associated with each feasible financing choice. The firm then chooses next-period capital, gross borrowing, and the maturity composition of new debt. If no feasible combination of investment and financing allows the firm to meet its obligations, it defaults. Otherwise, it issues new debt, repays inherited obligations, undertakes investment, distributes any residual cash flow, and transitions to the next period.

Continuation value. The time-varying component of discounting is common to shareholders and competitive lenders:

$$\mathcal{M}_{t,t+1} = e^{-r_t} \tag{22}$$

where r_t is the continuously compounded one-period risk-free rate. Shareholders additionally discount continuation payoffs by the time-invariant factor β, whereas lenders discount using $\mathcal{M}_{t,t+1}$ alone. Following Salomão and Varela (2022), we interpret $\beta < 1$ as a equity-discounting wedge that captures forces such as the tax advantage of debt or ownership turnover, which shorten shareholders' effective valuation horizon relative to creditors. This makes current borrowing proceeds relatively valuable to shareholders and is the model's principal motive for positive leverage.

[16] The empirical analysis includes all newly originated loans, including those associated with refinancing existing debt. Early refinancing is a distinct margin of maturity management: Mian and Santos (2018) show that firms refinance bank loans before maturity when credit conditions are favorable, allowing them to extend effective maturity and reduce future refinancing risk.

Conditional on continuation, the firm solves

$$V^C(x;A) = \max_{(K',m,\ell)\in\mathcal{F}(x;A)} \left\{ d + \beta \mathbb{E}_t \left[\mathcal{M}_{t,t+1} V(x';A') \right] \right\} \tag{23}$$

The firm values current distributions and future shareholder payoffs, taking into account how today's investment and financing decisions determine next period's capital, repayment obligations, and exposure to default risk. For firm state x and aggregate state A, define the feasible continuation set as

$$\mathcal{F}(x;A) = \left\{ (K',m,\ell) : K' \geq \underline{K},\ m \geq 0,\ \ell \in [0,1],\ d \geq 0 \right\} \tag{24}$$

Payout and financing constraints. Using the identity $B^L_{it} = (B_{it} - G_{it})/\rho_L$, shareholder payout is

$$d_{it} = \pi(z_{it}, K_{it}) - G_{it} - c_L \frac{B_{it} - G_{it}}{\rho_L} - I_{it} - \Phi_K(I_{it}, K_{it}) - H(m_{it}) + m_{it} \left[(1-\ell_{it}) Q^S_{it} + \ell_{it} Q^L_{it} \right] \tag{25}$$

Operating cash flow finances inherited principal and coupon payments, investment, capital adjustment costs, and issuance costs. The final term is the proceeds from new short- and long-term borrowing. The prices Q^S_{it} and Q^L_{it} depend on the firm's state and financing choices, as specified in Section 6.4. Note that increasing the face value of new borrowing does not generate proportional additional proceeds; in fact higher indebtedness can raise default risk and reduce debt prices.

Gross debt issuance entails the quadratic convex cost following Jungherr and Schott (2021)

$$H(m_{it}) = \frac{\eta}{2} m^2_{it} \tag{26}$$

which captures increasing marginal fees of raising new bank credit.[17] The cost depends on total gross issuance in face value. It is therefore maturity-neutral, since conditional on new borrowing m_{it} reallocating it between short- and long-term debt does not change the cost, and it is invariant to debt prices for a given plan, so rate, expected-rate, and term-premium shocks affect it only through the level of issuance the firm chooses. The issuance cost makes short-term debt unattractive because it needs to be constantly rolled over. Long-term debt matures slowly over time and therefore allows maintaining a given stock of debt at a lower level of issuance per period[18].

[17] Berg et al. (2016) document the importance of fees in corporate bank lending, while Altinkilic and Hansen (2000) find evidence of rising marginal underwriting costs in corporate bond and equity issuance. These findings motivate non-interest financing costs and a convex issuance-cost specification; the quadratic functional form remains a modeling assumption.

[18] We consider two alternative issuance-cost specifications to assess which features of the benchmark matter for the maturity responses. First, we replace the quadratic cost in face value with a cost quadratic in issuance proceeds,

$$H^{\text{prc}}(m_{it}, \ell_{it}) = \frac{\eta_{\text{prc}}}{2} \left[(1-\ell_{it}) Q^S_{it} + \ell_{it} Q^L_{it} \right]^2 m^2_{it}.$$

Because this specification depends directly on debt prices and the maturity mix, it provides a more demanding test of whether the opposing expected-rate and term-premium interactions arise from the benchmark face-value formulation. The simulated regressions continue to reproduce both interaction signs. Second, replacing convex issuance costs with

External equity is unavailable, so shareholder payout must satisfy

$$d_{it} \geq 0 \tag{27}$$

This restriction reflects the focus on private, bank-dependent firms and keeps the model centered on debt financing and maturity choice. Firms cannot cover a financing shortfall through an equity injection; instead, they must adjust investment, borrowing, or maturity subject to the prices offered by lenders. Together, the no-equity constraint, convex issuance costs, and default-sensitive debt prices generate endogenous debt capacity without an exogenous borrowing limit. These features make the timing of repayment obligations economically consequential, and also make debt maturity relevant for firm value. Longer maturity postpones principal repayment and can reduce exposure to future refinancing conditions and costly debt issuance. At the same time, obtaining longer-term financing may reduce current issuance proceeds when its relative price rises. Firms therefore choose maturity by trading off the current cost of long-duration financing against the continuation-value benefits of postponing repayment.

6.4 Default and debt pricing

Default and recovery. Default is liquidity based. A firm defaults when no feasible continuation policy allows it to meet its current obligations:

$$D(x; A) = \mathbf{1}\left\{\mathcal{F}(x; A) = \varnothing\right\}. \tag{28}$$

Default is therefore determined by financing feasibility rather than by a strategic comparison between continuation and default values. Because current payouts and continuation equity values are constrained to be nonnegative, shareholders continue whenever a feasible financing plan exists. When no such plan exists, they exercise limited liability and walk away, receiving $V^D = 0$ and bearing no responsibility for unpaid corporate debt. Future liquidity default is nevertheless an ex-ante risk to shareholders because it extinguishes positive continuation claims. A firm may remain economically valuable conditional on survival but be unable to meet current obligations because refinancing is infeasible in the current state. Debt maturity provides insurance against this risk: by reducing the amount that must be refinanced in future states, longer-maturity debt lowers exposure to rollover risk and the probability that adverse financing conditions lead to default (Diamond, 1991; He and Xiong, 2012). The model therefore captures liquidity default with collateral recovery rather than a comprehensive model of corporate bankruptcy.

a linear cost, $H^{\text{lin}}(m_{it}) = \eta_{\text{lin}} m_{it}$, preserves the expected-rate interaction but substantially changes equilibrium borrowing: issuance rises well above its target, exposed firms borrow at shorter maturities, and the term-premium interaction changes sign. These exercises indicate that the results are not specific to whether convex issuance costs are written in face value or proceeds, but that increasing marginal external-financing costs are quantitatively important for the maturity tradeoff.

Upon default, creditors recover the firm's depreciated capital. Recovery per unit of face value is

$$\mathcal{R}(x_{i,t+1}) = \min\left\{1,\ \frac{(1-\delta_K)K_{i,t+1}}{B_{i,t+1}}\right\}, \tag{29}$$

with $\mathcal{R}(x) = 1$ when $B = 0$. Short- and long-term creditors are pari passu and receive the same recovery per unit of face value. The benchmark imposes no additional liquidation haircut or deadweight bankruptcy cost beyond the losses that arise when recoverable collateral is insufficient to repay outstanding debt.

Default therefore extinguishes shareholders' continuation claims and exposes creditors to losses whenever recoverable assets fall short of promised repayment. Anticipating these losses, creditors price default risk into both short- and long-term debt, so the possibility of future liquidity default feeds back into firms' ex-ante borrowing, investment, and maturity choices.

Competitive debt pricing. Competitive lenders price new debt to break even in expectation, taking into account the firm's future default risk and recovery value. Prices depend on the current firm and aggregate states and on the firm's investment, borrowing, and maturity choices. For compactness, let $D_{i,t+1} \equiv D(x_{i,t+1}; A_{t+1})$ and write $Q^j_{it} \equiv Q^j(x_{it}, K_{i,t+1}, m_{it}, \ell_{it}; A_t)$ for $j \in \{S, L\}$. Next period's long-term price, $Q^L_{i,t+1} \equiv Q^L\left(x_{i,t+1}, K^*_{i,t+2}, m^*_{i,t+1}, \ell^*_{i,t+1}; A_{t+1}\right)$ is evaluated at the firm's optimal continuation policy. The price of one unit of newly issued short-term debt is

$$Q^S_{i,t} = \mathbb{E}_t\left[\mathcal{M}_{t,t+1}\left(1 - D_{i,t+1} + D_{i,t+1}\mathcal{R}(x_{i,t+1})\right)\right] \tag{30}$$

Short-term debt matures entirely in the following period. Creditors receive one unit of face value if the firm repays and the recovery value if it defaults. Its price therefore reflects the risk-free discount rate and the expected credit losses. The firm's current choices affect Q^S through their effect on next-period default risk and recovery.

Long-term debt is priced recursively because a fraction of each claim survives into subsequent periods.

$$Q^L_{i,t} = e^{-\omega_t}\mathbb{E}_t\left[\mathcal{M}_{t,t+1}\Big((1 - D_{i,t+1})\left[c_L + 1 - \rho_L + \rho_L Q^L_{i,t+1}\right] + D_{i,t+1}\mathcal{R}(x_{i,t+1})\Big)\right] \tag{31}$$

Conditional on repayment, creditors receive the coupon c_L, the amortizing fraction $1 - \rho_L$ of principal, and retain the surviving fraction ρ_L of the claim, valued at $Q^L_{i,t+1}$. Upon default, they receive the recovery value. Unlike short-term debt, the price of long-term debt therefore depends not only on next-period default risk but also on the future market value of the surviving claim. Because that value depends on the firm's future investment and financing policies, current long-term debt prices incorporate how today's choices affect future firm behavior and default risk. The wedge ω_t lowers the proceeds received from long-duration financing. A higher ω_t therefore raises the current cost of obtaining maturity protection without changing the contractual repayment schedule of the

claim. This recursive pricing structure gives rise to debt-overhang and dilution effects. Investment that lowers future default risk raises the value of debt, so part of the benefit of investment accrues to creditors rather than shareholders, generating the classic debt-overhang problem of Myers (1977). Conversely, additional borrowing increases future liabilities and can raise default risk, reducing the value of surviving pari passu long-term claims. Competitive lenders anticipate both effects when pricing debt today.

6.5 Firm Optimality Conditions

Because financial markets are incomplete and debt is subject to default risk, Modigliani–Miller irrelevance does not hold. Short- and long-term debt differ both in their current market prices and in the timing of future repayment obligations. The maturity choice therefore trades off the current cost of long-duration financing against the insurance benefit of reducing future refinancing needs. A larger long-term share ℓ lowers next-period principal due by the second property of (21), but long-term credit may provide fewer current proceeds than short-term credit. This trade-off arises even though issuance costs are maturity-neutral. For compactness, define $V_X(x_{i,t+1}; A_{t+1}) \equiv \frac{\partial V(x_{i,t+1};A_{t+1})}{\partial X_{i,t+1}}, X \in K, B, G$, with the continuation value evaluated at next period's state and optimal continuation policy.

Let $\mu_{i,t} \geq 0$ denote the multiplier on the no-equity constraint (27). For an issuing firm with an interior maturity choice, the first-order condition with respect to the long-term share ℓ is

$$Q^S_{it} - Q^L_{it} - (1-\ell_{it})\frac{\partial Q^S_{it}}{\partial \ell_{it}} - \ell_{it}\frac{\partial Q^L_{it}}{\partial \ell_{it}} = -\frac{\beta\rho_L}{1+\mu_{it}}\mathbb{E}_t\left[\mathcal{M}_{t,t+1}V_G(x_{i,t+1}; A_{t+1})\right] \tag{32}$$

Equation (32) equates the marginal cost of extending maturity today with the expected marginal benefit of reducing future principal repayments. The direct current cost is the price difference $Q^S - Q^L$: when long-term debt sells at a lower price than short-term debt, shifting borrowing toward longer maturity reduces current financing proceeds. The remaining terms on the left-hand side capture endogenous price feedback. By changing future repayment obligations, maturity affects default risk and the value of debt, which competitive lenders internalize in current prices. Because long-term claims survive into future periods, these price effects also incorporate the debt-overhang and dilution costs associated with persistent long-term liabilities: future investment and borrowing decisions affect the value of surviving claims, and forward-looking lenders price these effects at issuance [19]. The right-hand side is the intertemporal insurance benefit. Increasing ℓ leaves total borrowing unchanged but reduces next-period principal due by ρ_L. Since V_G is the marginal value of an additional unit of principal coming due, with $V_G < 0$, the term $-\frac{\beta\rho_L}{1+\mu}\mathbb{E}_t\left[\mathcal{M}_{t,t+1}V_G\right]$ measures the expected gain from lowering future refinancing needs. Longer maturity is therefore

[19] Appendix E.2 shows that long-term debt prices are more sensitive to gross issuance than short-term debt prices, particularly among firms with greater rollover exposure. These issuance-price sensitivities are distinct from the maturity-share derivatives in equation (32), but illustrate the endogenous financing costs associated with obtaining maturity protection.

more valuable when an additional unit of principal due has a large adverse effect on continuation value, as for firms with high refinancing exposure or limited capacity to absorb adverse idiosyncratic shocks. Maturity would become irrelevant if short- and long-term funding had the same price, debt prices were unaffected by maturity choices, and shifting principal across dates had no effect on continuation value.

The remaining first-order conditions govern investment and the amount of new borrowing. For an interior capital choice,

$$1 + \Phi_I(I_{it}, K_{it}) = m_{it}\left[(1-\ell_{it})\frac{\partial Q^S_{it}}{\partial K_{i,t+1}} + \ell_{it}\frac{\partial Q^L_{it}}{\partial K_{i,t+1}}\right] + \frac{\beta}{1+\mu_{it}}\mathbb{E}_t\left[\mathcal{M}_{t,t+1}V_K(x_{i,t+1}; A_{t+1})\right] \quad (33)$$

The left-hand side is the marginal resource cost of investment. The right-hand side captures two benefits. Additional capital raises continuation value directly through V_K and can improve current borrowing terms by increasing recovery values and reducing default risk. As in standard debt-overhang models, shareholders internalize the latter benefit only through the prices of newly issued debt; increases in the value of previously issued long-term claims accrue to existing creditors.

For an interior gross-borrowing choice,

$$\begin{aligned}&(1-\ell_{it})Q^S_{it} + \ell_{it}Q^L_{it} - H'(m_{it}) + m_{it}\left[(1-\ell_{it})\frac{\partial Q^S_{it}}{\partial m_{it}} + \ell_{it}\frac{\partial Q^L_{it}}{\partial m_{it}}\right] \\ &= -\frac{\beta}{1+\mu_{it}}\mathbb{E}_t\left[\mathcal{M}_{t,t+1}\left(V_B(x_{i,t+1}; A_{t+1}) + (1-\rho_L\ell_{it})V_G(x_{i,t+1}; A_{t+1})\right)\right]\end{aligned} \quad (34)$$

The left-hand side is the marginal amount of funds raised today, net of issuance costs and the endogenous effect of additional borrowing on debt prices. The continuation cost on the right-hand side has two components: an additional unit of borrowing raises next-period total debt by one, generating the marginal cost V_B, and raises next-period principal due by $1-\rho_L\ell$, generating the additional refinancing cost $(1-\rho_L\ell)V_G$. Maturity therefore determines how strongly additional borrowing translates into future refinancing exposure. Appendix B provides the full derivations.

6.6 Recursive equilibrium and stationarity

Given the exogenous productivity process in (9) and the processes for aggregate financing conditions in (13)–(15), these objects satisfy the following equilibrium conditions. Firms solve (23) subject to the feasible set $\mathcal{F}(x; A)$ in (24); default follows (28), with $V(x; A) = 0$ upon default and $V(x; A) = V^C(x; A)$ otherwise; debt-price schedules satisfy (30), (31); and expectations are consistent with the exogenous state transitions and firms' optimal policies. Let $\mathcal{X}$ denote the firm-state space, with $x = (z, K, B, G) \in \mathcal{X}$, and let $\mathcal{A}$ denote the state space for aggregate financing conditions. Let μ_t denote the cross-sectional distribution of firm states conditional on the realized history of aggregate financing conditions. For any measurable set $E \subseteq \mathcal{X}$, let $\Pi(E \mid x, A)$ denote the probability that a continuing firm currently in state x and facing aggregate state A has next-period state in E.

Let $\nu(E)$ denote the probability that a new entrant's initial state lies in E. Entrants begin with capital K_0, zero debt, and productivity drawn from its stationary distribution. The cross-sectional distribution then evolves according to:

$$\mu_{t+1}(E) = \int_{\mathcal{X}} \big[\big(1 - D(x; A_t)\big)\Pi(E \mid x, A_t) + D(x; A_t)\nu(E)\big]\, \mu_t(dx). \tag{35}$$

Thus, $\mu_{t+1}(E)$ is the mass of firms whose next-period state lies in E: continuing firms evolve according to their optimal policies and the productivity process, while defaulting firms are replaced one for one by entrants drawn from ν. Let $P(E' \mid A)$ denote the probability that next-period aggregate financing conditions lie in the measurable set $E' \subseteq \mathcal{A}$, conditional on current aggregate state A. Conditional on (x, A), idiosyncratic productivity innovations and entrants' initial states are independent of the next aggregate innovation. In addition, next-period inherited balance sheets are determined by current firm choices. The transition in firm states therefore factors from the transition in aggregate financing conditions. An invariant joint distribution Γ^* over firm and aggregate states, if it exists, satisfies

$$\Gamma^*(E \times E') = \int_{\mathcal{X}\times\mathcal{A}} \big[\big(1 - D(x; A)\big)\Pi(E \mid x, A) + D(x; A)\nu(E)\big]\, P(E' \mid A)\, \Gamma^*(dx, dA). \tag{36}$$

Unconditional model moments are computed under Γ^*.

6.7 Calibration

We calibrate the model annually to the matched AnaCredit–Orbis sample of Section 2; Table 10 summarizes the parameterization. Three parameters—the fixed operating cost F, the issuance cost η, and the capital adjustment cost ψ_K—are estimated jointly by simulated method of moments; the remaining parameters are measured outside the model. No maturity, rollover-exposure, or leverage moment enters the SMM objective, so those outcomes and the heterogeneous responses to financing shocks are untargeted tests of the mechanism.

Parameters determined outside the model. Profit curvature is $\alpha = 0.60$ and depreciation rate $\delta_K = 0.10$; productivity scale is normalized by $\bar{z} = [e^{\bar{r}} - (1 - \delta_K)]/\alpha$, so the frictionless capital stock of a median-productivity firm is one. The persistence and dispersion of idiosyncratic productivity are estimated from the Orbis panel $\rho_z = 0.85$, $\sigma_z = 0.15$ and the estimation procedure is described in Appendix D. Entrant capital is set to $K_0 = 0.30$ as in the Orbis panel data: excluding firms first observed in 2019 to avoid left censoring, median fixed assets among newly observed firms are about 30% of the median among firms observed for at least three prior years. Entrants start with no debt and draw productivity from its stationary distribution. Under geometric amortization, expected maturity is $1/(1-\rho_L)$, so $\rho_L = 1 - 1/5.19 = 0.807$, where 5.19 years is the credit-weighted mean original maturity of newly issued term loans in AnaCredit. The coupon is normalized to par at the mean risk-free rate, $c_L = e^{\bar{r}} - 1 = 0.00826$—a normalization, not a measurement: it places

Table 10: Calibration

Parameter	Description	Value	Source or target
Panel A. Externally assigned and normalizations			
α	Profit curvature	0.60	Cooper and Haltiwanger (2006)
δ_K	Depreciation rate	0.10	Literature
$\bar{z}$	Productivity scale	$[e^{\bar{r}} - (1 - \delta_K)]/\alpha$	Normalization, $K^{ss} = 1$
Panel B. Directly measured or estimated outside the model			
K_0	Entrant capital	0.30	Orbis: median fixed assets new firms
c_L	Long-debt coupon	0.0083	Par at the mean risk-free rate, $e^{\bar{r}} - 1$
κ	Equity-horizon wedge	0.05	Orbis: Annual exit rate
ρ_z	Productivity persistence	0.85	Firm-level panel estimates
σ_z	Productivity innovation s.d.	0.150	Firm-level panel estimates
$\bar{r}$	Mean short rate	0.0082	One-year OIS rate
ρ_r	Short-rate persistence	0.950	One-year OIS rate
σ_r	Short-rate innovation s.d.	0.0058	One-year OIS rate
ρ_f	Expected-rate news persistence	0.363	JSZ expectations component, normalized
σ_f	Expected-rate news innovation s.d.	0.0088	JSZ expectations component, normalized
$\bar{\omega}$	Term-premium wedge, level	0.0315	JSZ term premium, 2022–2024
ρ_ω	Term-premium wedge persistence	0.908	JSZ Term premium via Eq. (76)
σ_ω	Term-premium innovation s.d.	0.0160	JSZ Term premium via Eq. (76)
Panel C. Estimated by simulated method of moments			
F	Fixed operating cost	0.024	Annual SME default rate, 0.020
η	Debt issuance cost	0.32	Gross issuance rate, 0.267
ψ_K	Capital adjustment cost	0.02	Investment rate, 0.297

Notes: Aggregate processes are annual. The short-rate process is estimated on the one-year OIS rate alone; f_t is the normalized expected-rate-path component of Joslin et al. (2011), orthogonalized to the current rate; ω_t is recovered from the observed term premium through the measurement equation (76). Panel C values are selected on the star identification design (Section 7.1); the moments reported in Section 7 are computed at those values.

no credit spread in the coupon, so all compensation for default risk and duration enters through the issuance price. Away from the mean rate the claim trades away from par, and that valuation channel is exactly what maturity choice trades off against refinancing insurance. The equity-horizon wedge $\beta = 1-\kappa$ captures forces that make shareholders value current financing proceeds more than distant continuation payoffs. We set $\kappa = 0.05$, using the roughly 5% annual firm-exit rate in Orbis as a calibration anchor rather than a literal exit probability. The wedge provides a parsimonious motive for positive debt demand while keeping the model focused on maturity and refinancing decisions. The short-rate process in (13) is estimated using the one-year OIS rate. We discipline the expected-rate state f_t using the Joslin et al. (2011) expectations component orthogonalized to the current short rate. Normalizing its loading in the short-rate transition to one implies $\sigma_f = 0.0088$, so a one-standard-deviation news innovation raises the expected one-year-ahead short rate by about 0.9 percentage points. The wedge ω_t captures the price of duration on newly issued corporate loans. We set $\bar{\omega} = 0.0315$, corresponding to an annualized term premium of about 0.6 percent on

the model's long-term claim, equal to the 2004–2025 average five-year term premium in the Joslin et al. (2011) decomposition. We use this longer-run mean as the stationary reference point because the 2018–2024 AnaCredit window features an unusually low average term premium of -0.5 percent. Historical exercises instead feed the measured 2018–2024 changes in the wedge into the model. The same decomposition implies $\rho_\omega = 0.908$ and a stationary standard deviation of 0.038, corresponding to $\sigma_\omega = 0.016$. Equation (76) maps observed term-premium movements into the log-price wedge on the model's long-term claim.

Internally estimated parameters. The remaining parameters $\Theta = (F, \eta, \psi_K)$ are estimated jointly by SMM. With $m(x)$ the empirical moments and $\widehat{m}(\widetilde{x} \mid \Theta)$ their simulated counterparts, the estimator solves

$$\widehat{\Theta}_{\text{SMM}} = \arg\min_{\Theta} g(\Theta)' g(\Theta), \qquad g_j(\Theta) = \frac{\widehat{m}_j(\widetilde{x} \mid \Theta) - m_j(x)}{m_j(x)} \tag{37}$$

so the objective weights proportional deviations equally. Each target maps primarily to one parameter. The fixed operating cost F governs proximity to the liquidity-default boundary and is disciplined by the 2.0 percent annual default rate for euro-area SME exposures in EBA 2019 reporting. The issuance cost η is quadratic in gross issuance, $H(m_{it}) = \frac{\eta}{2} m_{it}^2$, it governs borrowing intensity. Its target is the aggregate gross issuance rate (gross borrowing during the year relative to the outstanding debt stock) from AnaCredit term-loan originations,

$$\frac{\sum_i m_{it}}{\sum_i B_{it}} = 0.267$$

A quadratic capital adjustment cost governs adjustment intensity rather than the arrival of expansion opportunities, so ψ_K is disciplined by investment conditional on a financed expansion: among firm-years with a net increase in bank debt,

$$\mathbb{E}\left[\frac{I_{it}}{K_{i,t-1}} \,\middle|\, B_{it} > B_{i,t-1} \right] = 0.297$$

This moment is monotone in ψ_K and leaves the other two targets nearly unchanged, so the moment–parameter mapping is close to diagonal.[20] The criterion is minimized in two stages under common random numbers. A scrambled-Sobol scan of the parameter box screens for competing basins; its leading candidates are re-evaluated at full accuracy, since the scan itself uses small panels and coarse tolerances, and none improves on the neighborhood of the initial calibration. A Newton step from one-at-a-time probe designs then locates the minimum, verified by re-solving at the selected vector; the probe table doubles as the identification exhibit for the moment–parameter Jacobian. At $\widehat{\Theta}$ the criterion is 0.0013. Moments are averaged across 50 simulated histories of 10,000 firms

[20] The unconditional mean absolute capital growth rate, the natural alternative, bundles this intensive margin with an extensive margin—the frequency of net debt expansions—that a convex issuance cost cannot move; asking ψ_K to fit it drives the parameter to a boundary. Both statistics are left outside the estimation.

after 300-period burn-in.

7 Quantitative Results

This section evaluates the model in three steps: untargeted cross-sectional moments, policy functions linking rollover exposure to the two components of long rates, and the paper's empirical regressions re-estimated on simulated panels.

7.1 Parameter estimates and targeted moments

Panel A of Table 11 reports the targeted moments. The estimated internal parameters are $F = 0.024$, $\eta = 0.32$, and $\psi_K = 0.02$. All three targets are matched closely: the default rate is 0.019 against 0.020, the gross issuance rate 0.272 against 0.267, and investment conditional on net debt expansion 0.281 against 0.297, with the small residual concentrated in the last moment.

7.2 Untargeted model fit

Panels B–D of Table 11 report untargeted moments. A measurement note is useful when comparing rollover exposure in the model and the data. The model state is

$$s_{it} = \frac{G_{it}}{B_{it}} = \frac{B^S_{it} + (1-\rho_L)B^L_{it}}{B_{it}},$$

the fraction of outstanding principal contractually due within the period. Because long-term claims amortize geometrically, s_{it} includes the scheduled repayment $(1-\rho_L)B^L_{it}$ on long-term debt and is bounded below by $1-\rho_L = 0.193$ for any indebted firm. The empirical rollover measure likewise captures debt whose principal comes due within the following twelve months. We therefore compare it directly with s_{it} in Table 11.[21] Maturity composition is reproduced closely. The mean long-term share of new issuance is 0.69 against 0.70 in the data, its standard deviation 0.37 (0.43), and the frequency of all-short issuers 0.17 (0.22). The model thus matches the level and the spread of maturity choice without any maturity moment in the SMM objective. Regarding inherited rollover exposure, the mean is 0.26 against 0.27, so the model reproduces how much principal comes due on average, but its dispersion is 0.10 against 0.32. With most issuance long-term and long-term debt amortizing smoothly, the model meets the average repayment burden through amortization rather than through short-term contracts, and it lacks the data's thick upper tail of predominantly short-term borrowers. This is a mechanical consequence of geometric amortization—

[21] The mapping is necessarily imperfect because AnaCredit reports contractual residual maturity rather than the full amortization schedule. The empirical measure includes the outstanding balance of long-term loans that have entered their final twelve months, but does not capture scheduled amortization on loans whose final maturity lies further in the future. Geometrically amortizing long-term claims in the model, by contrast, generate scheduled principal repayment every period but have no deterministic terminal maturity date. Thus, the two measures are not mechanically identical, but both are designed to capture the principal repayment burden falling due within the year.

Table 11: Model fit: targeted and untargeted moments

Moment	Data	Model
Panel A: Targeted		
Default rate	0.020	0.019
Gross issuance / debt	0.267	0.272
Investment rate	0.297	0.281
Panel B: Untargeted		
Mean long-term issuance share $\mathbb{E}[\ell]$	0.70	0.69
Dispersion long-term issuance share $\mathrm{sd}(\ell)$	0.43	0.37
All-short issuers $\Pr(\ell = 0)$	0.22	0.21
Mean rollover, $\mathbb{E}[s]$	0.27	0.26
Dispersion rollover $\mathrm{sd}(s)$	0.32	0.10
Leverage $\mathbb{E}[B/K \mid B > 0]$	0.85	0.90
Size–rollover $\mathrm{corr}(\log K, s)$	-0.15	-0.19
Debt expansion $\Pr(\Delta B > 0)$	0.47	0.23

Notes: Data moments from the matched AnaCredit–Orbis panel; construction in Section 2. Two conventions for refinancing needs are used and must not be crossed. Rollover exposure $s = G/B$ is the share of principal due within the year and includes the amortizing fraction of long-term debt, so $s \geq 1 - \rho_L$ for any indebted firm; its data counterpart is the raw maturity wall in the regression estimation sample ($N = 2.26$ million firm-years). The debt moments are measured among firms with lagged bank debt above its 5th percentile; conditional growth rates are winsorized at the 1st and 99th percentiles; the same sample restrictions are applied to the simulated panel. The capital-adjustment moments are winsorized fixed-asset growth rates. The targeted investment moment conditions on firm-years with a net increase in bank debt, with investment winsorized within the expander sample. Panel A moments enter the calibration objective; Panels B–D do not. The model column reports moments of the stationary distribution at the estimated parameters $\widehat{\Theta} = (0.024, 0.32, 0.02)$, averaged across 50 independently simulated histories of 10,000 firms after a 300-period burn-in; the criterion at $\widehat{\Theta}$ is 0.0013.

there are no discrete repayment dates to concentrate—and cannot be resolved by the calibration. The negative size–rollover exposure relation is matched closely ($\mathrm{corr}(\log K, s) = -0.19$ against -0.15). In addition, the frequency of net debt expansion is 0.23 against 0.47: the extensive margin of borrowing is understated, since a convex issuance cost prices intensity rather than participation. Leverage conditional on borrowing is close—0.90 against 0.85—so the model generates the observed intensity of debt usage without a tax advantage or a leverage target. Because the model understates both the dispersion of rollover exposure and the frequency of net debt expansion, its quantitative statements about heterogeneous refinancing needs and financing responses are conditional on the distribution it represents.

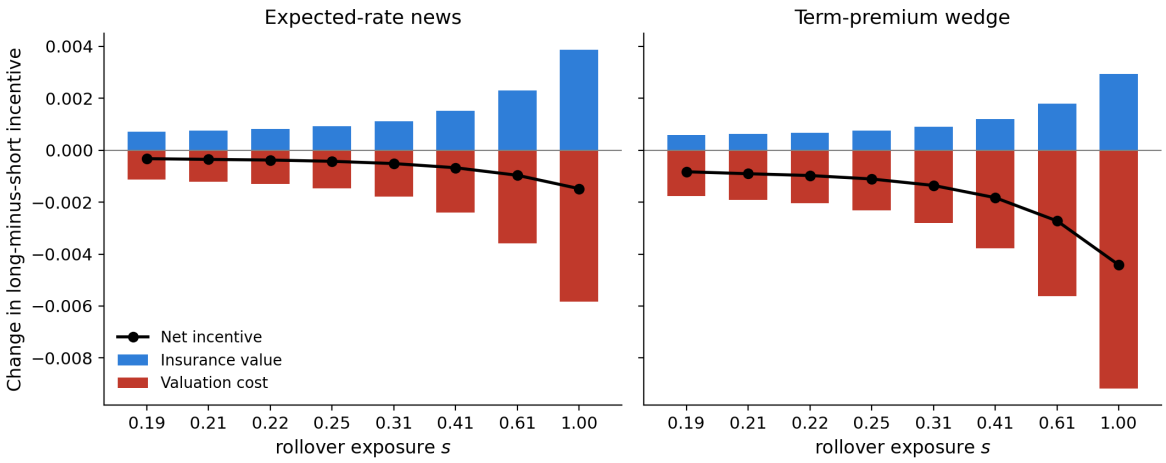


Figure 5: Maturity-insurance mechanism. The figure decomposes the change in the marginal incentive to issue long rather than short debt following one-standard-deviation shocks to expected-rate news (left) and to the term-premium wedge (right), relative to the mean aggregate state, by rollover exposure s. Bars report the change in the continuation-value benefit of postponing refinancing (insurance value) and in the valuation cost of long-duration funding; black dots report the change in the net incentive, which also includes the quantitatively smaller credit-price feedback. Higher expected-rate news and a higher term-premium wedge both reduce the net incentive to issue long debt. Expected-rate news, however, also increases the continuation value of postponing refinancing, substantially cushioning its valuation effect. This insurance offset grows with rollover exposure in proportion to the valuation cost, whereas the term-premium wedge generates a considerably larger deterioration in the net incentive at high exposure. The decomposition is evaluated at the rollover margin ($K' = K$, $B' = B$) for the median idiosyncratic state.

7.3 The Value and Price of Maturity Insurance

Equation (32) characterizes maturity choice as a trade-off between the insurance value of reducing future refinancing needs and the current cost of long-term financing. Both margins depend on firms' inherited rollover exposure. Firms with greater repayment obligations face higher refinancing and default risk, increasing the value of postponing repayment. At the same time, long-term debt prices are more sensitive to issuance among these firms, making maturity protection more costly to obtain. Appendix E.1 documents the relationship between rollover exposure and default risk, while Appendix E.2 examines the endogenous sensitivity of debt prices to issuance.

Expected-rate news and term-premium shocks affect this trade-off differently. Higher expected future short rates reduce the current price of long-term debt but also increase the value of avoiding refinancing under adverse future conditions. This insurance benefit is particularly important for firms with greater inherited rollover exposure. A higher term-premium wedge also reduces long-term debt prices, but without a comparable increase in the continuation value of maturity protection.

Figure 5 quantifies these forces by decomposing changes in the marginal incentive to issue long rather than short debt following one-standard-deviation shocks, evaluated at a common refinancing plan across rollover-exposure levels. Both shocks reduce the net incentive to issue long-term debt, but the insurance benefit offsets a substantially larger share of the current valuation cost under

expected-rate news.

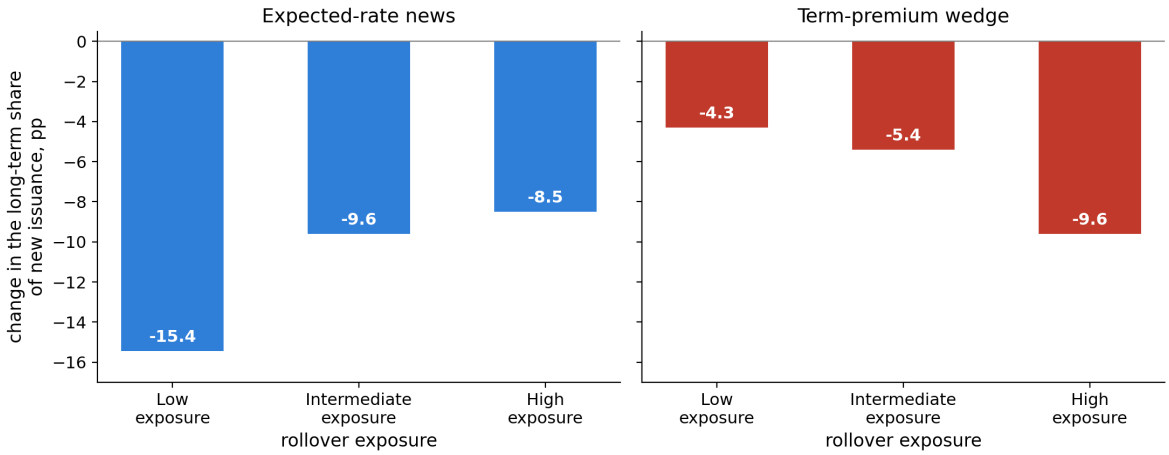


Figure 6: Conditional Maturity Responses by Rollover Exposure. Change in the optimal long-term share of new issuance, in percentage points, when expected-rate news (left) or the term-premium wedge (right) rises by one grid node from its mean, approximately one innovation standard deviation, with the other aggregate states at their means. Each bar averages responses at the balance-sheet positions observed in simulated histories, assigning equal weight to each exposure node within the group. Low exposure is the amortization floor, $s = 1 - \rho_L = 0.19$, containing 49 percent of issuing firms; intermediate exposure comprises nodes between 0.21 and 0.25, containing 25 percent; high exposure comprises nodes between 0.31 and 1, containing 26 percent. Means across 50 simulated histories.

Specifically, the increase in continuation value offsets approximately two-thirds of the valuation cost following expected-rate news, compared with less than one-third following a term-premium shock. Thus, expected-rate news generates a smaller deterioration in the net incentive to issue long-term debt despite raising its current cost.

These differences translate into opposite relative maturity responses across firms with different inherited rollover exposure. Although firms can issue both short- and long-term debt simultaneously, maturity choices exhibit substantial concentration at the two extremes: in the calibrated economy, 44% of issuers choose an interior maturity share, while 21% issue exclusively short-term debt and 35% exclusively long-term debt. Changes in borrowing incentives can therefore induce both adjustments in interior maturity shares and switching between maturity choices.

Figure 6 examines these responses using firms' optimal maturity policies. Holding each firm's initial state fixed, we compare the long-term issuance share at the mean aggregate state with its value following an increase in expected-rate news or the term-premium wedge. Following expected-rate news, the long-term issuance share declines by 15.4 percentage points among firms at the amortization floor, compared with 8.5 points among high-exposure firms. Thus, firms with greater refinancing needs shorten maturity by 6.9 percentage points less. A higher term-premium wedge instead reduces the long-term share by 4.3 percentage points at the amortization floor and 9.6 points among high-exposure firms, implying 5.3 percentage points greater maturity shortening.

These results establish that the two shocks generate opposite relative maturity responses even

though both reduce long-term borrowing in absolute terms. Expected-rate news raises the value of refinancing insurance, limiting maturity shortening among firms with greater repayment obligations. By contrast, a higher term-premium wedge raises the cost of maturity protection without a comparable increase in its insurance value. Appendix E.1 provides additional switching diagnostics holding firm composition fixed. The next section evaluates whether these mechanisms reproduce the empirical interactions in simulated equilibrium histories.

7.4 Model Counterparts to the Empirical Regressions

We evaluate the model's ability to reproduce the empirical maturity responses by estimating the same firm-year regression specification on simulated histories. Appendix Table 31 documents the mapping between empirical variables and their model counterparts. We estimate

$$\log(\text{Maturity}_{ft}) = \alpha_f + \alpha_t + \beta_s\, s_{f,t-1} + \beta_E \left(\text{ESR}^{5Y}_{t-1} \times s_{f,t-1}\right) + \beta_{TP} \left(\text{TP}^{5Y}_{t-1} \times s_{f,t-1}\right) + \gamma' X_{f,t-1} + \varepsilon_{ft} \tag{38}$$

where $s_{f,t-1} = G_{f,t-1}/B_{f,t-1}$ denotes inherited rollover exposure and $X_{f,t-1}$ includes lagged firm size and leverage. As in the data, the specification includes firm and time fixed effects, with the latter absorbing aggregate financing conditions. Since each simulation contains one realized aggregate history, we average the estimated coefficients across 50 independent histories and report Monte Carlo standard errors.

Table 12 compares the coefficients in the data and model, with interactions standardized by the respective standard deviations of the aggregate factors. The model reproduces the positive relationship between rollover exposure and maturity, although the estimated exposure coefficients are smaller than in the data. More importantly, it reproduces the opposite maturity responses to expected-rate and term-premium movements across rollover exposure. For expected-rate news, the log-maturity interaction is 0.424, compared with 0.430 in the data, while the corresponding long-term-share interactions are 0.245 and 0.201. For the term premium, the log-maturity interaction is -0.191, compared with -0.375 in the data, and the long-term-share interaction is -0.127, compared with -0.175. Thus, the model closely matches the expected-rate interactions and reproduces the signs and a substantial fraction of the term-premium effects.

These coefficients capture relative rather than aggregate maturity responses. Time fixed effects absorb common changes in financing conditions, so a positive interaction with expected-rate news does not imply that firms increase long-term borrowing on average. Consistent with the optimal policy responses in Section 7.3, firms with greater rollover exposure shorten maturity relatively less following expected-rate news, but relatively more when the term premium increases. Unlike the conditional policy-function exercises, the simulated regressions incorporate the evolution of firm characteristics and aggregate financing conditions over equilibrium histories.

Importantly, these differential maturity responses are not targeted in the calibration, which disciplines the model using default, issuance, and investment moments. The model's ability to

Table 12: Model Counterparts to the Empirical Estimates

	log(Maturity)		Long-Term Debt Share	
	Data	Model	Data	Model
Rollover Exposure$_{f,t-1}$	1.171	0.366	0.526	0.309
	(0.014)	(0.102)	(0.006)	(0.067)
Expected Rate$^{5Y}_{t-1}\times$ Rollover Exposure$_{f,t-1}$	0.430	0.424	0.201	0.245
	(0.009)	(0.054)	(0.004)	(0.037)
TP$^{5Y}_{t-1}\times$ Rollover Exposure$_{f,t-1}$	-0.375	-0.191	-0.175	-0.127
	(0.009)	(0.072)	(0.004)	(0.048)

Notes: Data: firm-year specification of Table 26, with standard errors in parentheses. Model: the same regression specification, equation (38), estimated on simulated firm-year panels at $\widehat{\Theta}$. Model coefficients are averaged across 50 independent histories, with Monte Carlo standard errors in parentheses. Entries are per unit of rollover exposure. Interaction coefficients are expressed per one-standard-deviation change in the lagged aggregate factor, standardized separately in the data and the model.

reproduce the opposite interactions with expected future rates and term premia therefore provides an untargeted quantitative validation of the maturity-insurance mechanism.

8 Counterfactuals

We use the model to quantify how endogenous debt maturity shapes the transmission of monetary shocks. Our main counterfactual removes the direct response of maturity choice to aggregate financing conditions while allowing maturity to adjust as firms' balance sheets evolve. We then examine how the value of this flexibility varies across firms and distinguish the insurance provided by contractual maturity from that provided by coupon fixation. Finally, we use the model to study how the value of maturity flexibility changes when higher expected refinancing costs coincide with a higher price of long-duration finance, and how the same expected-rate shock transmits across different term-premium environments. To quantify these effects, we compute generalized impulse response functions (GIRFs), following Koop et al. (1996). For each restricted economy, we re-solve firms' policies and lenders' price schedules. We simulate 1,000 economies of 10,000 firms, initialized from the benchmark stationary cross-section, and compare outcomes under a one-standard-deviation financing shock with those under a no-shock path. After the initial innovation, aggregate states follow (13)–(15) without further aggregate shocks. Under expected-rate news, the short rate is unchanged on impact and rises from the following year. Idiosyncratic productivity draws are held fixed across scenarios. For each economy, the GIRF is the difference between the shock and no-shock paths. We report averages across simulations and Monte Carlo standard errors. Because aggregate financing conditions are exogenous, these comparisons capture firm responses conditional on the specified financing shock.

8.1 Endogenous Maturity Adjustment Affects Monetary Transmission

We quantify the contribution of endogenous maturity adjustment to monetary transmission by restricting the long-term issuance share (ℓ) to its benchmark policy evaluated at the no-shock aggregate state:

$$\ell^{CF}(x_{it}, A_t) = \ell^{B}(x_{it}, \bar{A}) \tag{39}$$

The restriction removes the direct response of maturity choice to aggregate financing conditions without fixing maturity across firms or over time; firms can still change maturity as their balance sheets evolve. We re-solve investment, borrowing, and lenders' debt-price schedules under the restricted rule. Both economies begin from the benchmark stationary distribution, so firms enter each shock with identical inherited financial positions. We measure the contribution of maturity adjustment as the difference between the benchmark and counterfactual impulse responses, reported in Table 13.[22] The main findings are that maturity adjustment affects both real activity and firm survival and that the effects of monetary transmission are lower.

First, maturity flexibility substantially attenuates the investment contraction. Within three years, cumulative investment relative to capital is 4.4, 2.4, and 1.6 percentage points higher in the benchmark following current-rate, expected-rate news, and term-premium shocks, respectively. These effects are economically large relative to the corresponding benchmark contractions of 4.5, 8.4, and 1.1 percentage points. Their timing differs across shocks. Following a current-rate shock, the investment benefit emerges immediately. Following expected-rate news, investment initially falls slightly more in the benchmark, but the difference reverses as the anticipated increase in refinancing costs materializes. Following a term-premium shock the benefit is small and fades by year five: a higher price of duration by itself does not raise future refinancing costs, so there is little for maturity adjustment to insure against.

Second, the investment benefit does not require an aggregate extension of debt maturity. In the benchmark economy, firms predominantly shorten new borrowing following contractionary shocks. On impact, the long-term share of new issuance falls by 3.4, 8.8, and 4.8 percentage points more than in the restricted economy following current-rate, expected-rate news, and term-premium shocks, respectively. This adjustment allows firms to substitute away from costly long-duration funding while preserving financing capacity. Within three years, cumulative gross issuance is 5.1 percentage points higher in the benchmark following a current-rate shock and 2.2 points higher following a term-premium shock. The average credit-spread response over the first three years is also lower in the benchmark across all three shocks. Following expected-rate news, the financing benefit emerges later: gross issuance initially declines more in the benchmark, but the difference reverses as the anticipated tightening materializes.

Third, maturity flexibility provides substantial protection against default when the shock raises expected future refinancing costs. Following expected-rate news, cumulative default is 1.4 percentage

[22] This counterfactual isolates the contribution of maturity adjustment conditional on inherited benchmark balance sheets; it does not compare economies that have permanently adapted to different maturity regimes.

Table 13: Value of Maturity Flexibility

Horizon (years)	1	3	5
Investment (cumulative, pp)			
Current rate	+2.0 (0.5)	+4.4 (0.3)	+4.7 (0.4)
Expected-rate news	−0.9 (0.3)	+2.4 (0.3)	+1.9 (0.4)
Term-premium wedge	+1.0 (0.3)	+1.6 (0.2)	+0.6 (0.3)
Debt issuance (cumulative, pp)			
Current rate	+0.8 (0.5)	+5.1 (0.6)	+8.1 (0.8)
Expected-rate news	−2.5 (0.4)	+0.5 (0.5)	+3.1 (0.6)
Term-premium wedge	−0.6 (0.3)	+2.2 (0.4)	+3.3 (0.6)
Long-term issuance share (response at horizon, pp)			
Current rate	−3.4 (0.3)	−1.1 (0.4)	−1.2 (0.3)
Expected-rate news	−8.8 (0.6)	−8.5 (0.3)	−7.4 (0.3)
Term-premium wedge	−4.8 (0.3)	−2.0 (0.4)	+1.9 (0.3)
Credit spread (average through horizon, pp)			
Current rate	−4.5 (0.6)	−0.4 (0.2)	+0.9 (0.1)
Expected-rate news	−0.5 (0.4)	−3.4 (0.3)	−2.2 (0.3)
Term-premium wedge	−1.7 (0.2)	−0.7 (0.2)	−0.5 (0.1)
Default (cumulative, pp)			
Current rate	—	−0.5 (0.2)	+0.9 (0.3)
Expected-rate news	—	−1.4 (0.2)	−2.7 (0.3)
Term-premium wedge	—	−0.2 (0.2)	+0.7 (0.3)

Notes: Entries are flexible-minus-restricted differences in shock responses to one-standard-deviation innovations in the current rate, expected-rate news, or the term-premium wedge. Positive values indicate a larger response in the flexible economy. Investment, debt issuance, and default are cumulative through the reported horizon; the long-term issuance share is the response at that horizon; credit spreads are averaged through the horizon. Horizon 1 is the shock year. Default is omitted at horizon 1 because it does not respond on impact under the model's timing. The credit spread is the issuance-weighted yield spread of newly issued debt over its default-free counterpart. Because the restricted maturity policy does not respond directly to aggregate financing conditions, the horizon-1 entries for the long-term issuance share coincide with the flexible economy's own responses. After the initial innovation, aggregate states follow their estimated laws of motion without further aggregate shocks. Results average 1,000 simulated economies of 10,000 firms, initialized from the flexible economy's stationary cross-section. Idiosyncratic productivity draws are held fixed across scenarios. Monte Carlo standard errors of the mean are in parentheses.

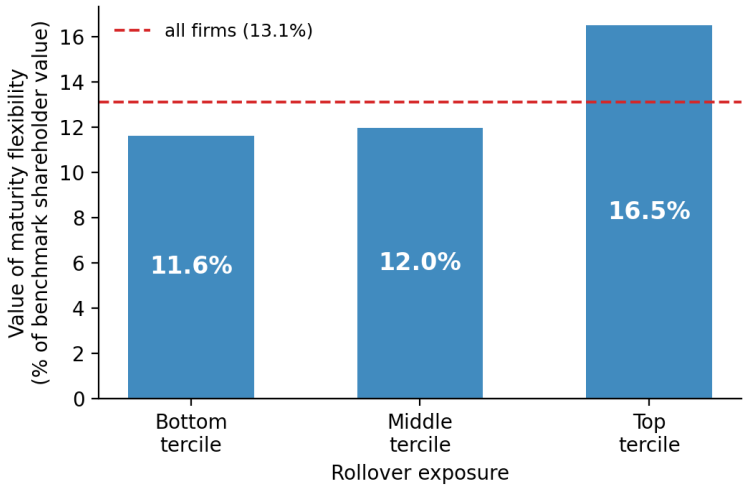


Figure 7: Value of Maturity Flexibility by Rollover Exposure. The figure reports the shareholder value preserved by maturity flexibility: benchmark minus counterfactual value, as a percentage of benchmark shareholder value, where the counterfactual permanently restricts maturity policy to its aggregate-state-invariant rule. Bars are terciles of rollover exposure among indebted firms. Values are evaluated at identical initial firm states drawn from the benchmark stationary distribution. The dashed line reports the average across indebted firms. The calculation is an ex ante value comparison over all future states, not the effect of a single financing shock.

points lower in the benchmark after three years and 2.7 points lower after five years. Thus, the ability to adjust maturity before higher expected short rates fully materialize reduces subsequent financial distress as firms encounter more costly refinancing conditions. The default effects are smaller and less persistent for the other shocks: following a current-rate shock, cumulative default is 0.5 percentage points lower after three years and 0.9 points higher by year five, and following a term-premium shock it is 0.7 points higher after five years. Following a current-rate shock, the reversal reflects the restricted economy's sharper initial default wave and deleveraging, followed by lower subsequent default rates, rather than rising defaults in the flexible economy. By contrast, expected-rate news allows firms to adjust their financing decisions ahead of higher refinancing costs, providing more persistent protection against financial distress.

Figure 7 reports the ex ante shareholder value of retaining maturity flexibility. Unlike the impulse responses, this calculation integrates over future financing shocks and firm dynamics rather than evaluating a single tightening episode. The shareholder value preserved averages 13.5% among indebted firms and increases from 11.8% in the bottom tercile of inherited rollover exposure to 16.9% in the top tercile. Firms facing greater refinancing needs therefore place a higher ex ante value on the ability to adjust repayment schedules, even though the realized investment and default effects depend on the shock.

8.2 Maturity and Interest-Rate Fixation

Long-term fixed-rate debt provides two forms of protection: a longer maturity postpones principal refinancing, while a fixed coupon insulates outstanding debt from changes in short-term interest rates. To separate these margins, we solve an alternative economy in which long-term debt retains the benchmark amortization schedule but its coupon resets with the short rate. We then repeat the maturity-flexibility counterfactual of Section 8.1, comparing economies with endogenous and restricted maturity adjustment under the same shocks and initial firm distribution.

Maturity flexibility continues to provide substantial survival protection even without coupon fixation. Restricting maturity adjustment raises five-year cumulative default by 1.2, 4.2, and 1.9 percentage points following current-rate, expected-rate news, and term-premium shocks, respectively. The effect is largest following expected-rate news, showing that firms can reduce refinancing risk by adjusting when principal must be repaid even when interest payments on outstanding debt respond to short rates. Maturity flexibility also lowers average credit spreads, by 3.1 percentage points over five years following expected-rate news. Its investment effects are less persistent: the initial investment gain dissipates and reverses over longer horizons, even as the survival benefit remains substantial.

The two forms of protection are also partially substitutable. Following expected-rate news, fixing coupons reduces five-year cumulative default by 2.0 percentage points when firms can adjust maturity, compared with 3.4 percentage points when maturity adjustment is restricted. Thus, coupon fixation provides greater protection when firms cannot restructure principal repayments, while maturity flexibility remains valuable when coupons float. These results establish that maturity adjustment provides insurance against refinancing risk beyond the protection afforded by fixed-rate contracts. Appendix F.2 reports the full investment, financing, and default responses.

8.3 Implications for Monetary Transmission

The preceding counterfactuals establish that endogenous debt maturity affects the investment and default responses to monetary shocks. We now examine two broader implications for monetary transmission. First, we show that higher term premia amplify the real effects of monetary shocks, particularly when tightening concerns expected future short rates. Second, we quantify how changes in term premia alter the investment protection provided by maturity flexibility. We then illustrate these mechanisms using the 2022–24 euro-area tightening sequence.

Term premia amplify monetary transmission. The real effects of monetary policy depend on the prevailing term-premium environment. To quantify this state dependence, we subject the benchmark economy to identical one-standard-deviation expected-rate news and current-rate shocks, starting from low and high term-premium wedge states, $\omega_0 \in \{0.005, 0.065\}$, corresponding to annualized yield premia of approximately 0.1 and 1.2 percent. We hold firms' initial balance sheets fixed across states and allow the term-premium wedge to revert toward its mean according to its

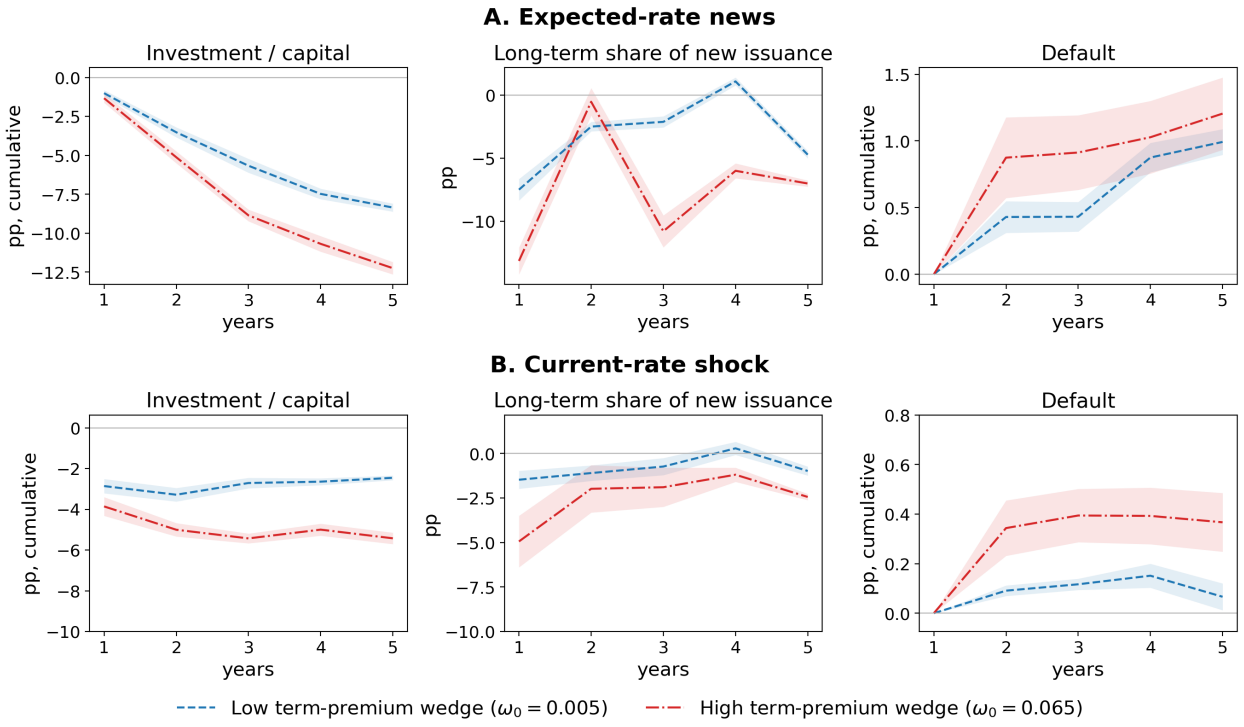


Figure 8: Monetary Transmission across Term-Premium States

Notes: Responses of the benchmark economy to a one-standard-deviation expected-rate news shock (Panel A) and current-rate shock (Panel B) arriving when the term-premium wedge is low ($\omega_0 = 0.005$, dashed) or high ($\omega_0 = 0.065$, dash-dotted). The economy starts in both cases from the same stationary cross-section of firms as in Table 13; the term premium wedge then reverts toward its mean along its estimated law of motion, and responses are measured relative to the same path without the innovation. Investment and default are cumulative; the long-term share of new issuance is the level response at each horizon. Shaded bands are 90 percent Monte Carlo confidence intervals of the mean across 1,000 simulated economies of 10,000 firms. Because the two states share the same simulated economies and idiosyncratic draws, state differences are tested as paired high-minus-low differences history by history (Appendix Figures 19 and 20); overlapping bands do not by themselves indicate an insignificant difference. The difference in cumulative investment is significant at the 1 percent level at every horizon under both shocks. The difference in the long-term issuance share is significant at the 1 percent level at every horizon under news, and under the current-rate shock except in years 2 and 3. The difference in cumulative default is significant at the 1 percent level in years 2 and 3 under news (0.45 and 0.48 percentage points), insignificant in years 4 and 5 as the low-cost state's refinancing wave arrives; under the current-rate shock it is significant at the 1 percent level at every horizon.

estimated law of motion. Responses are measured relative to each state's corresponding no-shock path.[23]

Figure 8 shows that higher term premia amplify investment contractions following identical monetary shocks. Under expected-rate news, three-year cumulative investment falls by 5.7 percentage points relative to capital in the low-premium state, compared with 8.9 points in the high-premium state. This additional 3.2-percentage-point contraction is accompanied by stronger maturity short-

[23]This exercise varies initial aggregate financing conditions while holding the monetary innovation and firms' inherited balance sheets fixed. It does not compare economies that have permanently adapted to different term-premium environments.

ening and higher subsequent refinancing exposure. On impact, the long-term share of new issuance declines by 13.1 percentage points in the high-premium state, compared with 7.5 points in the low-premium state. By the second year, rollover exposure increases by 3.1 percentage points in the high-premium state, compared with 1.6 points when the term premium is low. Higher term premia thus induce firms to reduce maturity protection precisely when expected future refinancing costs rise, leaving them more exposed to subsequent tightening.

Higher term premia also amplify the effects of current-rate shocks, but the associated maturity responses are less pronounced. The high-premium state generates an additional 2.7-percentage-point investment contraction over three years, while the corresponding difference in the long-term issuance share is only -1.2 percentage points. Three-year cumulative default increases by approximately 0.5 percentage points more under expected-rate news and 0.3 points more under current-rate shocks when the term premium is high. These differences are statistically significant in paired Monte Carlo comparisons, although the default difference under expected-rate news is no longer significant after five years. Appendix F.4 reports the full responses and paired comparisons.

These results establish state dependence in monetary transmission, but do not isolate the contribution of endogenous maturity adjustment to the amplification. We therefore compare the flexible- and restricted-maturity economies to quantify how the value of maturity flexibility changes when monetary tightening coincides with higher term premia.

Higher term premia increase the investment benefits of maturity flexibility under expected-rate news. We extend the counterfactual in Section 8.1 by combining either expected-rate news or a current-rate shock with a one-standard-deviation increase in the term-premium wedge. Unlike the preceding exercise, which varies the initial term-premium environment, this experiment introduces simultaneous innovations and compares the flexible- and restricted-maturity economies. Table 14 reports the flexible-minus-restricted differences in their responses.

When expected future short rates and term premia rise together, maturity flexibility provides greater protection against investment losses. Following expected-rate news alone, flexibility preserves 2.4 percentage points of cumulative investment over three years. When the same news coincides with a higher term premium, this benefit increases to 3.9 points. The additional 1.5-percentage-point benefit is statistically significant in paired Monte Carlo comparisons.

Importantly, the greater investment benefit does not arise from additional maturity extension. On impact, the long-term share of new issuance falls 16.4 percentage points more in the flexible than in the restricted economy under the joint shock, compared with 8.8 points under expected-rate news alone. Firms use maturity flexibility to substitute away from more expensive long-term financing, accepting greater refinancing exposure while limiting the immediate investment contraction. Thus, a higher term premium makes maturity insurance more costly, but also increases the investment value of being able to adjust the amount of protection obtained. This additional benefit is concentrated at the three-year horizon and is no longer statistically significant after five years. The five-year reduction in cumulative default remains similar, at 2.8 percentage points under

the joint shock compared with 2.7 points under expected-rate news alone.

The interaction is weaker following current-rate shocks. Adding a term-premium increase raises the three-year investment benefit of maturity flexibility by only 0.2 percentage points, a statistically insignificant difference. Moreover, maturity flexibility does not uniformly reduce default. Under the joint current-rate shock, it preserves 4.5 percentage points of cumulative investment after three years, but cumulative default after five years is 1.5 points higher than in the restricted economy. This reversal is consistent with earlier defaults and sharper subsequent deleveraging in the restricted economy, followed by lower default rates among surviving firms. Appendix Table 35 reports the corresponding experiment with a lower term premium.

Together, these experiments establish two dimensions of state dependence. Higher prevailing term premia amplify the real effects of monetary shocks, while the investment benefits of maturity flexibility also depend on financing conditions. In particular, when expected-rate news coincides with a higher term premium, the ability to adjust toward shorter maturities mitigates investment losses even as long-term refinancing insurance becomes more expensive.

The 2022–24 tightening through the model. We conclude by applying the model to the 2022–24 euro-area tightening cycle to examine how inherited liability structures and anticipatory maturity adjustment shape monetary transmission. We initialize the economy in 2018 and feed in the realized annual paths of short rates, expected-rate news, and the term-premium wedge, using the one-year OIS rate and the five-year yield decomposition of Joslin et al. (2011). During the preceding period of negative short rates and low term premia, firms accumulate longer-maturity liabilities before entering the tightening episode. Appendix F.5 reports the full simulation results.

Expected-rate news changes the timing of maturity adjustment and its consequences for financial distress. Under the realized short-rate path alone, cumulative default increases by 7.9 percentage points through 2024. Adding the realized sequence of expected-rate news brings forward the shortening of new-debt maturity without additional deleveraging relative to the short-rate-only path, reducing the cumulative default response to 3.8 percentage points. In the restricted-maturity economy, by contrast, the same news raises cumulative default from 9.2 to 9.9 percentage points. Thus, anticipated monetary tightening can reduce subsequent financial distress when firms can adjust the timing of their refinancing obligations.

Maturity flexibility also mitigates the real effects of the full tightening sequence. Relative to the restricted-maturity economy, endogenous maturity adjustment mitigates cumulative investment losses by 5.0 percentage points and reduces the cumulative default response by 4.9 percentage points, with both comparisons measured relative to each economy's corresponding no-shock path. These results illustrate how inherited liabilities and anticipatory maturity adjustment shape monetary transmission over a sequence of shocks. The exercise provides conditional model predictions rather than estimates of the realized aggregate effects of ECB policy.

Table 14: Value of Maturity Flexibility under Joint Monetary and Term-Premium Shocks

Horizon (years)	1	3	5
Investment (cumulative, pp)			
Expected-rate news + higher term premium	−0.5 (0.3)	+3.9 (0.6)	+2.2 (0.6)
Current rate + higher term premium	+1.9 (0.6)	+4.5 (0.5)	+4.7 (0.4)
Debt issuance (cumulative, pp)			
Expected-rate news + higher term premium	−2.6 (0.4)	+4.3 (0.7)	+7.0 (0.9)
Current rate + higher term premium	−0.3 (0.7)	+7.6 (0.8)	+13.3 (0.8)
Long-term issuance share (response at horizon, pp)			
Expected-rate news + higher term premium	−16.4 (0.8)	−12.8 (0.4)	−7.6 (0.3)
Current rate + higher term premium	−9.6 (0.7)	−4.6 (0.5)	+1.1 (0.3)
Credit spread (average through horizon, pp)			
Expected-rate news + higher term premium	−1.1 (0.8)	−5.3 (0.4)	−2.7 (0.2)
Current rate + higher term premium	−6.5 (0.6)	−2.6 (0.3)	−0.5 (0.2)
Default (cumulative, pp)			
Expected-rate news + higher term premium	—	−1.2 (0.2)	−2.8 (0.4)
Current rate + higher term premium	—	−0.5 (0.2)	+1.5 (0.4)

Notes: Entries are flexible-minus-restricted differences in shock responses, as in Table 13, to a one-standard-deviation expected-rate news shock or current-rate shock combined with a one-standard-deviation increase in the term-premium wedge in the shock year; each factor subsequently follows its estimated law of motion. Responses to each monetary shock in isolation are reported in Table 13. Restricted firm policies and debt-price schedules are the same as in that table, as are units, horizons, and the simulation design. Investment, debt issuance, and default are cumulative through the reported horizon; the long-term issuance share is the response at that horizon; credit spreads are averaged through the horizon. Horizon-1 default is omitted because default does not respond on impact under the model's timing. Monte Carlo standard errors of the mean are in parentheses.

8.4 Additional Counterfactuals and Diagnostics

Appendix F reports additional exercises that clarify the interpretation of the main counterfactual. First, we verify that its investment effects do not arise mechanically from changes in the composition of surviving firms: the investment-preservation result remains when outcomes are computed for the initial cohort and when the capital of defaulting firms is counted as destroyed. We then trace the opposite maturity responses to the model's switching threshold (Appendix E.1). Rollover exposure is itself an equilibrium outcome; holding productivity, capital, and debt fixed, refinancing risk rises sharply with the share of principal coming due, even though the corresponding relationship is not monotone in the equilibrium cross-section, and a larger share of high-exposure firms than of firms at the amortization floor sits on the long-maturity side of the threshold. Expected-rate news lowers the incentive to issue long debt by little and moves only firms close to the threshold, which are

concentrated at low exposure; the term premium lowers it by much more and moves most initially long-maturity firms at every level of exposure, so its effect reflects the mass initially on the long side. Placing the same firms at every exposure level reproduces both patterns. Finally, we document the endogenous price of obtaining maturity insurance. Long-term debt prices are substantially more sensitive to additional issuance than short-term debt prices, and this sensitivity rises with refinancing needs. Thus, the firms for whom long maturity provides the greatest protection also face a steeper price schedule for obtaining that protection.

Taking stock.

The quantitative analysis delivers three main findings. *First*, endogenous maturity adjustment mitigates the real effects of monetary tightening. Following expected-rate news, maturity flexibility preserves 2.4 percentage points of cumulative investment and reduces cumulative default by 1.4 points over three years. *Second*, maturity insurance is distinct from interest-rate fixation: even when coupons reset with the short rate, flexibility reduces default by limiting exposure to principal refinancing risk. *Third*, term premia shape both monetary transmission and the value of maturity flexibility. A higher prevailing term premium amplifies the three-year investment contraction following identical expected-rate news by 3.2 percentage points. Yet when expected-rate news coincides with a higher term premium, the investment benefit of maturity flexibility rises from 2.4 to 3.9 percentage points. Thus, maturity flexibility matters not only because firms can postpone refinancing, but also because they can avoid costly long-term borrowing when term premia rise.

9 Conclusion

This paper identifies a maturity-insurance channel of monetary transmission through which nonfinancial corporations adjust their exposure to future refinancing conditions. Using euro-area credit-registry data matched to firm balance sheets, we show that firms with greater inherited refinancing needs obtain longer maturities on new borrowing, but their relative maturity responses depend on the determinants of long-term interest rates. Higher expected future short rates strengthen relative maturity extension among exposed firms, whereas higher term premia weaken it. These opposing responses reflect the distinction between the value of reducing future refinancing risk and the cost of obtaining maturity protection.

A heterogeneous-firm model shows that endogenous maturity adjustment has important consequences for investment and default. Following news of higher future rates, maturity flexibility mitigates investment contractions and reduces cumulative default by allowing firms to adapt their financing decisions to changing monetary conditions. Importantly, this protection does not always require longer maturity: when long-term financing becomes expensive, firms can benefit from shortening new borrowing rather than paying for costly refinancing insurance. The model also shows that higher term premia amplify the investment effects of expected-rate news, while increasing the

investment benefits of maturity flexibility. Applying the model to the 2022–24 euro-area tightening illustrates how inherited liability structures and successive monetary shocks shape the timing of maturity adjustment. Expected-rate news brings maturity shortening forward without inducing additional deleveraging relative to the realized short-rate path alone, while maturity flexibility mitigates investment losses and cumulative default.

These findings have implications for assessing monetary transmission. The expected path of policy rates alone is insufficient to characterize how monetary shocks affect corporate financing and investment. Their real effects also depend on the term-premium environment and firms' ability to adjust inherited refinancing obligations. For monetary policymakers, this highlights the importance of considering corporate repayment schedules and maturity choices alongside the decomposition of long-term interest rates into expected future short rates and term premia. More broadly, the transmission of monetary policy depends not only on the financial obligations firms inherit, but also on how they adapt those obligations as financing conditions change.

The maturity-insurance channel raises broader questions for monetary transmission and even international spillovers. In small open economies, changes in global interest rates and risk premia may affect both the cost of long-term borrowing and firms' incentives to manage refinancing exposure, with potential implications for domestic investment and international capital flows. An important direction for future research is to examine how banks' funding structures and balance-sheet constraints shape the supply of long-term credit, potentially limiting firms' access to maturity protection precisely when refinancing risks increase.

References

Acharya, V. V., Byoun, S., and Xu, Z. (2020). The sensitivity of cash savings to the cost of capital. NBER Working paper No. w27517.

Adrian, T., Crump, R. K., and Moench, E. (2013). Pricing the term structure with linear regressions. *Journal of Financial Economics*, 110(1):110–138.

Almeida, H., Campello, M., Laranjeira, B., and Weisbenner, S. (2012). Corporate debt maturity and the real effects of the 2007 credit crisis. *Critical Finance Review*, 1(1):3–58.

Altavilla, C., Brugnolini, L., Gürkaynak, R. S., Motto, R., and Ragusa, G. (2019). Measuring euro area monetary policy. *Journal of Monetary Economics*, 108:162–179.

Altavilla, C., Gürkaynak, R. S., Kind, T., and Laeven, L. (2026). Monetary transmission with frequent policy events. *Journal of Monetary Economics*, 163:103986.

Altinkilic, O. and Hansen, R. S. (2000). Are there economies of scale in underwriting fees? evidence of rising external financing costs. *The Review of Financial Studies*, 13(1):191–218.

Anderson, G. and Cesa-Bianchi, A. (2024). Crossing the credit channel: Credit spreads and firm heterogeneity. *American Economic Journal: Macroeconomics*, 16(3):417–446.

Arellano, C. and Ramanarayanan, A. (2012). Default and the maturity structure in sovereign bonds. *Journal of Political Economy*, 120(2):187–232.

Baker, M., Greenwood, R., and Wurgler, J. (2003). The maturity of debt issues and predictable variation in bond returns. *Journal of Financial Economics*, 70(2):261–291.

Bauer, M. D., Lakdawala, A., and Mueller, P. (2022). Market-based monetary policy uncertainty. *The Economic Journal*, 132(644):1290–1308.

Berg, T., Saunders, A., and Steffen, S. (2016). The total cost of corporate borrowing in the loan market: Don't ignore the fees. *The Journal of Finance*, 71(3):1357–1392.

Berger, D., Milbradt, K., Tourre, F., and Vavra, J. (2021). Mortgage prepayment and path-dependent effects of monetary policy. *American Economic Review*, 111(9):2829–2878.

Bernanke, B. S. and Gertler, M. (1995). Inside the black box: The credit channel of monetary policy transmission. *Journal of Economic Perspectives*, 9(4):27–48.

Bolton, P., Chen, H., and Wang, N. (2013). Market timing, investment, and risk management. *Journal of Financial Economics*, 109(1):40–62.

Bräuning, F., Fillat, J. L., and Wang, J. C. (2020). Corporate debt maturity and monetary policy. Current policy perspectives, Federal Reserve Bank of Boston.

Brunnermeier, M. K. and Oehmke, M. (2013). The maturity rat race. *Journal of Finance*, 68(2):483–521.

Chatterjee, S. and Eyigungor, B. (2012). Maturity, indebtedness, and default risk. *American Economic Review*, 102(6):2674–2699.

Cooley, T. F. and Quadrini, V. (2001). Financial markets and firm dynamics. *American Economic Review*, 91(5):1286–1310.

Cooper, R. W. and Haltiwanger, J. C. (2006). On the nature of capital adjustment costs. *Review of Economic Studies*, 73(3):611–633.

Core, F., De Marco, F., Eisert, T., and Schepens, G. (2025). Inflation and floating-rate loans: Evidence from the euro area. Working paper, European Central Bank. Available at SSRN.

Crouzet, N. (2017). Default, debt maturity and investment dynamics. Working Paper, Northwestern University.

Degryse, H., De Jonghe, O., Jakovljević, S., Mulier, K., and Schepens, G. (2019). Identifying credit supply shocks with bank-firm data: Methods and applications. *Journal of Financial Intermediation*, 40:100813.

Deng, M. and Fang, M. (2022). Debt maturity heterogeneity and investment responses to monetary policy. *European Economic Review*, 144:104095.

Diamond, D. W. (1991). Debt maturity structure and liquidity risk. *Quarterly Journal of Economics*, 106(3):709–737.

Duchin, R., Ozbas, O., and Sensoy, B. A. (2010). Costly external finance, corporate investment, and the subprime mortgage credit crisis. *Journal of Financial Economics*, 97(3):418–435.

Eichenbaum, M., Rebelo, S., and Wong, A. (2022). State-dependent effects of monetary policy: The refinancing channel. *American Economic Review*, 112(3):721–761.

Eisfeldt, A. L. and Muir, T. (2016). Aggregate external financing and savings waves. *Journal of Monetary Economics*, 84:116–133.

Ericson, K. M. and Sydnor, J. (2026). Liquidity constraints and the value of insurance. *American Economic Journal: Microeconomics*, 18(3):426–453.

Fabiani, A., Heineken, J., and Falasconi, L. (2024). Monetary policy and the maturity structure of corporate debt. Unpublished manuscript.

Faulkender, M. (2005). Hedging or market timing? selecting the interest rate exposure of corporate debt. *The Journal of Finance*, 60(2):931–962.

Ferrante, F., Prestipino, A., and Schott, I. (2025). Corporate debt maturity and business cycle fluctuations. Technical Report 1409, Board of Governors of the Federal Reserve System.

Flannery, M. J. (1986). Asymmetric information and risky debt maturity choice. *Journal of Finance*, 41(1):19–37.

Gertler, M. and Gilchrist, S. (1994). Monetary policy, business cycles, and the behavior of small manufacturing firms. *Quarterly Journal of Economics*, 109(2):309–340.

Gomes, J. F. (2001). Financing investment. *American Economic Review*, 91(5):1263–1285.

Greenwood, R., Hanson, S. G., and Stein, J. C. (2010). A gap-filling theory of corporate debt maturity choice. *Journal of Finance*, 65(3):993–1028.

Greenwood, R. and Vayanos, D. (2014). Bond supply and excess bond returns. *The Review of Financial Studies*, 27(3):663–713.

Gürkaynak, R. S., Sack, B., and Swanson, E. T. (2005). Do actions speak louder than words? the response of asset prices to monetary policy actions and statements. *International Journal of Central Banking*, 1(1):55–93.

Gürkaynak, R. S., Sack, B., and Wright, J. H. (2010). The TIPS yield curve and inflation compensation. *American Economic Journal: Macroeconomics*, 2(1):70–92.

Hanson, S. G. and Stein, J. C. (2015). Monetary policy and long-term real rates. *Journal of Financial Economics*, 115(3):429–448.

He, Z. and Xiong, W. (2012). Rollover risk and credit risk. *Journal of Finance*, 67(2):391–430.

Hennessy, C. A. and Whited, T. M. (2005). Debt dynamics. *Journal of Finance*, 60(3):1129–1165.

Ippolito, F., Ozdagli, A. K., and Pérez-Orive, A. (2018). The transmission of monetary policy through bank lending: The floating rate channel. *Journal of Monetary Economics*, 95:49–71.

Jarociński, M. and Karadi, P. (2020). Deconstructing monetary policy surprises. *American Economic Journal: Macroeconomics*, 12(2):1–43.

Jarociński, M. and Karadi, P. (2020). Deconstructing Monetary Policy Surprises—The Role of Information Shocks. *American Economic Journal: Macroeconomics*, 12(2):1–43.

Jiménez, G., Ongena, S., Peydró, J.-L., and Saurina, J. (2012). Credit supply and monetary policy: Identifying the bank balance-sheet channel. *American Economic Review*, 102(5):2301–2326.

Joslin, S., Singleton, K. J., and Zhu, H. (2011). A new perspective on gaussian dynamic term structure models. *Review of Financial Studies*, 24(3):926–970.

Jungherr, J., Meier, M., Reinelt, T., and Schott, I. (2024). Corporate debt maturity matters for monetary policy. Working paper.

Jungherr, J. and Schott, I. (2021). Optimal debt maturity and firm investment. *Review of Economic Dynamics*, 42:110–132.

Kalemli-Özcan, Ş., Laeven, L., and Moreno, D. (2022). Debt overhang, rollover risk, and corporate investment: Evidence from the european crisis. *Journal of the European Economic Association*, 20(6):2353–2395.

Kalemli-Özcan, Ş., Sørensen, B. E., Villegas-Sanchez, C., Volosovych, V., and Yeşiltaş, S. (2024). How to construct nationally representative firm-level data from the orbis global database: New facts on smes and aggregate implications for industry concentration. *American Economic Journal: Macroeconomics*, 16(2):353–374.

Kashyap, A. K. and Stein, J. C. (2000). What do a million observations on banks say about the transmission of monetary policy? *American Economic Review*, 90(3):407–428.

Keen, B. D., Richter, A. W., and Throckmorton, N. A. (2017). Forward guidance and the state of the economy. *Economic Inquiry*, 55(4):1593–1624.

Khwaja, A. I. and Mian, A. (2008). Tracing the impact of bank liquidity shocks: Evidence from an emerging market. *American Economic Review*, 98(4):1413–1442.

Koop, G., Pesaran, M. H., and Potter, S. M. (1996). Impulse response analysis in nonlinear multivariate models. *Journal of Econometrics*, 74:119–147.

Kozlowski, J. (2021). Long-term finance and investment with frictional asset markets. *American Economic Journal: Macroeconomics*, 13(4):411–448.

Leland, H. E. (1994). Corporate debt value, bond covenants, and optimal capital structure. *Journal of Finance*, 49(4):1213–1252.

Lunsford, K. G. (2020). Policy language and information effects in the early days of federal reserve forward guidance. *American Economic Review*, 110(9):2899–2934.

McKay, A., Nakamura, E., and Steinsson, J. (2016). The power of forward guidance revisited. *American Economic Review*, 106(10):3133–3158.

Mian, A. and Santos, J. A. C. (2018). Liquidity risk and maturity management over the credit cycle. *Journal of Financial Economics*, 127(2):264–284.

Myers, S. C. (1977). Determinants of corporate borrowing. *Journal of Financial Economics*, 5(2):147–175.

Ottonello, P. and Winberry, T. (2020). Financial heterogeneity and the investment channel of monetary policy. *Econometrica*, 88(6):2473–2502.

Poeschl, J. (2023). Corporate debt maturity and investment over the business cycle. *European Economic Review*, 152:104348.

Ropele, T., Gorodnichenko, Y., and Coibion, O. (2022). Inflation expectations and corporate borrowing decisions: New causal evidence. NBER Working Paper 30537, National Bureau of Economic Research.

Salomão, J. and Varela, L. (2022). Exchange rate exposure and firm dynamics. *The Review of Economic Studies*, 89(1):481–514.

Vayanos, D. and Vila, J.-L. (2021). A preferred-habitat model of the term structure of interest rates. *Econometrica*, 89(1):77–112.

Xu, Q. (2018). Kicking maturity down the road: Early refinancing and maturity management in the corporate bond market. *The Review of Financial Studies*, 31(8):3061–3097.

A Empirics Appendix

A.1 Tables

Table 15: Summary statistics

Variable	Obs.	Mean	Std. Dev.	Min	p25	Median	p75	Max
Loan-level variables (monthly)								
Loan maturity, days	14,792,489	1,895	1,525	29	113.42	737	1,836	7,330
Annualized interest rate	9,317,565	0.033	0.023	0	0.015	0.030	0.049	0.137
Log(Total credit)	11,319,955	11.12	1.92	3.26	10.05	11.00	12.21	16.32
Probability of default	4,765,303	0.050	0.146	0	0.004	0.011	0.031	1
Short-Term Debt (monthly)	11,319,955	0.28	0.34	0	0	0.12	0.50	1
Maturity Wall (monthly)	11,319,955	0.34	0.37	0.00	0.18	0.66	0.80	1
Firm-level variables (yearly)								
Short-Term Debt (annual)	10,457,475	0.18	0.31	0.00	0.00	0.00	0.20	1.00
Interest expense/assets	6,364,645	0.01	0.01	0.00	0.00	0.01	0.01	0.08
Debt-service ratio	6,364,462	0.44	0.28	0.01	0.21	0.40	0.62	1.24
Interest coverage ratio	6,087,148	40.74	110.66	-55.96	2.44	8.54	28.38	640.43
Δ Investment rate	7,374,655	0.19	1.03	-0.93	-0.11	-0.02	0.10	7.70
Δ Bank Maturity, years	3,478,903	-0.10	4.25	-24.85	-1.97	0.00	1.94	24.83
Maturity extension dummy	10,457,475	0.67	0.47	0.00	0.00	1.00	1.00	1.00
Δ long-term bank loans	3,653,379	0.01	0.57	-1.00	0.00	0.00	0.06	1.00
Δ bank loans	1,585,898	2.59	8.61	-1.00	-0.68	-0.04	1.59	50.30
Δ net total debt	6,957,930	0.00	1.43	-7.08	-0.24	-0.01	0.18	7.45
Log total assets	10,452,781	13.72	1.55	10.33	12.66	13.60	14.65	18.26
Tangibility	10,247,641	0.30	0.30	0.00	0.05	0.20	0.50	0.99
Cash/assets	9,853,643	0.15	0.18	0.00	0.02	0.09	0.23	0.78
Sales growth	4,721,460	0.23	1.24	-0.99	-0.21	0.00	0.25	9.26
Leverage	10,452,781	0.70	0.36	0.05	0.47	0.69	0.88	2.48
Time-level variables (monthly)								
Yield Curve 5Y–1Y	76	-0.16	0.55	-1.51	-0.42	-0.05	0.17	1.02
Yield Curve 10Y–2Y	76	0.18	0.53	-0.94	-0.18	0.25	0.55	1.10
Expected Rate Path 5Y, JSZ	76	1.133	1.005	-0.033	0.336	0.495	2.217	3.020
Expected Rate Path 5Y, ACM	76	1.047	0.965	-0.023	0.269	0.413	2.136	2.954
Expected Rate Path 10Y, JSZ	76	1.159	0.567	0.422	0.721	0.842	1.770	2.278
Expected Rate Path 10Y, ACM	76	1.041	0.525	0.392	0.641	0.738	1.618	2.141
Term Premium 5Y, JSZ	76	-0.531	0.506	-1.296	-1.039	-0.433	-0.134	0.455
Term Premium 5Y, ACM	76	-0.423	0.532	-1.182	-0.938	-0.354	-0.035	0.713
Term Premium 10Y, JSZ	76	-0.276	0.801	-1.411	-1.073	-0.249	0.459	1.172
Term Premium 10Y, ACM	76	-0.178	0.847	-1.335	-1.005	-0.223	0.617	1.305
Target shock	76	0.13	1.09	-1.88	-0.15	0.00	0.00	8.00
Timing shock	76	0.02	0.34	-1.14	-0.03	0.00	0.15	1.32
Forward guidance shock	76	0.05	0.87	-2.61	-0.26	0.00	0.10	2.96
Quantitative policy shock	76	-0.06	0.73	-3.13	-0.23	0.00	0.14	1.88

Notes: This table reports the summary statistics of the variables used in the empirical analysis. Loan-level and firm-level variables are winsorized at the 1st and 99th percentiles.

Table 16: Firm Characteristics and Rollover Exposure

	Low exposure $\leq p25$	High exposure $\geq p75$	Mean Difference High − Low
Fixed assets$_{f,t-1}$	12.325	12.266	-0.059^{***}
Tangibility$_{f,t-1}$	0.347	0.233	-0.114^{***}
Sales growth$_{f,t-1}$	20.569	18.106	-2.463
Leverage$_{f,t-1}$	0.715	0.673	-0.041^{***}
Interest coverage$_{f,t-1}$	50.808	61.709	10.901^{***}

Notes: This table compares firms in the bottom and top quartiles ($\leq$p25 and $\geq$p75) of the within-year distribution of rollover exposure proxied by short-term bank debt, calculated from AnaCredit. Firm characteristics are measured in the previous year; the final column reports the difference in group means, with standard errors clustered at the firm level. Sample period 2018–2023. *, **, and *** denote significance at the 10%, 5%, and 1% levels, respectively.

Table 17: Rollover Exposure and Floating-Rate New Loans

Floating-rate Share of New Loans	(1) 5Y–1Y	(2) 10Y–2Y
ST Debt$_{f,t-1}$	0.0246***	0.0283***
	(0.0011)	(0.0012)
YC$_{t-1}\times$ ST Debt$_{f,t-1}$	-0.0121***	-0.0137***
	(0.0015)	(0.0017)
Observations	1,674,254	1,674,254
Adjusted R^2	0.696	0.696
Bank × Month FE	Yes	Yes
Firm × Bank FE	Yes	Yes
Firm Controls	Yes	Yes

Notes: The dependent variable is the share of new loans with a floating rate over total new loans. Column (1) uses the 5Y–1Y yield curve slope and column (2) uses the 10Y–2Y slope as robustness. All specifications include lagged firm controls, bank×month fixed effects, and firm×bank fixed effects. Standard errors are clustered at the firm and month level.
*, **, and *** denote significance at the 10%, 5%, and 1% levels, respectively.

Table 18: Robustness of the baseline result

$\log(\text{Maturity}_{bft})$	(1) Excl. COVID period	(2) Richer supply FE	(3) New credit	(4) Alt. slope	(5) Policy rate	(6) Bank controls	(7) Firm × year FE
ST $\text{debt}_{f,t-1}$	0.131***	0.242***	0.230***	0.252***	0.226***	−0.427***	0.438***
	(0.00489)	(0.00795)	(0.00436)	(0.0059)	(0.00432)	(0.00342)	(0.00733)
Yield Curve Slope_{t-1}× ST $\text{debt}_{f,t-1}$	−0.0224***	−0.0219***	−0.0175***	−0.0812***	−0.0180**	−0.150***	−0.0163**
	(0.00463)	(0.00811)	(0.00427)	(0.00503)	(0.00704)	(0.00611)	(0.00745)
DFR_{t-1}× ST $\text{debt}_{f,t-1}$				0.0050	0.00152		
				(0.0086)	(0.00283)		
Firm × bank FE	Yes	Yes	Yes	Yes	Yes	Yes	No
Firm × year FE	No	No	No	No	No	No	Yes
Bank × month FE	Yes	No	Yes	Yes	Yes	No	Yes
Bank × ind. × loc. × month FE	No	Yes	No	No	No	No	No
Firm controls	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Bank controls	No	No	No	No	No	Yes	No
New credit control	No	No	Yes	No	No	No	No
yield curve slope	5Y–1Y	5Y–1Y	5Y–1Y	10Y–2Y	5Y–1Y	5Y–1Y	5Y–1Y
Observations	1,139,504	992,254	1,674,254	1,674,254	1,674,254	1,765,750	1,624,347
Adjusted R^2	0.613	0.583	0.620	0.620	0.620	0.056	0.607

Notes: Column (1) estimates the baseline specification excluding the COVID period, defined as 2019Q4–2021Q1. Column (2) replaces bank × month fixed effects with bank × industry × location × month fixed effects, so that identification comes from firms borrowing from the same lender, in the same industry and region, in the same month. Column (3) adds total new credit as a control. Column (4) uses the 10Y–2Y slope instead of the 5Y–1Y slope and adds the lagged deposit facility (DFR) rate interacted with rollover exposure. Column (5) adds the lagged deposit facility rate interacted with rollover exposure. Column (6) replaces bank × month fixed effects with observable bank controls. Column (7) is the most saturated specification: firm × year fixed effects absorb annual investment plans, asset composition, and cash-flow expectations at the firm level, so identification comes from within-year monthly variation in the term structure among firms borrowing from the same bank in the same month. Standard errors are clustered at the firm and month level. *, **, *** denote significance at the 10%, 5%, and 1% levels.

Table 19: High-Frequency Monetary Policy Factors across the Yield Curve

	1M	3M	6M	1Y	2Y	5Y	10Y
Target	3.083***	2.638***	2.241***	1.560***	0.860	0.282	-0.047
	(0.071)	(0.228)	(0.216)	(0.383)	(0.562)	(0.527)	(0.284)
Timing	0.823***	1.225***	1.413***	1.704***	1.520***	1.095***	0.490**
	(0.109)	(0.107)	(0.097)	(0.176)	(0.205)	(0.197)	(0.221)
Forward Guidance	0.028	0.092	1.036***	2.108***	3.011***	3.349***	2.656***
	(0.083)	(0.287)	(0.283)	(0.406)	(0.410)	(0.375)	(0.274)
Quantitative Policy	-0.054	-0.429***	-1.213***	-2.025***	-2.258***	-1.206***	0.024
	(0.081)	(0.100)	(0.139)	(0.296)	(0.305)	(0.313)	(0.249)
Observations	128	128	128	128	128	127	128
R^2	0.965	0.831	0.801	0.659	0.589	0.579	0.618

Notes: The table validates the high-frequency monetary policy factors of Altavilla et al. (2019) by relating them to changes in OIS yields at different maturities. Each entry reports the response of the corresponding OIS yield to a one-standard-deviation monetary policy factor. Target primarily loads on the short end of the yield curve, Timing affects short- and intermediate-maturity rates, Forward Guidance loads most strongly on intermediate maturities, and Quantitative Policy primarily affects intermediate-maturity yields. The sample covers ECB monetary policy events from July 2011 to October 2025. Standard errors are reported in parentheses. *, **, and *** denote significance at the 10%, 5%, and 1% levels, respectively.

Table 20: Rollover Exposure and Protection Value

$\log(\text{Protection Value})_{bft}$	(1)
$\text{ST Debt}_{f,t-1}$	0.306***
	(0.0105)
$\text{Yield Curve}_{t-1} \times \text{ST Debt}_{f,t-1}$	-0.00113
	(0.0133)
Observations	675,691
Adjusted R^2	0.706
Bank × Month FE	Yes
Firm × Bank FE	Yes
Firm Controls	Yes

Notes: The dependent variable is the logarithm of the protection value reported in AnaCredit, which measures the monetary value of collateral or guarantees securing bank credit, based on their reported notional amount or most recent valuation. Short-Term Debt proxy rollover exposure as the share of the firm's outstanding bank debt maturing within the following 12 months. Yield Curve is the lagged 5Y–1Y yield curve slope. The specification includes bank×month fixed effects, firm fixed effects, and lagged firm controls. The main effect of the yield curve is absorbed by bank×month fixed effects. Standard errors are reported in parentheses. ***, **, and * denote statistical significance at the 1%, 5%, and 10% levels, respectively.

Table 21: Rollover Exposure and New Loan Maturity by contract rate type

$\log(\text{Maturity}_{bft})$	(1) Floating rate	(2) Fixed rate
ST $\text{Debt}_{f,t-1}$	0.0883***	0.154***
	(0.00740)	(0.00439)
Yield Curve $\text{Slope}_{t-1}\times$ ST $\text{debt}_{f,t-1}$	−0.0165**	−0.0108**
	(0.00668)	(0.00444)
Firm FE	Yes	Yes
Bank × month FE	Yes	Yes
Firm controls	Yes	Yes
Observations	359,175	1,179,346
Adjusted R^2	0.753	0.616

Notes: The sample is split by the interest-rate type of the newly originated loan. Column (1) uses new loans with a floating rate at origination , column (2) new loans with fixed-rate at origination. Firm FE are used to maximize the number of observations. Firm controls are lagged log total assets, cash over assets, interest coverage, tangibility and sales growth. The slope is the lagged 5Y–1Y spread. Standard errors clustered at the firm and month level. *, **, *** denote significance at the 10%, 5% and 1% levels.

Table 22: Rollover Exposure and New Loan Maturity across the maturity distribution

$\mathbf{1}\{\text{Maturity}_{bft} \geq h\}$	≥1Y	≥2Y	≥3Y	≥5Y	≥7Y
ST $\text{Debt}_{f,t-1}$	0.0763***	0.0761***	0.0664***	0.0323***	0.00799***
	(0.00159)	(0.00143)	(0.00135)	(0.00104)	(0.000512)
$\text{Slope}_{t-1}\times$ ST $\text{debt}_{f,t-1}$	−0.0108***	−0.00685***	−0.00427**	0.00206	−0.00114
	(0.00242)	(0.00204)	(0.00192)	(0.00143)	(0.000821)
Bank × month FE	Yes	Yes	Yes	Yes	Yes
Firm × bank FE	Yes	Yes	Yes	Yes	Yes
Firm controls	Yes	Yes	Yes	Yes	Yes
Observations	1,414,098	1,414,098	1,414,098	1,414,098	1,414,098
Adjusted R^2	0.564	0.543	0.537	0.470	0.569

Notes: Each dependent variable is an indicator equal to one if the new loan has maturity at least the indicated threshold; the yield curve slope is 5Y–1Y. Standard errors are clustered at the firm and month level. *, **, *** denote significance at the 10%, 5% and 1% levels.

Table 23: Extensive margin: selection into new borrowing

Dependent variable: $\mathbf{1}\{\text{New loan}_{f,t}\}$	(1) 10Y–2Y	(2) 5Y–1Y
ST Debt$_{f,t-1}$	0.0665***	0.0693***
	(0.000291)	(0.000291)
YC$_{t-1}$ × ST Debt$_{f,t-1}$	0.0122***	0.00678***
	(0.000431)	(0.000364)
Firm FE	Yes	Yes
Country × industry × month FE	Yes	Yes
Firm controls	Yes	Yes
Observations	53,200,020	53,200,020
Adjusted R^2	0.301	0.301

Notes: Firm-month linear probability models. The dependent variable equals one if firm f obtains at least one new loan in month t. The yield curve slope is 10Y-2Y in column (1) and 5Y-1Y in column (2). Firm controls include size, cash over assets, interest coverage, and tangibility. Standard errors, in parentheses, are clustered at the firm and month level. *, **, *** denote significance at the 10%, 5%, and 1% level.

Table 24: Real-rate and inflation-compensation components

Dependent variable: log(Loan maturity)$_{bft}$	(1) 10Y–2Y	(2) 5Y–1Y
ST Debt$_{f,t-1}$	0.258***	0.223***
	(0.00491)	(0.00450)
YC$^{real}_{t-1}$ × ST Debt$_{f,t-1}$	−0.0792***	−0.0707***
	(0.00648)	(0.00652)
YC$^{inflation}_{t-1}$ × ST Debt$_{f,t-1}$	−0.114***	−0.0975***
	(0.00857)	(0.00832)
Bank × month FE	Yes	Yes
Firm × bank FE	Yes	Yes
Firm controls	Yes	Yes
Observations	1,674,254	1,674,254
Adjusted R^2	0.620	0.620

Notes: The nominal yield curve slope is decomposed into real-rate and inflation-compensation components following Gürkaynak et al. (2010), using inflation swaps. Firm controls are included. Standard errors, in parentheses, are clustered at the firm and month level. *, **, *** denote significance at the 10%, 5%, and 1% level.

Table 25: Extended sample using balance-sheet exposure

Dependent variable: $\log(\text{Loan maturity})_{bft}$	(1) 5Y–1Y	(2) 10Y–2Y	(3) 5Y–1Y
ST Debt$_{f,t-1}$	−0.200***	−0.166***	−0.198***
	(0.00879)	(0.00936)	(0.0109)
YC$_{t-1}$ × ST Debt$_{f,t-1}$	−0.0750***	−0.115***	−0.0683*
	(0.0138)	(0.0118)	(0.0363)
Post2020$_t$ × ST Debt$_{f,t-1}$			−0.00234
			(0.00962)
Post2020$_t$ × YC$_{t-1}$ × ST Debt$_{f,t-1}$			−0.00985
			(0.0393)
Bank × month FE	Yes	Yes	Yes
Firm × bank FE	Yes	Yes	Yes
Firm controls	Yes	Yes	Yes
Trade-credit controls	Yes	Yes	Yes
Observations	1,523,207	1,523,207	1,523,207
Adjusted R^2	0.636	0.636	0.636

Notes: The dependent variable is new loan maturity. Rollover exposure is proxied by ST Debt measured as current liabilities over total debt, which is coarser than the loan-level measure and includes trade credit and other short-term obligations. Columns (1)–(2) report the extended-sample estimates using the 5Y–1Y and 10Y–2Y yield curve slopes. Column (3) tests for a structural break after 2020 by allowing both the level effect of refinancing exposure and its interaction with the yield curve to differ in the post-2020 period. Firm controls include size, cash over assets, sales growth, tangibility, and receivables over assets. Standard errors, in parentheses, are clustered at the firm and month level. *, **, *** denote significance at the 10%, 5%, and 1% level.

Table 26: Yield Decomposition and New Loan Maturity at year level

	(1) log(Maturity)	(2) Maturity, years	(3) LT debt share
ST Debt$_{f,t-1}$	1.171***	3.975***	0.526***
	(0.0135)	(0.751)	(0.00585)
ESR^{5Y}_{t-1} (1 s.d.) × ST Debt$_{f,t-1}$	0.430***	1.921***	0.201***
	(0.0085)	(0.489)	(0.0036)
TP^{5Y}_{t-1} (1 s.d.) × ST Debt$_{f,t-1}$	−0.375***	−1.925***	−0.175***
	(0.0085)	(0.492)	(0.0036)
Firm controls	Yes	Yes	Yes
Observations	1,232,782	1,243,865	1,261,899
Adjusted R^2	0.588	0.262	0.459

Notes: The dependent variables are the volume-weighted maturity of newly originated bank loans and the long-term share of bank debt in columns 1-3 respectively. ST Debt is the lagged share of outstanding bank debt with residual maturity of twelve months or less. ESR^{5Y} and TP^{5Y} are the lagged five-year expected-rate and term-premium components of the Joslin et al. (2011) decomposition. Interaction coefficients are standardized: each reports the effect of a one-standard-deviation change in the lagged factor. Controls include lagged log assets and lagged outstanding loan volume. Standard errors in parentheses. *, **, *** denote significance at the 10%, 5%, and 1% levels.

Table 27: Rollover Exposure, the Term Structure, and Firm Size

Panel A. Maturity and Liability Restructuring

	(1) log(Maturity) Loan level	(2) Δ Bank-Debt Maturity	(3) Δ Long Term Bank Loans	(4) Δ Total Debt
ST Debt$_{f,t-1}$	0.200***	2.390***	0.577***	-0.0343***
	(0.00450)	(0.0280)	(0.00484)	(0.00230)
YC (5Y–1Y)$_{t-1}$× ST Debt$_{f,t-1}$	-0.0309***	-1.701***	-0.535***	-0.0382***
	(0.00496)	(0.0742)	(0.0125)	(0.00632)
YC (5Y–1Y)$_{t-1}$× ST Debt$_{f,t-1}$× Low TA$_f$	-0.0387***	-1.128***	-0.216***	-0.0472***
	(0.0149)	(0.164)	(0.0308)	(0.0156)
Observations	1,674,254	594,168	594,168	594,168
Adjusted R^2	0.587	0.084	0.044	0.098
Bank × Time FE	Yes	No	No	No
Firm × Bank FE	Yes	No	No	No
Industry × Country × Time FE	No	Yes	Yes	Yes
Firm FE	No	Yes	Yes	Yes
Firm Controls	Yes	Yes	Yes	Yes

Panel B. Liquidity, Interest Expenses, and Investment

	Δ Cash /TA	Interest Exp. / EBITDA	Debt-Service Ratio	Investment Rate
ST Debt$_{f,t-1}$	0.140***	-0.00520***	-0.00846***	0.103***
	(0.0523)	(0.000943)	(0.000576)	(0.00961)
YC (5Y–1Y)$_{t-1}$× ST Debt$_{f,t-1}$	-0.442***	0.00920***	0.0554***	-0.124***
	(0.134)	(0.00261)	(0.00158)	(0.0242)
YC (5Y–1Y)$_{t-1}$× ST Debt$_{f,t-1}$× Low TA$_f$	-0.0615	0.0195***	0.0155***	-0.137**
	(0.280)	(0.00650)	(0.00421)	(0.0674)
Observations	594,168	594,168	594,168	594,168
Adjusted R^2	0.029	0.490	0.821	0.137
Industry × Country × Time FE	Yes	Yes	Yes	Yes
Firm FE	Yes	Yes	Yes	Yes
Firm Controls	Yes	Yes	Yes	Yes

Notes: This table reports how the interaction between rollover exposure and the term structure varies with firm size. Low TA is an indicator for firms in the bottom quartile of total assets. Column (1) is the loan-level maturity regression of Table 1; columns (2)–(8) are firm-level regressions with the outcomes of Table 8 and the investment rate, the latter in the full firm-year sample (7) and in the sample with observed bank-debt maturity changes (8). All specifications include the lower-order interactions of Low TA with rollover exposure and with the slope, lagged firm controls, and the fixed effects indicated. Short-Term Debt is the lagged share of outstanding bank debt with residual maturity within 12 months. YC (5Y–1Y) is the lagged difference between the 5-year AAA euro-area government bond yield and the 1-year OIS rate. Standard errors are clustered at the firm level and reported in parentheses. *, **, and *** denote statistical significance at the 10%, 5%, and 1% levels, respectively.

Table 28: Rollover Exposure, Yield Curve, and New Credit

Log New Credit$_{f,b,t}$	(1) Total Credit	(2) Term Loans
ST Debt$_{f,t-1}$	0.381***	0.363***
	(0.00661)	(0.00618)
Yield Curve$_{t-1}^{5Y-1Y}$ × ST Debt$_{f,t-1}$	0.00026	-0.00668
	(0.00827)	(0.00772)
Observations	1,822,621	1,776,785
Adjusted R^2	0.697	0.719

Notes: The dependent variable is the logarithm of newly originated credit at the firm–bank–month level. Column (1) considers total new credit, while column (2) restricts the outcome to new term loans. ST Debt is the lagged share of outstanding bank debt maturing within twelve months. The Yield Curve (5Y–1Y) is the difference between the five-year and one-year yields. Both specifications include firm and bank-by-month fixed effects and firm-level controls. Standard errors are reported in parentheses. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.10$.

A.2 Figures

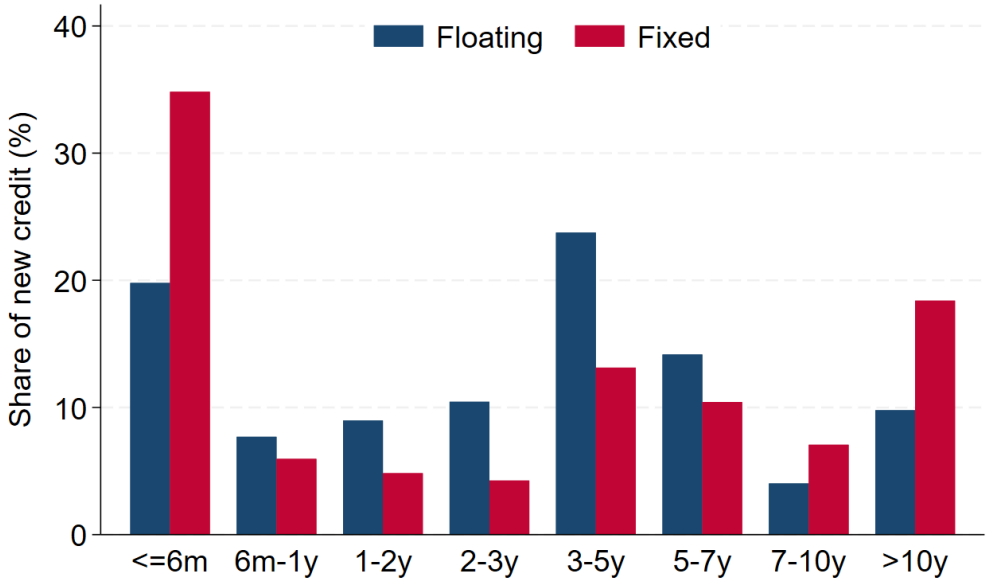
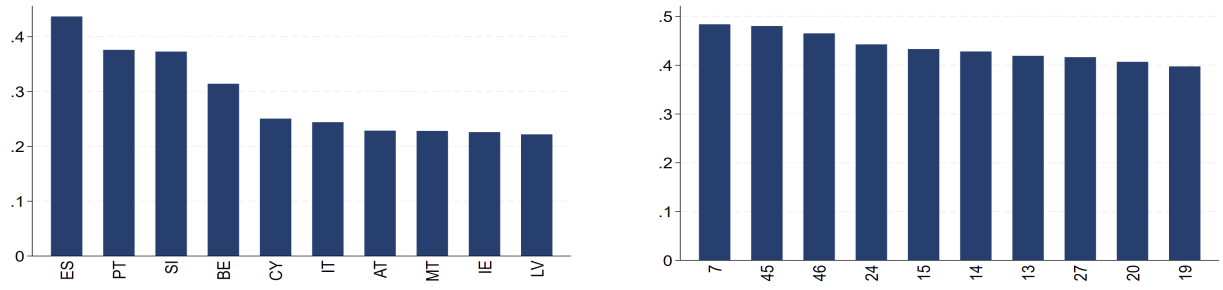


Figure 9: New Loan Maturity by rate type

Notes: Distribution of new credit across original-maturity buckets, separately for floating-rate and fixed-rate originations, September 2018 to December 2024. Maturity is the new-credit-weighted average original maturity of originations within a firm–bank–month; shares are computed over new credit and sum to 100 within each rate type. Fixed-rate credit is bimodal, concentrated at maturities of six months or less (30 percent) and beyond ten years (24 percent), while floating-rate credit is concentrated between one and seven years (57 percent against 32 percent for fixed-rate credit). Duration and coupon fixation are therefore not the same contractual margin.



(b) By country **(c) By sector (NACE2)**

Figure 10: Rollover Exposure by Country and Sector

Notes: Median short-term debt, the share of outstanding debt with residual maturity up to 12 months excluding credit lines, by country (left) and by sector, NACE2 (right), in 2018. Source: AnaCredit.

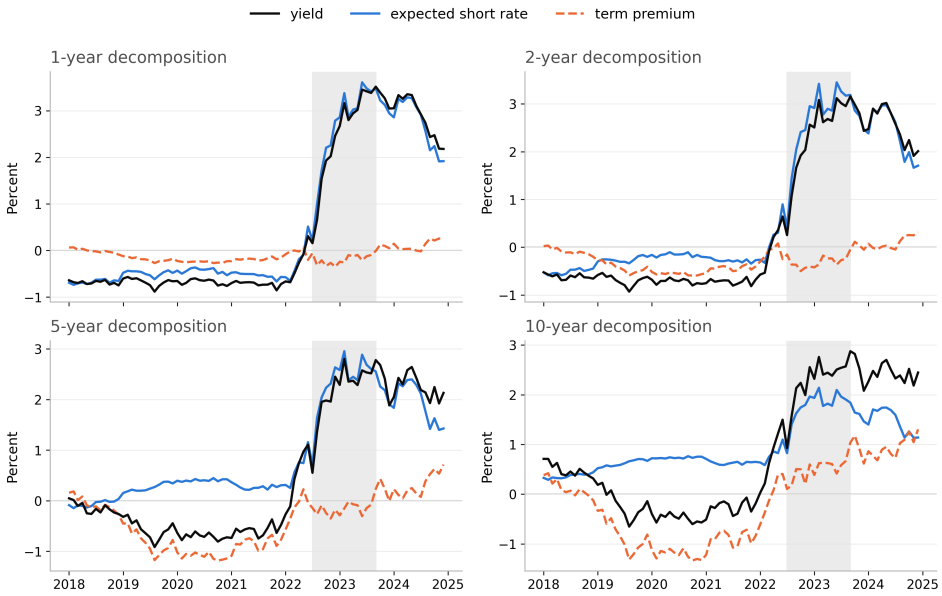
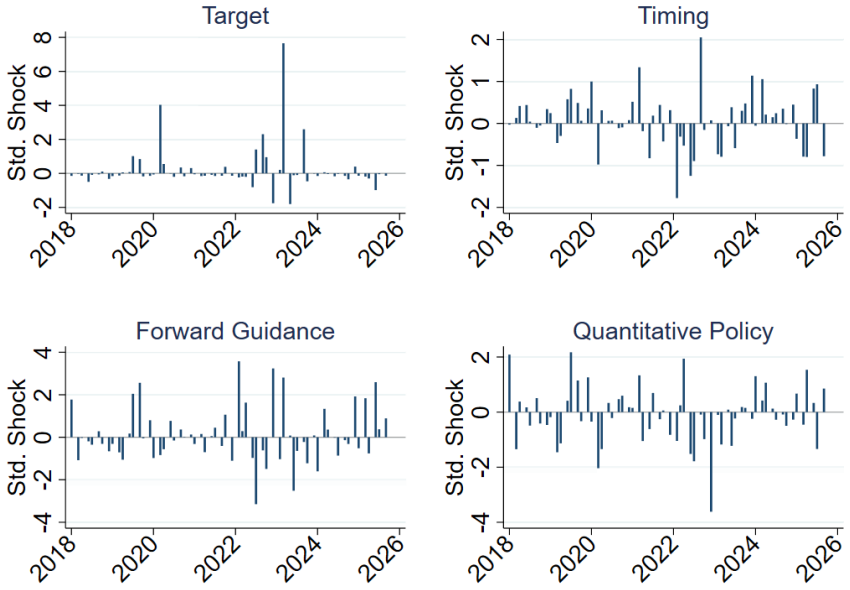


Figure 11: Yield Decomposition: Adrian, Crump, and Moench (2013)

Notes: Same decomposition as Figure 2, using the methodology of Adrian et al. (2013) in place of the affine model of Joslin et al. (2011). Euro-area OIS zero-coupon yields, monthly, September 2018 to December 2024; the shaded region denotes the ECB tightening cycle beginning in 2022.

Panel A. High-frequency monetary-policy surprises



Panel B. Pure monetary policy and information shocks

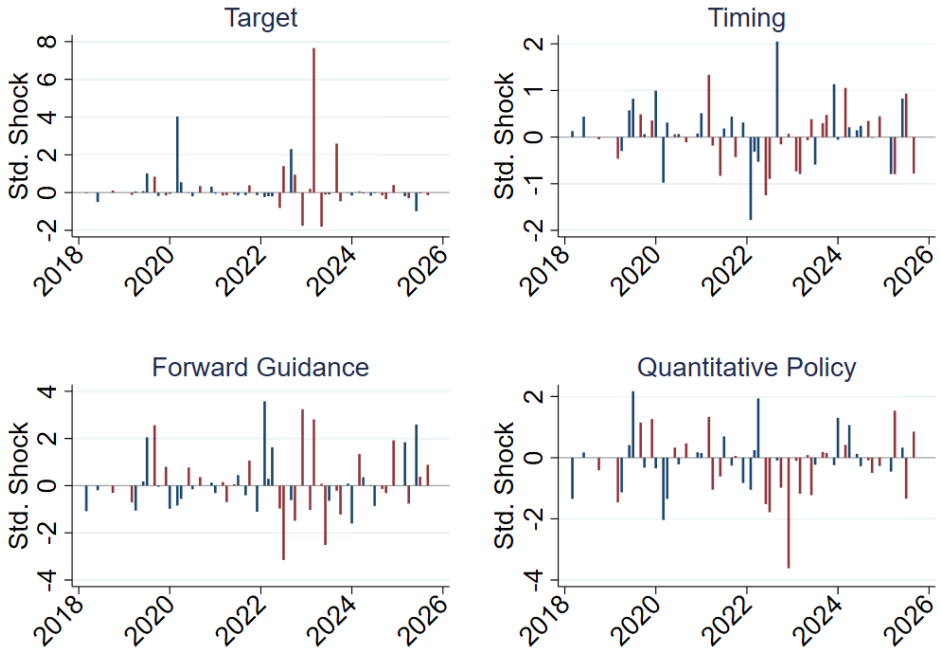


Figure 12: High-Frequency Monetary Policy Surprises

Notes: The figure reports monthly high-frequency monetary-policy surprises around narrow ECB announcements, following Altavilla et al. (2019). Panel A reports the Target, Timing, Forward Guidance, and Quantitative Policy factors. The *Target* factor captures unexpected changes in the current policy rate; *Timing* and *Forward Guidance* capture revisions to the expected policy path at short and medium horizons; and *Quantitative Policy* captures variation associated with asset- purchase policies and longer-horizon yields. Panel B separates monetary-policy events into *pure monetary policy shocks* (in blue) and *central-bank information shocks* (in red) (Jarociński and Karadi, 2020). Events are classified as pure monetary policy shocks when the 1-month OIS surprise and Euro Stoxx 50 equity prices move in opposite directions, and as central-bank information shocks when they move in the same direction. Positive monetary-policy surprises correspond to unexpected policy tightening.

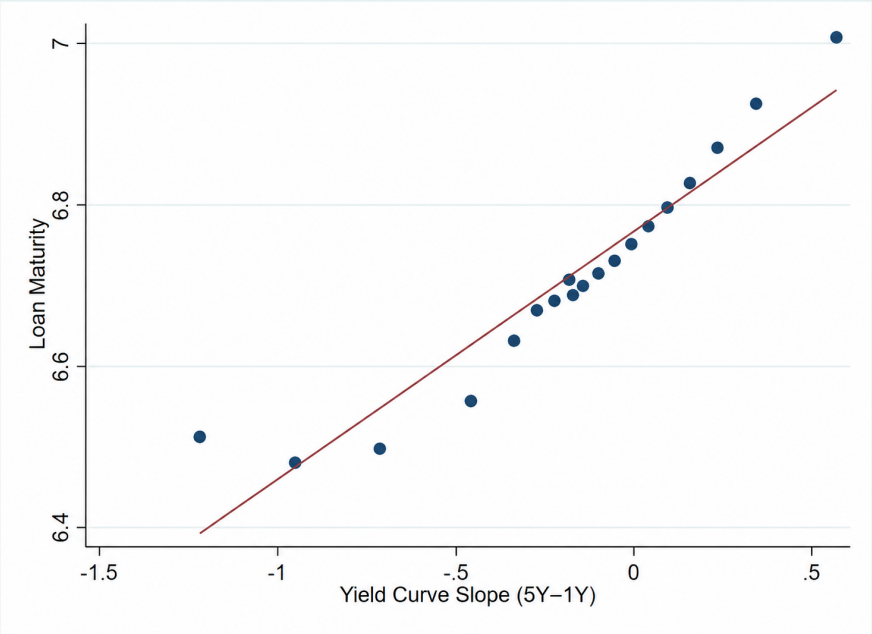


Figure 13: yield curve slope 5Y-1Y and new-loan maturity

Notes: Binscatter of average new loan maturity against the 5Y–1Y euro-area yield curve slope. Loan maturity is residualized on firm and bank fixed effects. AnaCredit, 2018–2024.

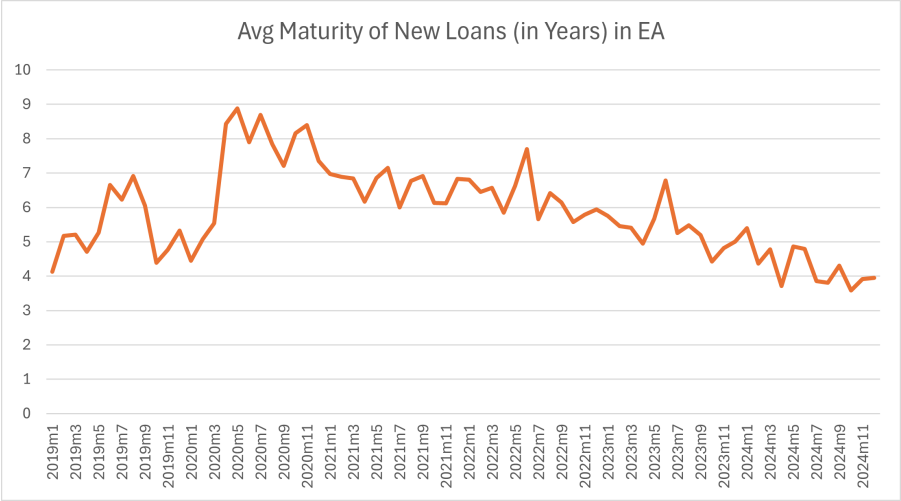


Figure 14: New Loan Maturity in the Euro Area

Notes: The figure reports the average maturity (in years) of newly originated bank loans to non-financial corporations in the euro area. The sample covers September 2018 through December 2024 and is based on AnaCredit.

A.3 Robustness to Frequent Policy Events

This appendix repeats the high-frequency analysis using the extended event-study database of Altavilla et al. (2026). Relative to the baseline EA-MPD, the extended database incorporates a broader set of monetary-policy events, including policymakers' communication between scheduled Governing Council meetings. We construct four mutually orthogonal components of monetary-

policy news. *Target* is the 1-month OIS surprise; *Timing* is the component of the 6-month OIS surprise orthogonal to Target; *Forward Guidance* is the component of the 2-year OIS surprise orthogonal to Target and Timing; and *Quantitative Policy* is the residual 10-year surprise orthogonal to the three shorter-horizon components. Announcement-level surprises are aggregated to the monthly frequency before being merged with the loan-level data.

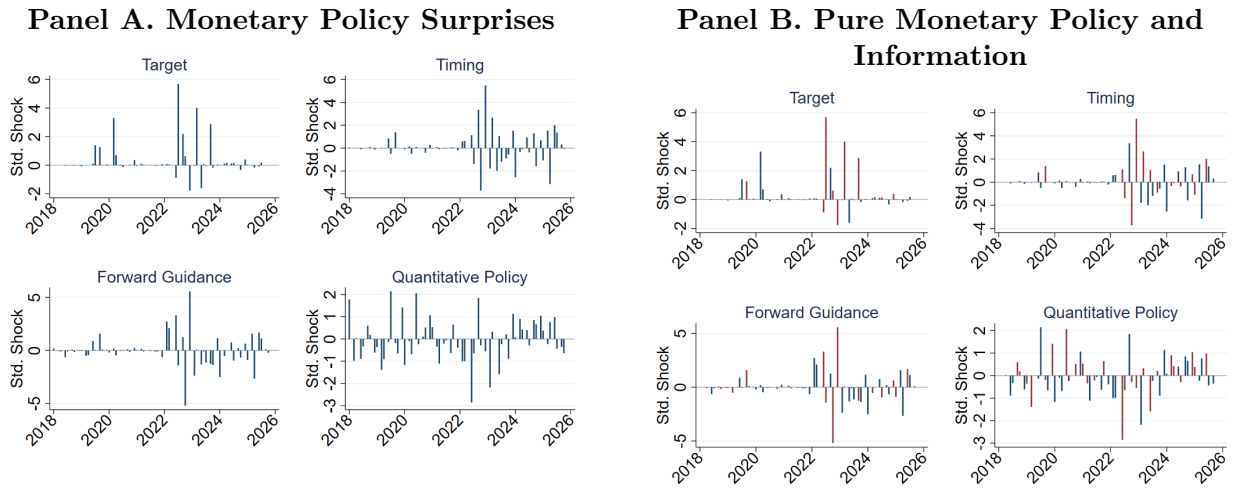


Figure 15: High-Frequency Monetary Policy Surprises: Extended Event Sample

Notes: Panel A reports the monthly Target, Timing, Forward Guidance, and Quantitative Policy surprises constructed from the extended event-study database of Altavilla et al. (2026). Panel B separates these surprises into pure monetary-policy and central-bank information components using the high-frequency co-movement of the 1-month OIS surprise and Euro Stoxx 50 equity prices (Jarociński and Karadi, 2020).

Table 29: Validation of High-Frequency Policy Factors

	1M	3M	6M	1Y	2Y	5Y	10Y
Target	3.190***	2.953***	2.640***	2.242***	1.792***	0.847***	0.147***
	(0.000)	(0.043)	(0.000)	(0.117)	(0.000)	(0.077)	(0.000)
Timing	-0.000	1.378***	1.993***	1.286***	0.000*	-0.647***	-0.490***
	(0.000)	(0.111)	(0.000)	(0.250)	(0.000)	(0.191)	(0.000)
Forward Guidance	-0.000	-0.642***	-0.000	2.274***	4.345***	4.632***	3.202***
	(0.000)	(0.127)	(0.000)	(0.226)	(0.000)	(0.193)	(0.000)
Quantitative Policy	0.000	0.078*	0.000	-0.120***	0.000	1.191***	2.074***
	(0.000)	(0.046)	(0.000)	(0.043)	(0.000)	(0.099)	(0.000)
Observations	128	128	128	128	128	128	128
R^2	1.000	0.984	1.000	0.983	1.000	0.959	1.000

Notes: The table reports the response of OIS yields at different maturities to a one-standard-deviation innovation in each high-frequency policy factor. The sample covers July 2011 through October 2025 and contains 128 ECB policy events. Target loads primarily on the short end of the curve. Timing is concentrated at short and intermediate maturities. Forward Guidance loads most strongly at intermediate and long maturities, while Quantitative Policy primarily raises longer-maturity yields. Standard errors are reported in parentheses. *, **, and *** denote significance at the 10%, 5%, and 1% levels, respectively.

Table 30: High-Frequency Policy Shocks and Loan Maturity: Extended Event Sample

$\log(\text{Maturity})_{b,f,t}$	(1) All Shocks	(2) Pure MP	(3) Information
$\text{Target}_{t-1} \times$ ST $\text{Debt}_{f,t-1}$	-0.00942***	-0.0363***	0.00249
	(0.00248)	(0.00651)	(0.00205)
$\text{Timing}_{t-1} \times$ ST $\text{Debt}_{f,t-1}$	0.00725***	0.0184***	0.00661***
	(0.00153)	(0.00373)	(0.00188)
Forward $\text{Guidance}_{t-1} \times$ ST $\text{Debt}_{f,t-1}$	-0.000200	-0.000974	0.00241
	(0.00159)	(0.00223)	(0.00209)
Quantitative $\text{Policy}_{t-1} \times$ ST $\text{Debt}_{f,t-1}$	-0.00756***	-0.0129***	-0.00199
	(0.00238)	(0.00366)	(0.00356)
Observations	1,674,254	1,674,254	1,674,254
Adjusted R^2	0.620	0.620	0.620
Bank × Month FE	Yes	Yes	Yes
Firm × Bank FE	Yes	Yes	Yes
Firm Controls	Yes	Yes	Yes

Notes: The dependent variable is the logarithm of maturity of newly originated loans. Short-term Debt is the lagged share of firm bank debt maturing within 12 months. Monetary-policy surprises are standardized by their monthly standard deviation. Column (1) uses the full set of high-frequency surprises. Columns (2)–(3) separate them into pure monetary-policy and central-bank information components using the high-frequency co-movement between the 1-month OIS surprise and Euro Stoxx 50 equity prices. All specifications include lagged firm controls, bank×month fixed effects, and firm×bank fixed effects. Standard errors are clustered at the firm and month level. *, **, and *** denote significance at the 10%, 5%, and 1% levels, respectively.

B Firm Problem and Optimality Conditions

This appendix states the firm's continuation problem and derives the first-order conditions for capital, total issuance, and debt maturity. Firm and time subscripts are suppressed. The arguments of debt-price schedules are also suppressed when there is no ambiguity.

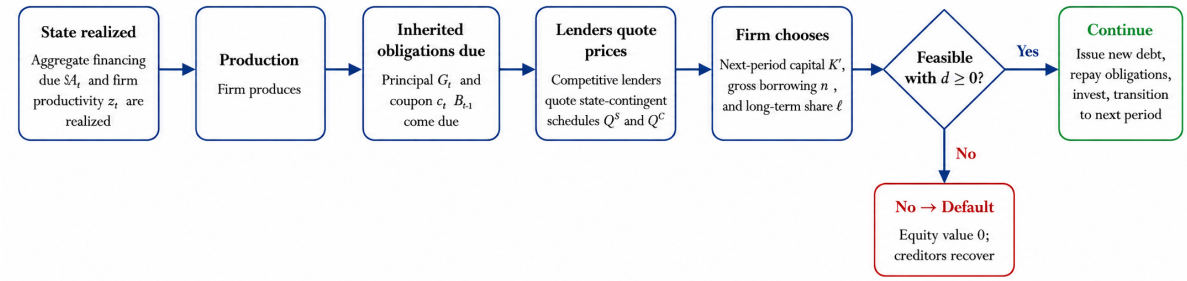


Figure 16: Model Timing

Notes: The figure summarizes the within-period timing of decisions and shocks in the model.

B.1 Firm problem

Given the firm state

$$x = (z, K, B, G) \tag{40}$$

and aggregate financing conditions

$$A = (r, f, \omega) \tag{41}$$

a continuing firm chooses next-period capital, gross new borrowing, and the long-term share of new borrowing:

$$K' \geq 0, \qquad m \geq 0, \qquad \ell \in [0, 1] \tag{42}$$

New short- and long-term debt are

$$M^S = (1 - \ell)m, \qquad M^L = \ell m \tag{43}$$

Thus, $M^S \geq 0$ and $M^L \geq 0$, and the benchmark does not allow early repayment of surviving long-term debt. When $m = 0$, the value assigned to ℓ is immaterial.

Investment is

$$I = K' - (1 - \delta_K)K \tag{44}$$

Total debt evolves according to

$$B' = B - G + m \tag{45}$$

Next-period principal due is

$$G' = (1 - \rho_L)B' + \rho_L(1 - \ell)m \tag{46}$$

Equivalently,

$$G' = (1 - \rho_L)(B - G) + (1 - \rho_L\ell)m \tag{47}$$

The relevant derivatives of the state transition are

$$\frac{\partial B'}{\partial m} = 1, \qquad \frac{\partial G'}{\partial m} = 1 - \rho_L\ell \tag{48}$$

and

$$\frac{\partial B'}{\partial \ell} = 0, \qquad \frac{\partial G'}{\partial \ell} = -\rho_L m \tag{49}$$

Dividends are

$$d = \pi(z, K) - G - c_L \frac{B - G}{\rho_L} - I - \Phi_K(I, K) - H(m) + m\left[(1 - \ell)Q^S + \ell Q^L\right] \tag{50}$$

The coupon term follows from

$$B^L = \frac{B - G}{\rho_L} \tag{51}$$

The firm cannot issue external equity:

$$d \geq 0 \tag{52}$$

Conditional on continuation, the value function is

$$V(x; A) = \max_{K', m, \ell} \left\{ d + \beta \mathbb{E}_t \left[\mathcal{M}_{t,t+1} V(z', K', B', G'; A') \right] \right\} \tag{53}$$

subject to (42), (45), (46), and (52). The benchmark sets

$$\beta = 1 - \kappa, \qquad \mathcal{M}_{t,t+1} = e^{-r_t} \tag{54}$$

where β is the reduced-form equity-horizon wedge of Section 6: shareholders and lenders use the same stochastic discount factor, and shareholders additionally discount continuation payoffs by β. The wedge is a valuation object only—it generates no exit transition in the firm distribution and does not enter creditor pricing.

Let $\mu \geq 0$ denote the multiplier on the no-equity constraint. For an interior solution, the Lagrangian is

$$\mathcal{L} = (1 + \mu)d + \beta \mathbb{E}_t \left[\mathcal{M}_{t,t+1} V(z', K', B', G'; A') \right] \tag{55}$$

with complementary slackness

$$\mu d = 0 \tag{56}$$

The conditions below apply at differentiable interior choices with

$$K' > 0, \qquad m > 0, \qquad \ell \in (0, 1) \tag{57}$$

At a boundary, the corresponding equality is replaced by the appropriate Kuhn–Tucker inequality.

For compactness, define

$$\Phi_I(I, K) \equiv \frac{\partial \Phi_K(I, K)}{\partial I} = \psi_K \frac{I}{K} \tag{58}$$

and

$$H'(m) = \eta m \tag{59}$$

B.2 Capital choice

An increase in K' raises investment one-for-one and may change debt prices through its effect on next-period default risk. The first-order condition for capital is

$$0 = (1+\mu)\left[-1 - \Phi_I(I,K) \; + m\left((1-\ell)\frac{\partial Q^S}{\partial K'} + \ell\frac{\partial Q^L}{\partial K'}\right)\right] + \beta\mathbb{E}_t\left[\mathcal{M}_{t,t+1}V_K(z',K',B',G';A')\right] \tag{60}$$

Equivalently,

$$1 + \Phi_I(I,K) = m\left[(1-\ell)\frac{\partial Q^S}{\partial K'} + \ell\frac{\partial Q^L}{\partial K'}\right] + \frac{\beta}{1+\mu}\mathbb{E}_t\left[\mathcal{M}_{t,t+1}V_K(z',K',B',G';A')\right] \tag{61}$$

The left-hand side is the current marginal resource cost of investment. The right-hand side contains the effect of capital on current borrowing terms and the discounted continuation value of additional installed capital.

B.3 Total issuance

An additional unit of gross borrowing raises total debt by one and next-period principal due by $1-\rho_L\ell$. The first-order condition for gross issuance is

$$\begin{aligned} 0 = (1+\mu)\left[(1-\ell)Q^S + \ell Q^L - H'(m) \; + m\left((1-\ell)\frac{\partial Q^S}{\partial m} + \ell\frac{\partial Q^L}{\partial m}\right)\right] \\ + \beta\mathbb{E}_t\left[\mathcal{M}_{t,t+1}\left\{V_B(z',K',B',G';A') + (1-\rho_L\ell)V_G(z',K',B',G';A')\right\}\right] \end{aligned} \tag{62}$$

The firm equates the current proceeds from additional borrowing, net of issuance costs and endogenous price effects, to the continuation cost of higher total debt and higher future principal obligations.

If debt prices are locally insensitive to the size of the candidate issuance, the condition simplifies to

$$(1-\ell)Q^S + \ell Q^L - H'(m) = -\frac{\beta}{1+\mu}\mathbb{E}_t\left[\mathcal{M}_{t,t+1}\left\{V_B + (1-\rho_L\ell)V_G\right\}\right] \tag{63}$$

B.4 Maturity choice

Holding gross issuance fixed, a higher ℓ replaces short-term borrowing with long-term borrowing. It leaves total debt unchanged but lowers next-period principal due by $\rho_L m$. The first-order condition for maturity is

$$0 = (1+\mu)m\left[Q^L - Q^S \; + (1-\ell)\frac{\partial Q^S}{\partial \ell} + \ell\frac{\partial Q^L}{\partial \ell}\right] - \beta\rho_L m\mathbb{E}_t\left[\mathcal{M}_{t,t+1}V_G(z',K',B',G';A')\right] \tag{64}$$

For $m > 0$, this can be written as

$$Q^S - Q^L - (1-\ell)\frac{\partial Q^S}{\partial \ell} - \ell\frac{\partial Q^L}{\partial \ell} = -\frac{\beta\rho_L}{1+\mu}\mathbb{E}_t\left[\mathcal{M}_{t,t+1}V_G(z', K', B', G'; A')\right] \tag{65}$$

The left-hand side is the current marginal funding cost of replacing short-term debt with long-term debt. The right-hand side is the continuation value of the associated reduction in next-period principal due.

If debt prices are locally insensitive to the maturity composition of the candidate issuance, the condition simplifies to

$$Q^S - Q^L = -\frac{\beta\rho_L}{1+\mu}\mathbb{E}_t\left[\mathcal{M}_{t,t+1}V_G(z', K', B', G'; A')\right] \tag{66}$$

When inherited principal obligations reduce firm value,

$$V_G < 0 \tag{67}$$

the right-hand side is positive. The firm then accepts a lower current price for long-term debt in exchange for lower future refinancing needs.

A higher expected-rate state f may increase this insurance value by raising expected future refinancing costs. It can also affect the current valuation of long-duration claims through expected future discount rates. By contrast, the long-duration funding component directly affects the borrower price

$$\left.\frac{\partial Q^L}{\partial \omega}\right|_{Q^L} = -Q^L < 0 \tag{68}$$

The maturity response to either aggregate financing shock is therefore an equilibrium outcome rather than an imposed sign restriction.

C Numerical Algorithm

The model is solved on a discretized state space $x = (z, K, B, s)$, $A = (r, f, \omega)$, where $s = G/B$ replaces G as the debt-structure coordinate (equation (16)). Choices of (K', B', ℓ) are restricted to grid points, so continuation values are interpolated only in the exogenous states.

Grids. Productivity uses a five-node Rouwenhorst discretization. Capital and debt are log-spaced, with 11 nodes on $[0.20, 2.00]$ for capital and 21 nodes for debt including an exact zero, giving uniform relative steps of 26 and 21 percent; log spacing is used because the estimated adjustment and issuance costs are identified by relative growth moments, which a uniform grid would resolve well at the top of the grid and poorly at the bottom. Rollover exposure s has eight log-spaced nodes on $[1-\rho_L, 1]$ and the maturity share ℓ nine uniform nodes on $[0, 1]$. The aggregate states use uniform grids with seven nodes on $[0.001, 0.045]$ for r, five on $[-0.018, 0.018]$ for f, and seven on $[0.005, 0.095]$ for ω.

Expectations. Expectations over next-period aggregate states use Gauss–Hermite quadrature with three nodes per innovation in (13)–(15), 27 joint nodes per state, with next-period states clipped to the grid and continuation objects interpolated. The same rule is used for every result in the paper.

Backward induction. Because default is determined by feasibility and debt prices depend on next-period default, prices and default are coupled within a period. We solve a long finite-horizon economy backward from a terminal date at which $V = 0$, default probabilities are zero, and long-debt values equal their risk-free counterparts. At each earlier date we construct Q^S, $\widetilde{Q}^L$, and Q^L from the already determined date-t+1 value function, default rule, and claim values using the lenders' break-even conditions (30) and (31); solve the date-t firm problem state by state, with default when no choice satisfies (24); and carry $(V, D, \widetilde{Q}^L)$ back one period as a damped average with the previous iterate, re-optimizing policies against the damped continuation objects. Iteration stops when values, prices, the default rule, and every cross-sectional policy aggregate are unchanged, after which the long-debt price recursion is re-solved with policies and the default rule frozen to drive the residual to tolerance. The terminal condition dies out geometrically at rate βe^{-r} per step, and lengthening the horizon leaves policies and moments unchanged.

Simulation. The stationary distribution is computed by Monte Carlo panel simulation, with exposure tracked continuously between grid nodes and policies evaluated at the nearest node. Aggregate states are drawn from their estimated processes, clipped to the grid, and mapped to the nearest node. Defaulting firms are replaced by entrants with capital K_0, zero debt, and productivity drawn from its invariant distribution. Counterfactual economies are re-solved in full, prices included, and simulated under common shock paths from the benchmark stationary distribution.

D Measurement of Aggregate Processes

D.1 Yield Decomposition

To distinguish changes in long-term yields due to expected future short rates from changes in term premia, we decompose euro-area zero-coupon yields using the affine term-structure model of Joslin et al. (2011). We use the Adrian et al. (2013) decomposition as an alternative measure. The two approaches differ in how they identify risk compensation, but both express an n-period yield as

$$y_t^{(n)} = ER_t^{(n)} + TP_t^{(n)} \tag{69}$$

where

$$ER_t^{(n)} = \frac{1}{n} \sum_{j=0}^{n-1} \mathbb{E}_t^P[r_{t+j}] \tag{70}$$

is the average expected short rate over the life of the bond and $TP_t^{(n)}$ is the corresponding term premium.

JSZ decomposition. The Joslin et al. (2011) approach summarizes the zero-coupon yield curve with a small set of pricing factors, X_t. Factor dynamics under the physical measure follow

$$X_{t+1} = \mu^P + \Phi^P X_t + \Sigma \varepsilon_{t+1} \tag{71}$$

and the short rate is affine in the factors,

$$r_t = \delta_0 + \delta_1' X_t \tag{72}$$

No-arbitrage restrictions map these factor dynamics into zero-coupon bond prices. The expected-rate component is obtained from forecasts of future short rates under the physical measure, and the term premium is the portion of the observed long yield not accounted for by those expected short rates.

ACM decomposition. As a robustness check, we use the methodology of Adrian et al. (2013). ACM also summarizes the yield curve using a small number of factors but identifies time-varying prices of risk from the predictability of excess bond returns. For an n-period bond, excess returns are given by

$$rx_{t+1}^{(n)} = p_{t+1}^{(n-1)} - p_t^{(n)} - r_t \tag{73}$$

and are related to innovations in the yield factors and to their lagged values. These estimates identify the compensation investors require for bearing yield curve risk. The expected-rate component is then the portion of the yield implied by expected future short rates, with the residual attributed to the term premium.

Despite their different identification strategies, the JSZ and ACM decompositions produce very similar expected-rate and term-premium series in our sample. Table 2 reports the corresponding maturity estimates under both decompositions.

Table 31: Mapping the Model to the Data

Empirical Object	Model Object	Mapping
Rollover exposure Debt due within 1y / total bank debt	Principal due, $s = G/B$	Both count principal due within the year;
Expected-rate component 5Y JSZ	Rate states (r_t, f_t)	$\mathrm{ESR}_t^{5Y} = \frac{1}{5}\sum_{j=1}^{5} \mathbb{E}_t[r_{t+j}]$
Term premium 5Y JSZ	Term-premium wedge ω_t	$TP_t = y(e^{-\omega_t} Q_t^E) - y(Q_t^E)$; ω_t is inverted period by period to match the observed term premium
Maturity outcomes Log original maturity; LT share of new loans	LT share of new issuance, ℓ	$\mathrm{Mat} = (1-\ell)\cdot 1 + \ell \cdot \frac{1}{1-\rho_L}$; model outcomes are log(Mat) and ℓ

Notes: For the long-term claim, $y(Q) = \log\left(\rho_L + \frac{c_L + 1 - \rho_L}{Q}\right)$. Both maturity outcomes are transformations of the same underlying maturity choice and are reported to match the two empirical specifications rather than as independent model moments. Differences in log-maturity units up to an additive constant are absorbed by fixed effects.

D.2 Model financing states

The model's aggregate block is estimated at annual frequency from monthly series. The current-rate process is an AR(1) estimated on the one-year OIS rate. The news state is the five-year

expected-rate-path component orthogonalized to the current rate; monthly persistence is converted to annual as $\rho_a = \rho_m^{12}$, with the innovation variance point-sampled accordingly.

D.3 The term-premium wedge measurement equation

The empirical long-duration component is observed in yield units, whereas ω_t is a price wedge. We map the yield decomposition into the model using the duration of the model's long-term claim. Let Q_t^E denote the price of its risk-free counterpart when only the expected path of short rates varies:

$$Q_t^E = e^{-r_t}\mathbb{E}_t\left[c_L + 1 - \rho_L + \rho_L Q_{t+1}^E\right] \tag{74}$$

Q_t^E is the expectations-only counterpart of the claim value $\widetilde{Q}^L$ in (31); the corresponding issuance price after the wedge is $Q_t^D = e^{-\omega_t}Q_t^E$, as in Section 6.4. For a geometrically amortizing claim, the continuously compounded yield associated with price Q is

$$y(Q) = \log\left(\rho_L + \frac{c_L + 1 - \rho_L}{Q}\right) \tag{75}$$

The measurement equation sets the observed long-duration yield component equal to the yield differential generated by the model wedge:

$$TP_t = y\big(e^{-\omega_t}Q_t^E\big) - y(Q_t^E) \tag{76}$$

We apply (76) to deviations of the term premium from its sample mean, which identifies movements in ω_t and hence its law of motion (15); the level $\bar{\omega}$ is preset, as discussed in Section 7. Because (74) is linear in $c_L + 1 - \rho_L$, Q_t^E is proportional to this cash-flow term, which cancels from the yield differential. Hence the recovered ω_t depends on the duration parameter ρ_L and the expected short-rate path, but not on the coupon normalization c_L.

D.4 Firm-level productivity

Firm-level productivity is estimated using the matched AnaCredit–Orbis panel. Given the production function

$$Y_{fct} = \bar{z} z_{fct} K_{fct}^{\alpha}$$

we measure productivity using Orbis sales for output and fixed assets for capital. With $\alpha = 0.60$, log productivity is recovered from

$$\log(\text{Sales}_{fct}) = \alpha \log(K_{fct}) + \delta_c + \delta_s + \log z_{fct}$$

where δ_c and δ_s denote country and sector fixed effects.

We then estimate

$$\log z_{fct} = \rho_z \log z_{fc,t-1} + \varepsilon_{fct}^z$$

Following the literature, we instrument lagged productivity with its second lag to correct for transitory measurement error. The estimated coefficient gives the persistence ρ_z, while σ_z is the standard deviation of the residuals ε_{fct}^z.

E Debt Maturity Choice: Mechanisms and Diagnostics

This appendix provides additional evidence on the mechanisms underlying firms' maturity choices. We first examine how inherited rollover exposure affects refinancing risk, then investigate the maturity-switching incentives generated by expected-rate news and term-premium shocks. Finally, we document how the sensitivity of debt prices to issuance varies with rollover exposure.

E.1 Rollover Exposure and Maturity Switching

Rollover exposure and refinancing risk. Because rollover exposure is endogenous, its unconditional relationship with default risk also reflects firms' characteristics and past financing choices. Figure 17 shows that this relationship is non-monotonic in the equilibrium distribution (Panel A). However, holding productivity, capital, debt, and the refinancing plan fixed, next-period default risk increases sharply with inherited rollover exposure (Panel B). This confirms that greater repayment obligations increase the underlying value of refinancing insurance, even when the unconditional relationship is obscured by firm selection.

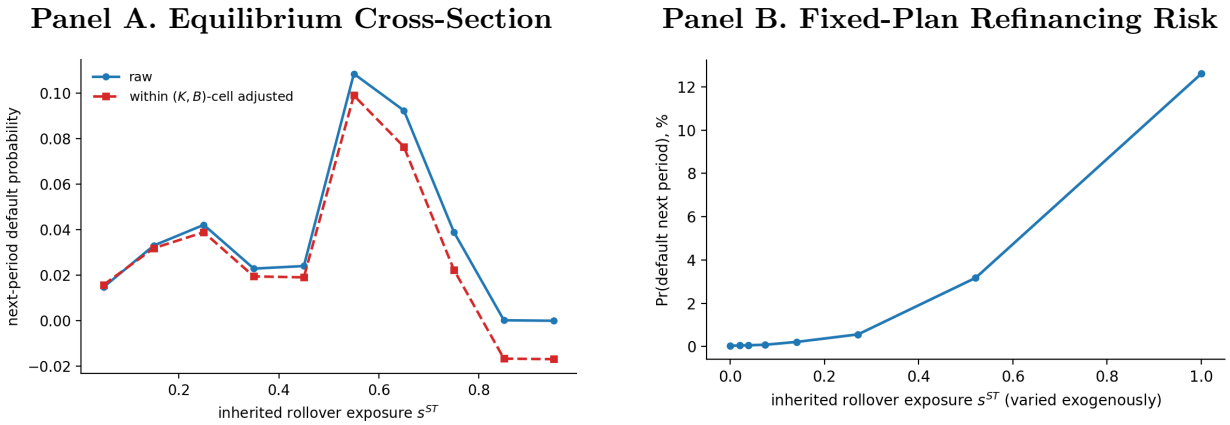


Figure 17: Rollover Exposure, Refinancing Risk, and Equilibrium Selection. Panel A reports default risk across the equilibrium distribution of rollover exposure. Because exposure reflects both refinancing needs and firms' past financing choices, its unconditional relationship with default need not be monotone. The adjusted series residualizes composition within (K, B) cells. Panel B averages next-period default probabilities at a fixed refinancing plan over the stationary composition of firm states, holding each firm's (z, K, B) fixed as inherited exposure varies, at the mean aggregate financing state. Exposure is measured on the horizontal axis as the share of debt in one-period claims, s^{ST}, which maps into $s = (1-\rho_L)+\rho_L s^{ST}$. Refinancing risk rises sharply with inherited exposure.

From marginal incentives to maturity switching. Section 7.3 shows that expected-rate news and term-premium shocks both increase the current cost of long-term borrowing, but differ in their effects on the insurance value of postponing refinancing. To examine how these differences affect maturity incentives, we compare the values of all-long and all-short issuance at a common refinancing plan ($K' = K$, $B' = B$). A firm is classified as switching when it prefers long-term issuance before the shock but short-term issuance afterward.

Table 32 reports the results. Panel A evaluates firms at their observed equilibrium exposure levels, where maturity preferences also reflect differences in firm composition. Panel B instead places the same firms at each exposure level, holding productivity, capital, and debt fixed. Following

expected-rate news, the share switching from long to short declines from 20.6% at the amortization floor to 17.9% at high exposure. Following a term-premium shock, it instead increases from 34.5% to 42.1%.

These opposite patterns arise even with the same underlying firm composition, reflecting the interaction between inherited rollover exposure, changes in relative borrowing incentives, and firms' initial maturity preferences. Because this exercise compares the two maturity corners at a fixed refinancing plan, rather than firms' optimal maturity shares, it serves as a diagnostic of the mechanism. The corresponding responses of the solved maturity policies are reported in Section 7.3.

Table 32: Maturity Switching by Rollover Exposure

	Mass	Long preferred	Share switching (%)		Of initially long (%)	
Exposure group	(%)	at baseline (%)	News	Term premium	News	Term premium
Panel A. Firms observed at their equilibrium exposure						
Floor	49.6	49.5	23.0	43.2	46.4	87.2
Intermediate	26.8	30.9	17.9	24.7	57.5	78.9
High	23.6	54.5	10.6	35.4	19.7	64.7
Panel B. The same firms placed at every exposure level						
Floor	–	43.4	20.6	34.5	47.5	79.5
Intermediate	–	43.6	20.9	34.9	47.8	80.0
High	–	55.4	17.9	42.1	32.8	76.3

Notes: The underlying incentive is the value of all-long relative to all-short issuance at the common refinancing plan ($K' = K$, $B' = B$), evaluated at the mean aggregate state and after an increase of one unconditional standard deviation in expected-rate news or the term-premium wedge, with other aggregate states at their means. A firm switches when the long-minus-short incentive is positive before the shock and negative afterward. "Share switching" is measured relative to all firms in the exposure group, while "Of initially long" conditions on initially preferring long-term issuance. Panel A uses issuing, non-defaulting firm-years from simulated histories at their observed exposure nodes. Panel B places the same pooled firms at every exposure node, holding productivity, capital, and debt fixed. Floor denotes $s = 1 - \rho_L = 0.19$; Intermediate denotes $0.21 \leq s \leq 0.25$; High denotes $s \geq 0.31$. Intermediate and High statistics aggregate the underlying exposure nodes using stationary issuing mass. The exercise evaluates maturity-corner values at a common refinancing plan, not the firm's solved maturity policy.

E.2 Debt-Price Sensitivity to Issuance

The cost of maturity protection also depends on endogenous debt pricing. Because long-term claims are priced recursively, additional borrowing affects both future default risk and the value of surviving claims. To quantify this sensitivity, we perturb gross issuance around the stationary policy by varying next-period debt one grid point above and below its optimal level, holding capital and the maturity share fixed. We compute the finite-difference elasticity

$$\varepsilon^j = \frac{\Delta \log Q^j}{\Delta \log m}, \qquad j \in \{S, L\}, \tag{77}$$

where m denotes gross issuance.

Table 33 shows that long-term debt prices are substantially more sensitive to issuance than short-term prices. The average absolute elasticity is 0.092 for long-term debt, compared with 0.014 for short-term debt, and the long-term elasticity exceeds the short-term elasticity for 98% of stationary issuing mass. Moreover, long-term price sensitivity increases from 0.063 at the amortization floor

to 0.182 among firms with high rollover exposure, compared with 0.006 and 0.040 for short-term debt, respectively.

Thus, firms with greater refinancing needs face steeper endogenous financing costs when expanding long-term borrowing. These issuance-price elasticities reinforce the cost side of maturity insurance, but are distinct from the maturity-share derivatives governing optimal maturity choice.

Table 33: Debt-Price Sensitivity to Gross Issuance

	Issuing mass	Mean $-\varepsilon^S$	Mean $-\varepsilon^L$	Share $\lvert\varepsilon^L\rvert > \lvert\varepsilon^S\rvert$
All issuers	1.00	0.014	0.092	0.98
At amortization floor	0.46	0.006	0.063	0.99
High rollover exposure	0.27	0.040	0.182	0.94

Notes: $-\varepsilon^j = -\Delta \log Q^j / \Delta \log m$ measures the sensitivity of the price of claim $j \in \{S, L\}$ to an increase in gross issuance. Statistics are weighted by the stationary distribution among issuing, non-defaulting firms at the mean aggregate state. "At amortization floor" denotes $s = 1-\rho_L$; "High rollover exposure" denotes $s > 0.22$, the upper half of the above-floor exposure distribution. Because adjacent debt-grid points imply sizable changes in issuance, the reported objects are secant rather than local elasticities.

F Additional Model Counterfactuals and Diagnostics

This appendix supports the counterfactual exercises in Section 8. It reports the full dynamics of the main counterfactual and the complete results for the separation of maturity from interest-rate fixation. It then examines the scope of the main result: how the comparison changes when each economy starts from its own stationary distribution, whether the investment effects reflect survivorship, and the switching evidence behind the opposite interaction signs together with selection into exposure states. It closes with the sensitivity of debt prices to issuance, which underlies the endogenous cost of obtaining duration.

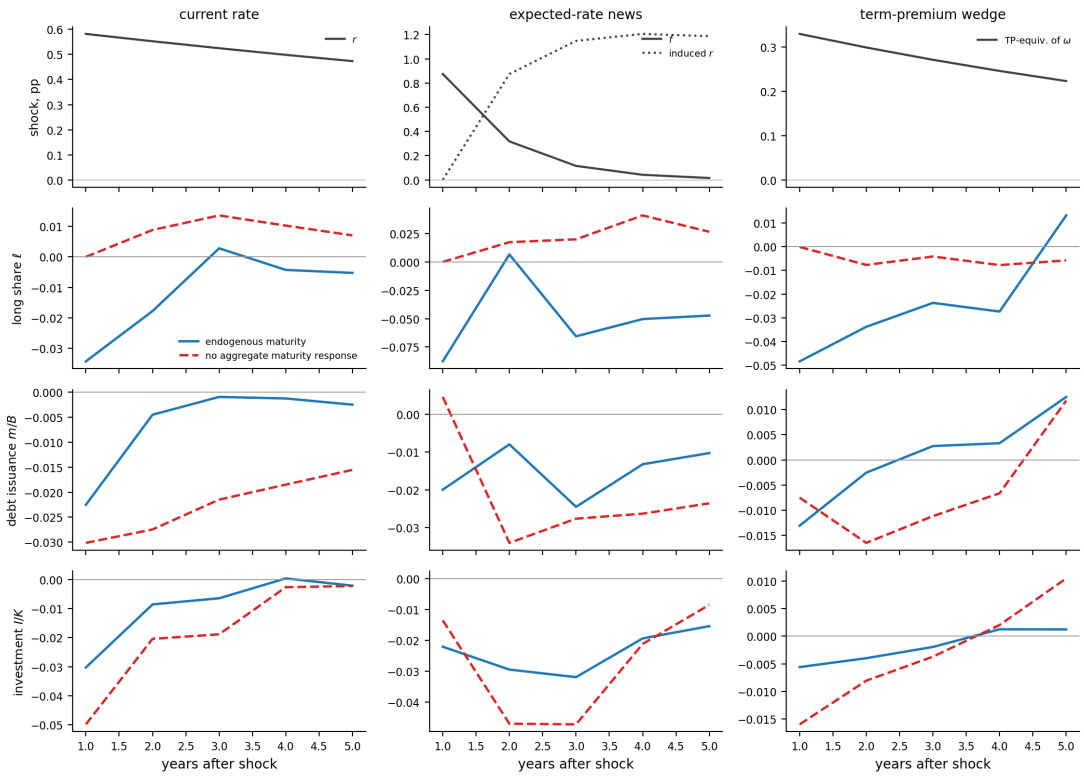


Figure 18: Aggregate Maturity Adjustment: Full Dynamics. The figure reports shock responses in the benchmark economy and the counterfactual that removes the direct dependence of maturity choice on aggregate financing conditions. The top row reports the aggregate shocks, including the short-rate path induced by expected-rate news; ω is expressed in term-premium-equivalent yields. The remaining rows report the long-term share of new issuance, gross debt issuance, and investment. Solid lines denote the benchmark; dashed lines denote the restricted economy. Results are averaged across 1,000 simulated economies.

F.1 Maturity Adjustment: Full Dynamics

Figure 18 reports the impulse responses underlying Table 13. Investment and debt issuance diverge immediately between the benchmark and restricted economies following current-rate and term-premium shocks; following the term-premium shock the investment gap closes by year four, as the wedge decays and the restricted economy's long-dated issuance stops being repriced. Following expected-rate news, the investment difference emerges primarily as the anticipated increase in short rates materializes.

F.2 Maturity and Interest-Rate Fixation

This appendix reports the full results for the floating-coupon experiment in Section 8.2. Long-term debt retains the benchmark amortization schedule, but its coupon resets with the short rate. We repeat the maturity-flexibility counterfactual of Section 8.1, comparing economies with endogenous and restricted maturity adjustment while allowing coupons to float.

Table 34 shows that maturity flexibility continues to protect firms against default even without coupon fixation. Relative to restricted maturity, endogenous maturity adjustment reduces five-year cumulative default by 1.2, 4.2, and 1.9 percentage points following current-rate, expected-rate news,

Table 34: Maturity Flexibility in the Floating-Coupon Economy

Horizon (years)	1	3	5
Investment (cumulative investment / capital, pp)			
Current rate	+0.5 (0.2)	+0.3 (0.2)	+0.1 (0.2)
Expected-rate news	+4.2 (0.3)	−1.1 (0.4)	−1.4 (0.3)
Term-premium wedge	+3.2 (0.3)	+0.5 (0.3)	+0.1 (0.1)
Debt issuance (cumulative gross issuance / outstanding debt, pp)			
Current rate	−0.1 (0.3)	+4.5 (0.5)	+10.1 (0.7)
Expected-rate news	+0.9 (0.4)	+2.9 (0.7)	+7.2 (0.9)
Term-premium wedge	+1.9 (0.3)	+8.2 (0.6)	+11.8 (0.7)
Credit spread (average through horizon, pp)			
Current rate	−0.2 (0.7)	−1.6 (0.3)	−1.7 (0.3)
Expected-rate news	−3.9 (1.1)	−4.5 (0.3)	−3.1 (0.2)
Term-premium wedge	−2.9 (0.4)	−2.6 (0.1)	−2.4 (0.1)
Long-term issuance share (level response, pp)			
Current rate	−16.8 (1.0)	−10.8 (0.5)	−7.1 (0.3)
Expected-rate news	−29.8 (1.4)	−10.9 (0.4)	−9.4 (0.3)
Term-premium wedge	−20.9 (1.1)	−8.8 (0.6)	+0.7 (0.5)
Default (cumulative incidence, pp)			
Current rate	—	−0.8 (0.2)	−1.2 (0.2)
Expected-rate news	—	−2.4 (0.3)	−4.2 (0.3)
Term-premium wedge	—	−0.7 (0.2)	−1.9 (0.1)

Notes: Entries report differences in shock responses between the economies named in each panel (first minus second). Floating-coupon debt retains the benchmark amortization schedule but reprices with the short rate. Investment, gross issuance, and default are cumulative; credit spreads are averaged through each horizon, while the long-term issuance share is measured at each horizon. Default is recorded beginning in the period after the shock. The horizon-1 default entries are therefore omitted. All economies start from the fixed-coupon benchmark's stationary cross-section and face identical idiosyncratic draws. Policies and debt prices are re-solved in each economy. Results average 1,000 economies of 10,000 firms following one-standard-deviation shocks with no subsequent aggregate innovations. Monte Carlo standard errors are in parentheses.

and term-premium shocks, respectively. The effect is particularly pronounced following expected-rate news: although existing floating-rate debt reprices as short rates rise, firms can still adjust when principal must be refinanced.

The financing and investment effects are more nuanced. Maturity flexibility induces larger adjustments in new borrowing when coupons float. In the first year, the long-term issuance share falls by 16.8, 29.8, and 20.9 percentage points more under endogenous than restricted maturity following current-rate, expected-rate news, and term-premium shocks, respectively. Gross issuance

is also higher under endogenous maturity at the five-year horizon across all three shocks. Average credit spreads are lower under maturity flexibility, with five-year differences of 1.7, 3.1, and 2.4 percentage points, respectively. The investment response is less persistent than the survival benefit. Following expected-rate news, maturity flexibility raises cumulative investment by 4.2 percentage points in the first year, but the difference reverses to -1.4 percentage points by year five. Thus, the reduction in default does not necessarily imply higher cumulative investment over longer horizons.

A complementary comparison holds the maturity regime fixed and varies coupon fixation. Following expected-rate news, fixed coupons reduce five-year cumulative default by 2.0 percentage points when firms can adjust maturity, but by 3.4 percentage points when maturity adjustment is restricted. The corresponding reductions following a current-rate shock are 1.1 and 2.5 percentage points. Coupon fixation therefore provides greater survival protection when firms have less scope to restructure principal repayments.

Together, these results establish that maturity adjustment and coupon fixation provide distinct but partially substitutable forms of insurance. Maturity determines when principal must be refinanced, whereas coupon fixation determines how outstanding debt payments respond to changes in short-term rates.

Table 35: Value of Maturity Flexibility: Expected-Rate and Current-Rate Shocks with a Lower Term Premium

Horizon (years)	1	3	5
Investment (cumulative investment / capital, pp)			
News with lower term premium	+0.2 (0.2)	+2.7 (0.2)	+3.1 (0.2)
Current rate with lower term premium	+0.5 (0.3)	+3.3 (0.2)	+4.0 (0.2)
Debt issuance (cumulative gross issuance / outstanding debt, pp)			
News with lower term premium	−0.9 (0.3)	−0.3 (0.5)	+1.2 (0.7)
Current rate with lower term premium	+0.5 (0.4)	+2.7 (0.4)	+4.7 (0.3)
Long-term issuance share (level response, pp)			
News with lower term premium	−4.0 (0.3)	−5.4 (0.2)	−4.3 (0.3)
Current rate with lower term premium	+2.2 (0.7)	−2.2 (0.2)	−3.1 (0.2)
Credit spread (annual yield, pp; average response through horizon)			
News with lower term premium	−1.0 (0.4)	−2.9 (0.3)	−1.8 (0.3)
Current rate with lower term premium	−2.3 (0.2)	−0.3 (0.1)	+0.8 (0.1)
Default (cumulative incidence, pp)			
News with lower term premium	—	−1.5 (0.2)	−3.7 (0.3)
Current rate with lower term premium	—	−0.8 (0.1)	−0.6 (0.1)

Notes: Entries are the benchmark-minus-restricted difference in shock responses, as in Table 13, to a one-standard-deviation expected-rate news shock or current-rate shock combined with a one-standard-deviation increase or decrease in the term-premium wedge in the shock year; each factor then follows its estimated law of motion. The responses to each monetary shock alone are the corresponding rows of Table 13, and the rows with a higher term premium repeat Table 14. Restricted firm policies and debt-price schedules, units, horizons, and the simulation design are as in Table 13. Horizon-1 default entries are omitted because default does not respond on impact under the model's timing. Monte Carlo standard errors of the mean are in parentheses.

F.3 Joint Tightening: Lower Term Premium and Current-Rate Shocks

Table 35 completes the joint tightening experiment of Section 8.3 with the mirror case of a one-standard-deviation *decrease* in the term-premium wedge and with the same three cases for a current-rate shock. Two facts stand out. First, when long-duration financing becomes cheaper as expected refinancing costs rise, flexible firms shorten considerably less than under news alone (4.0 rather than 8.8 percentage points on impact) and retain more maturity protection; the investment cushion after three years is 2.7 percentage points, close to the news-only value, and the five-year reduction in default rises from 2.7 to 3.7 points. Second, the interaction between the price of duration and the value of flexibility is specific to expected refinancing risk. Adding the same term-premium wedge increase to a current-rate shock leaves the three-year investment value of flexibility unchanged (4.5 against 4.4 percentage points), whereas it raises it from 2.4 to 3.9 points under expected-rate news; a lower wedge reduces it under the current-rate shock (3.3 points) but not under news. The incremental maturity response to the wedge is similar for the two monetary shocks (shortening of

9.6 against 3.4 points for the current-rate shock, and 16.4 against 8.8 for news), so the difference lies in the payoff of that adjustment: when expected future refinancing costs rise, maturity choice is an insurance decision and the price of duration changes the value of having it; under a current-rate shock the same repricing of duration adds a similar cost to both economies.

F.4 Monetary Transmission across Term-Premium Aggregate States

Figures 19 and 20 report the full responses underlying Figure 8, including gross issuance, credit spreads, and paired high-minus-low differences across simulation histories. The exercise uses the flexible-maturity economy throughout and therefore characterizes state-dependent monetary transmission rather than isolating the contribution of maturity flexibility.

Two additional financing responses are noteworthy. First, following expected-rate news, the contraction in gross issuance is initially stronger when the term premium is high. After three years, cumulative issuance falls by 3.5 percentage points of debt relative to the no-shock path, compared with 1.9 points in the low-premium state. By year five, however, the corresponding declines are 4.0 and 4.3 points, respectively. Outstanding debt nevertheless contracts more in the high-premium state, by 9.1% compared with 6.0% after five years. Second, credit-spread responses differ in timing. Spreads increase more on impact when the term premium is high (3.3 versus 1.9 percentage points), whereas by year three the increase is larger in the low-premium state (2.7 versus 1.1 points). These patterns indicate that the prevailing term premium affects both the timing of borrowing adjustments and the evolution of financing costs.

The paired comparisons confirm that these differences are statistically meaningful. Following expected-rate news, the high-minus-low differences in cumulative investment are significant at the 1% level at every horizon. Differences in the long-term issuance share are also significant at the 1% level except in year two, when the difference briefly reverses and is significant at the 5% level. The difference in cumulative default is significant in years two and three (0.45 and 0.48 percentage points, respectively), but not in years four and five. Following current-rate shocks, investment differences are significant at the 1% level throughout the horizon, while default differences are significant from year two onward. Differences in the long-term issuance share are not significant at the 5% level in years two and three. Thus, higher term premia amplify the real effects of both shocks, while differences in maturity adjustment are generally more pronounced following expected-rate news.

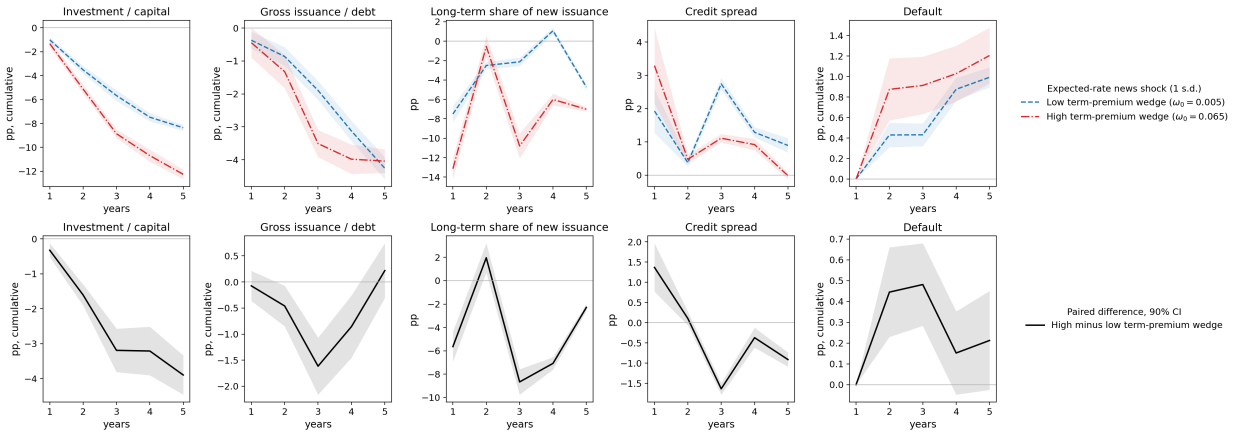


Figure 19: Expected-Rate News across Aggregate Term-Premium States

Notes: Top row: responses of the benchmark economy to a one-standard-deviation expected-rate news shock arriving when the term-premium wedge is low ($\omega_0 = 0.005$, dashed) or high ($\omega_0 = 0.065$, dash-dotted). The economy starts in both cases from the same stationary cross-section of firms as in Table 13; the wedge then reverts toward its mean along its estimated law of motion, and responses are measured relative to the same path without the innovation. Investment, gross issuance, and default are cumulative; the long-term share of new issuance and the credit spread are level responses at each horizon. Bottom row: the high-minus-low difference in responses, computed history by history, since the two states share the same simulated economies and idiosyncratic draws. Shaded bands are 90 percent Monte Carlo confidence intervals of the mean across 1,000 simulated economies of 10,000 firms; in the bottom row they are intervals for the mean paired difference, so a band that excludes zero indicates a significant state difference.

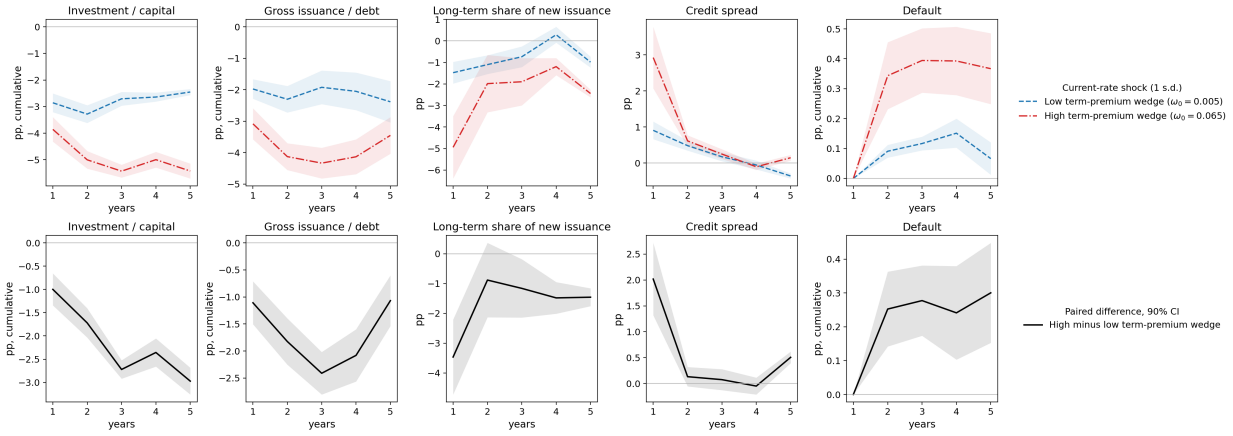


Figure 20: Current-Rate Shocks across Aggregate Term-Premium States

Notes: Top row: responses of the benchmark economy to a one-standard-deviation current-rate shock arriving when the term-premium wedge is low ($\omega_0 = 0.005$, dashed) or high ($\omega_0 = 0.065$, dash-dotted). Bottom row: the high-minus-low difference computed history by history. Initial conditions, the wedge path, the definition of responses, and the bands are as in Figure 19; the horizon is five years for comparability.

F.5 The 2018–24 Path: Inherited Liabilities and Anticipatory Maturity Adjustment

We initialize the economy in 2018 and feed in the realized annual paths of short rates, expected-rate news, and the term-premium wedge. This initialization allows firms to accumulate longer-maturity liabilities during the preceding period of negative short rates and low term premia. To

Table 36: The 2022–24 Tightening: Maturity Adjustment and Real Effects

Cumulative through 2024, pp	LT share	Issuance	Investment	Default	Credit spread
Benchmark economy					
Realized short-rate path	−23.5 (0.1)	−12.6 (0.1)	−27.4 (0.1)	+7.9 (0.1)	+17.6 (0.2)
Short-rate path + expected-rate news	−23.2 (0.1)	−10.8 (0.1)	−25.6 (0.1)	+3.8 (0.1)	+8.7 (0.2)
Full realized path	−35.7 (0.1)	−11.8 (0.1)	−31.1 (0.1)	+5.2 (0.1)	+11.2 (0.2)
Full path, term premium fixed at 2021	−9.2	−12.4	−23.7	+5.0	+12.9
Restricted-maturity economy					
Realized short-rate path	+0.7 (0.0)	−18.5 (0.2)	−34.8 (0.2)	+9.2 (0.1)	+24.5 (0.2)
Short-rate path + expected-rate news	+1.7 (0.0)	−17.2 (0.2)	−28.5 (0.1)	+9.9 (0.1)	+20.0 (0.1)
Full realized path	+1.7 (0.0)	−18.4 (0.2)	−36.1 (0.3)	+10.1 (0.0)	+23.9 (0.2)
Full path, term premium fixed at 2021	+0.0	−15.4	−29.4	+7.6	+16.5

Notes: Responses to the realized 2022–24 monetary path and selected counterfactual paths, relative to each economy's no-shock path from the 2018-initialized distribution. The long-term issuance share and credit spread are averages of annual responses; issuance, investment, and default are cumulative through 2024. The fixed-term-premium experiment holds the wedge at its 2021 level from 2022 onward, retaining the realized short-rate and expected-rate news paths. Means across 10 simulated economies of 10,000 firms; Monte Carlo standard errors are reported in parentheses where available.

accommodate these conditions, we extend the short-rate and term-premium grids by one lower node, leaving the calibrated parameters unchanged. Responses are measured relative to each economy's corresponding no-shock path.

Table 36 reports the effects of the historical monetary sequence and selected counterfactual paths. Along the full realized path, maturity flexibility mitigates cumulative investment losses by 5.0 percentage points and reduces cumulative default by 4.9 percentage points relative to the restricted-maturity economy. Holding the term-premium wedge at its 2021 level reduces the cumulative investment contraction by 7.4 percentage points in the flexible economy and 6.7 points in the restricted economy. Thus, higher term premia amplify investment losses along the historical tightening path.

Expected-rate news also changes the timing of maturity adjustment. As shown in Figure 21, firms extend new-debt maturity during 2019–21 and enter tightening with longer-maturity liabilities. Under the realized short-rate path alone, the long-term share of new issuance falls from 69% in 2022 to 25% in 2023. Adding expected-rate news brings shortening forward: the corresponding shares are 50% and 34%.

This anticipatory adjustment is not accompanied by additional deleveraging. Outstanding debt and leverage remain virtually unchanged between the two paths, while the share of debt due in 2023 increases from 21.7% to 24.2%. Nevertheless, the path with expected-rate news exhibits less severe distressed borrowing in 2023: the issuance-weighted mean spread is 21 percentage points, compared with 55 points under the short-rate-only path. Cumulative default through 2024 is also lower, at 3.8 rather than 7.9 percentage points in the flexible economy, whereas adding the same news increases default from 9.2 to 9.9 points under restricted maturity.

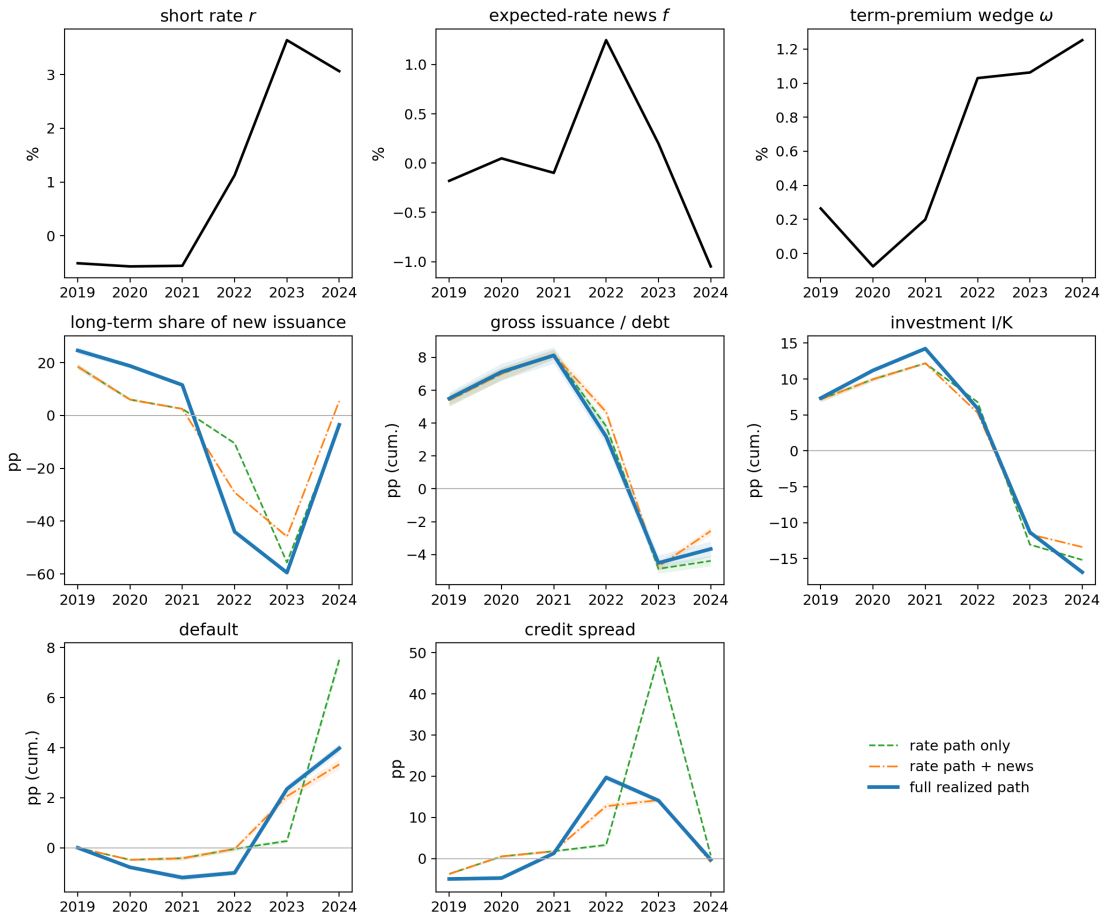


Figure 21: The 2018–24 Path: Shocks and Benchmark Responses. Top row: the realized annual path of the short rate, expected-rate news, and the term-premium wedge (in term-premium-equivalent yield) fed to the model from 2019, with the 2018 state used to condition the initial cross-section. Other panels: benchmark responses to the realized rate path alone, to the rate path with expected-rate news, and to the full path, relative to the no-shock path; gross issuance, investment, and default are cumulated from 2019. Means across 10 simulated economies; shaded bands are 95 percent Monte Carlo intervals.

Table 37: Investment Effects and Survivorship

	Current Rate	Expected-Rate News	Term-Premium Wedge
Survivor mean	+4.2	+2.9	+3.5
Initial-cohort mean	+4.1	+3.6	+3.1
Cohort aggregate	+5.3	+3.9	+3.5

Notes: Entries report the benchmark-minus-counterfactual difference in cumulative investment responses over five years, in percentage points of capital, as in Table 13. Positive values indicate that endogenous maturity flexibility preserves investment. "Survivor mean" compares the surviving cross-section in each economy, including replacement entrants. "Initial-cohort mean" restricts the sample to surviving firms of the cohort alive at the shock date. "Cohort aggregate" retains the full initial cohort, counting the capital of firms that subsequently default as destroyed.

Results are averages across 1000 simulated economies.

F.6 Investment Effects and Survivorship

Because the maturity restriction changes default, differences in average investment could reflect changes in the composition of surviving firms rather than changes in firms' investment decisions. Table 37 recomputes the five-year investment gap of the main counterfactual on the surviving cross-section, on the cohort of firms alive at the shock date, and on the full initial cohort with the capital of defaulting firms counted as destroyed. The gap remains positive in every case and is larger, not smaller, for the cohort aggregate after current-rate and expected-rate shocks, so the main counterfactual does not arise mechanically from differential survivorship.

F.7 Contract Structure: Short-Only and Long-Only Economies

The main counterfactual removes the response of maturity to aggregate conditions. A complementary experiment removes the contract menu itself. Holding the calibrated parameters fixed, we re-solve the firm problem in a short-only economy, in which all new borrowing is one-period debt ($\ell_{it} = 0$), and in a long-only economy, in which all new borrowing is the amortizing long-term claim ($\ell_{it} = 1$), with prices re-solved in each. Each economy is simulated from its own stationary distribution under common random numbers, so the comparison is between economies that have adapted to their contract structure. It is therefore a different experiment from the main counterfactual, which holds inherited balance sheets fixed at the shock date. The adaptation is large: the short-only economy holds about one-third of the benchmark's debt, all of it due within the year, and the long-only economy about two-thirds, with rollover exposure pinned at the amortization floor. Panel A of Table 38 reports the levels of the three economies along their no-shock paths, and Panel B the cumulative responses to one-standard-deviation shocks at three and five years, as benchmark-minus-alternative differences. The stationary comparisons differ from the levels reported by Deng and Fang (2022). In their model, the investment rate is lower when firms can issue only short-term debt (20.0% against 20.5% in their benchmark). In ours, the short-only economy has a higher investment-to-capital ratio than the benchmark (11.8% versus 10.9%), but a 21% smaller capital stock, substantially less debt relative to capital (0.35 versus 0.88), and a lower default rate (0.3% versus 1.7%). The absence of long-term borrowing is therefore associated with a smaller and less indebted economy, rather than a lower investment rate. The long-only economy is closer to the benchmark in capital and leverage, although its default rate is higher (2.1% versus 1.7%). These level comparisons characterize economies that have adapted to different contract menus; they do not isolate the contribution of maturity adjustment for firms entering a shock with identical balance sheets. The two restricted economies reveal why endogenous maturity choice is valuable. Under current-rate tightening, both contract extremes generate substantially larger investment losses than the benchmark: short-only firms are fully exposed to refinancing, while long-only firms cannot economize on long-duration funding when financing conditions deteriorate. Expected-rate news exposes a different trade-off. The long-only economy suffers larger investment losses, higher default, and wider spreads because firms are forced to purchase duration even as its current price deteriorates. Short-only firms initially lose more investment than in the benchmark, but the difference largely disappears by five years because the economy has adapted to its contract structure by carrying substantially less debt. A term-premium shock reverses the comparison: the short-only economy is mechanically insulated from the direct increase in the price of duration, whereas the long-only economy experiences lower investment and issuance. Thus, no fixed contract structure dominates across financing shocks. The value of the benchmark contract menu is that firms can adjust the amount of refinancing insurance they purchase to the source of the change in financing conditions.

Table 38: Short-Only and Long-Only Economies

Panel A. Levels along the no-shock path (averages over years 3–10; MC s.e.)	Benchmark	Short debt only	Long debt only
Investment rate I/K (%)	10.9 (0.1)	11.8 (0.1)	11.0 (0.1)
Mean capital	0.511 (0.002)	0.406 (0.001)	0.497 (0.002)
Debt to capital	0.88 (0.00)	0.35 (0.00)	0.68 (0.00)
Default rate (%)	1.7 (0.0)	0.3 (0.0)	2.1 (0.0)
Rollover exposure s	0.23	1.00	0.19
Long-term issuance share	0.79	0.00	1.00

Panel B. Shock responses, benchmark minus alternative

	Current rate		Expected-rate news		Term-premium wedge	
Horizon (years)	3	5	3	5	3	5
Investment (cumulative I/K, pp)						
Short debt only ($\ell = 0$)	+2.7 (0.2)	+4.1 (0.1)	+1.7 (0.2)	+0.1 (0.1)	−1.1 (0.1)	−0.9 (0.0)
Long debt only ($\ell = 1$)	+2.9 (0.2)	+4.5 (0.2)	+0.8 (0.3)	+1.4 (0.3)	+1.3 (0.2)	+0.9 (0.2)
Debt issuance (cumulative m/B, pp)						
Short debt only ($\ell = 0$)	+5.5 (0.2)	+4.3 (0.2)	+5.3 (0.3)	+0.6 (0.2)	−1.3 (0.2)	+0.3 (0.1)
Long debt only ($\ell = 1$)	+2.7 (0.2)	+2.8 (0.3)	−0.6 (0.2)	−1.3 (0.3)	+3.8 (0.2)	+4.0 (0.3)
Default (cumulative, pp)						
Short debt only ($\ell = 0$)	+0.1 (0.1)	+0.1 (0.1)	+0.4 (0.1)	+0.4 (0.1)	0.0	0.0
Long debt only ($\ell = 1$)	−0.5 (0.1)	−0.7 (0.1)	−1.6 (0.2)	−2.1 (0.2)	0.0	+0.3 (0.0)
Credit spread (average annual, pp)						
Short debt only ($\ell = 0$)	+0.6 (0.2)	+0.2 (0.1)	+1.5 (0.2)	+0.6 (0.1)	−0.4 (0.0)	−0.7 (0.0)
Long debt only ($\ell = 1$)	−0.4 (0.1)	−0.5 (0.1)	−0.3 (0.1)	−0.7 (0.1)	−0.4 (0.1)	−0.5 (0.1)

Notes: Panel A reports averages over years three to ten of the no-shock paths of the three economies, each initialized from its own stationary distribution with aggregate states held at their means. Because the stationary distributions reflect fluctuating aggregate states, all three economies adjust slowly along this path (capital rises by 2–4% and the investment rate falls by about one percentage point over the window); the differences between economies are stable across it. Panel B reports the benchmark-minus-alternative difference in cumulative responses to a one-standard-deviation shock (shock minus no-shock path in each economy); positive values indicate a higher response in the benchmark. Positive investment and issuance entries therefore mean that the benchmark loses less than the alternative economy, and negative default and spread entries mean smaller increases in default and financing costs in the benchmark. Both alternative economies use the calibrated parameters, with prices re-solved, and are simulated from their own stationary distribution, unlike the main counterfactual, which starts every economy from the benchmark cross-section; all economies face common random numbers. The short-only economy issues no long-term debt and is therefore mechanically unaffected by the term-premium wedge. Means across 1000 simulated economies of 10,000 firms; Monte Carlo standard errors of the mean in parentheses.